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Master thesis

Energy-Guided Sampling for Controllable Text Generation: From Langevin Dynamics to Amortized Inference

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Stuttgart, 29. Juli 2026

(Sarthak Singh)

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Abstract

Controllable text generation asks a language model for text satisfying a property of the sequence as a whole, which cannot be verified until the sequence is complete.[2] One family of controllable generation methods pursues this without retraining the model, by treating its frozen likelihood as an *energy function*, i.e. a score that is low for text the model considers good, and sampling from that score with Langevin dynamics, which repeatedly nudges the text in the direction the gradient recommends, with an additive constraint supplied at generation time. This thesis tests the assumption on which that construction rests, namely that the gradient indicates which token should be placed where.

The evaluation uses masked token recovery, in which one token of a real sentence is replaced at random and the sampler must restore it, over 145 configurations and five frozen energy functions: a GPT-2 Large fine-tuned on stories, three variants of it tuned further with a GFlowNet, which learns to generate in proportion to a reward, and a Llama-3 8B as a cross-architecture control.

Results show that the direction of the gradient is not usable: replacing it with a random direction of the same magnitude produced no reliable difference, an equivalence certified over 1000 paired sequences, the correct token was recovered in 0.0% of attempts in 139 of the 145 configurations, and a sweep over the settings that make the gradient decisive found none at which it helped. The failure is attributable to the coordinate in which the derivative is taken, since differentiating the same likelihood with respect to the token's indicator over the vocabulary rather than its input embedding introduces the candidate's own conditional log-probability, to which that gradient is structurally blind; in the same sampler and on the same weights that derivative recovers 40% of tokens on GPT-2 and 41% on Llama-3, where the input-embedding gradient never exceeds 2%. The gap that remains is associated with conditioning rather than training, as a single chain accepting by the exact energy and varying only the proposal recovers 0.5% from a uniform draw, 23.5% from the model's own left-to-right conditional, 39% from a diffusion model trained to estimate scores, and 44.5% from a masked language model never trained on scores. These findings indicate that inference-time control on a frozen autoregressive model should differentiate the token indicator, or draw proposals from the output distribution, rather than from the input embeddings.

[2] A length constraint is the simplest example: whether a sentence stays under twenty words is settled only once the sentence is finished.

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1 Introduction

The dominant paradigm behind modern **language models**, by which this thesis throughout means a neural network that assigns a probability to a sequence of tokens, is **autoregressive generation**: the model factorizes the probability of a sequence into a product of next-token conditionals and produces text one token at a time, from left to right (Radford et al., 2019; Brown et al., 2020). This factorization is what makes these models trainable at scale and fast at inference, and it is one reason among several that they work as well as they do; the volume of text they are trained on and the Transformer architecture that consumes it are the other principal ingredients (Vaswani et al., 2017; Brown et al., 2020). The factorization is also, as this thesis argues, the reason one popular approach to controlling them does not work.

That same factorization is the source of a practical limitation. Because generation proceeds strictly left to right and each token is fixed the moment it is emitted, the model has no mechanism for revising an earlier decision in light of a later one: if the second half of a sentence reveals that the first half has drifted in logic or tone, the model cannot go back, only continue or discard and restart. This is no obstacle when the goal is a fluent continuation, which is what these models are trained to produce, but it becomes one the moment we want to *control* what they produce.

The reason control is hard has a precise structure. Many of the properties we care about most, the overall sentiment of a passage, its topical focus, its adherence to a factual constraint, the absence of toxic content, are **global properties** of a completed sequence rather than local properties of any single token. A greedy, left-to-right decision about the next token cannot see such a property, because the object that would be evaluated against it, the finished text, does not yet exist when the decision is made. Enforcing a global constraint autoregressively therefore requires generating many complete sequences and keeping those that happen to satisfy it, which is wasteful when the constraint is easy to satisfy and effectively impossible when it is not. Steering a language model toward such global objectives while keeping its output fluent is the problem of **controllable text generation**, and it is the subject of this thesis.

1.1 Established Routes to Control Generation and Their Costs

There are, broadly, two established ways of imposing control on a language model, and each carries a cost that motivates the search for a third.

The first is to bake the desired behaviour into the parameters, through supervised fine-tuning on curated data or, more commonly today, reinforcement learning from human feedback (Ouyang et al., 2022). This is effective and now standard, and the aligned language models in wide use are its product, but the control it provides is *static*: a model tuned to be helpful and non-toxic embodies those constraints and no others, and a genuinely new constraint requires new preference data and a new and expensive training run.

The second is to leave the model untouched and filter or re-rank its outputs, generating many independent candidates and discarding those that violate the constraint. This preserves the model but inherits the intractability described above, since for a constraint a fluent generation satisfies only rarely by chance, the number of candidates one must draw grows without practical bound.

What both approaches lack, and what would be genuinely valuable, is the ability to introduce a fresh constraint at the moment of generation, cheaply, without touching the underlying model and without generating a haystack to find a needle. That capability is **inference-time control**: the model's parameters are fixed once and never revisited, and the only thing that differs between one constraint and the next is a term handed to the decoding procedure while it runs. This is the capability that the energy-based approach promises, and it is the promise that this thesis sets out to examine.

1.2 Energy-Based Control and Its Central Assumption

An elegant alternative reframes decoding itself. Rather than generating a sequence token by token, one can define a probability distribution over the space of all possible sequences and *sample* from it directly. Following the standard formulation of **energy-based models** (LeCun et al., 2006; Hinton, 2002), the distribution is written in the Boltzmann form

$$p(\boldsymbol{x}) = \frac{1}{Z}\exp\bigl(-E(\boldsymbol{x})\bigr), \tag{1}$$

where $\boldsymbol{x} = (x_1, \ldots, x_T)$ is a sequence of T tokens drawn from a vocabulary V, $E(\boldsymbol{x}) \in \mathbb{R}$ is the **energy** assigned to that sequence, and $Z \in \mathbb{R}$ is a normalizing constant. The minus sign in the exponent fixes the direction of the correspondence between the two quantities, and it is worth reading off explicitly: since $\exp(-E)$ falls as E rises, *low energy corresponds to high probability and high energy to low probability*, so every statement about one is the mirror image of a statement about the other, and a sequence the model considers good is one of low energy. Any scalar-valued function of a complete sequence can serve as E; the choice this thesis studies, the negative log-likelihood of a language model, is made in Section 2.1. The constant Z is intractable to compute, because it sums over every possible sequence, but the great advantage of the sampling framework is that it need never be computed: the methods used to draw samples from (1) depend only on *ratios* of probabilities or on gradients of the log-probability, in both of which the constant cancels.

The attraction of this view for controllable generation is that the energy can be assembled additively from independent pieces. One writes

$$E(\boldsymbol{x}) = -\log p_{\mathrm{LM}}(\boldsymbol{x}) + \lambda\, C(\boldsymbol{x}), \tag{2}$$

where the first term is the negative log-likelihood of a frozen, pretrained language model and supplies *fluency*, pulling samples toward text the model considers natural, and the second term is an arbitrary differentiable *constraint* $C : V^T \to \mathbb{R}$, which scores a complete sequence for the property wanted, weighted by a scalar coefficient $\lambda > 0$ that trades off fluency against constraint satisfaction. Because the constraint is simply added on, it can be specified at the moment of generation and swapped freely. A practitioner who wants positive sentiment today writes C as the negative log-probability of the positive class under a sentiment classifier; one who wants factual consistency tomorrow writes a different C; and in both cases the language model is untouched. This is the promise of plug-and-play, inference-time controllable generation, and it is an appealing one. A substantial and growing body of recent work pursues it, through gradient-guided decoders such as PPLM (Dathathri et al., 2020), FUDGE (Yang and Klein, 2021), and GeDi (Krause et al., 2021), and, most directly related to this thesis, through sampling-based methods such as COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), which draw samples from the energy in (2) using continuous relaxations and dynamics derived from Langevin sampling.

Every method in this family rests on one assumption, which is worth isolating. To sample from the energy in (2) using a gradient-guided method, two things must be true. First, one must be able to compute a gradient of $-\log p_{\mathrm{LM}}(\boldsymbol{x})$ that points, at least on average, in the direction of better sequences. Second, one must be able to *follow* that gradient across the space of possible texts, moving from a worse sequence to a better one by steps the gradient recommends. Taken together, these amount to the presupposition that the frozen likelihood of an autoregressive language model, viewed as a function over whole sequences, defines a *landscape that a sampler can actually navigate*: a surface with meaningful slopes, whose downhill direction can be computed locally and trusted to lead somewhere good. This presupposition is intuitive, and it is almost never tested directly in the literature that relies on it. Testing it, and explaining what is found, is the subject of this thesis. In the settings tested here that presupposition does not hold. Following the **input-embedding gradient**, the derivative of the likelihood with respect to the vector the model looks up for the token at the position being changed, proves no more effective than following a **norm-matched random direction**, a direction drawn at random and then rescaled to the exact length of the true gradient, so that the two differ in direction and in nothing else. Section 5.3 states the finding precisely. This outcome is stated here, before the research questions are posed, because the thesis is organized as an explanation of that result rather than as a reveal.

That explanation has a definite shape, and setting it out now is what makes the order of the experiments legible. A null of this kind admits three candidate causes, and they are not equivalent. Under **Hypothesis A** the target is at fault: the frozen likelihood, evaluated on the counterfactual sequences an in-place revision must consider, is not an object any local search could profit from, in which case the choice of proposal mechanism is beside the point. Under **Hypothesis B** the target is serviceable but the model does not contain what a local proposal needs, because teacher forcing, the training scheme that always supplies the true prefix and asks only for the next token, shapes those next-token predictions and never asks the likelihood of a whole sequence to rank substitutions inside it; on this reading the deficiency is one of training objective, and only a differently trained model could repair it. Under **Hypothesis C** the information is present in the frozen model and the parameterization discards it: differentiating with respect to the input embedding registers only part of what a substitution changes, while a proposal that reads the model's output side recovers the discarded part, and one that may additionally condition on the right context recovers more still. The results chapter is arranged as an elimination in that order. Gradient-free rescoring and Gibbs sampling on the identical energy dispose of A; re-analysing the same measurements under a derivative taken with respect to the token's identity rather than its embedding, and substituting that alternative into the same sampler on the same frozen weights, disposes of B; and a ladder of proposals run inside one exact-energy chain confirms and refines C.

1.3 From Sampling to Amortization

The natural response to a gradient that will not cooperate is to stop using the gradient. There is a version of the energy-based idea that does not differentiate the energy at all. If a per-sequence energy can still be *evaluated*, that is, if one can take a finished sequence and compute its score, even when one cannot usefully *differentiate* that score to decide how to improve a sequence, then one can treat the language model as a black box that ranks finished sequences and learn a separate policy that generates in proportion to that ranking. This is the idea of **amortized inference**: rather than paying the cost of a search procedure afresh for every sample, one pays a one-time training cost to learn a generator that produces good samples directly.

The most principled recent realization of this idea for sequence generation is the **Generative Flow Network**, or GFlowNet (Bengio et al., 2021; 2023), which trains a generative policy so that the probability of producing a complete sequence is proportional to its reward. Applied to language models, GFlowNets have been proposed as a way to sample diverse, high-reward text without the mode-collapse that afflicts reward-maximizing reinforcement learning (Hu et al., 2024). The second half of this thesis asks a specific question about this approach: does amortizing the search, and thereby never touching the gradient, escape the difficulties that defeat the gradient-guided samplers? The answer, established in Section 5, is that it does not, and it separates into a statement about the learned policy and a statement about the energy that policy leaves behind. Because the initial sampling experiments failed, the study developed into a diagnostic investigation of the underlying mechanisms.

1.4 Research Questions and Contributions

The work is organized around three research questions, the third of which separates into two that the earlier drafts of this study conflated. The results section answers them in turn and the discussion section revisits each explicitly.

RQ1. Does the input-embedding gradient of a frozen autoregressive sequence likelihood provide a useful proposal direction for discrete token revision, and if not, which alternative local quantities do?

RQ2. What is the role of the Metropolis–Hastings correction in each of the discrete and continuous samplers, and does equipping them with the exact accept-reject correction for the proposal they implement also make them work?

RQ3a. Does a GFlowNet trained on the frozen likelihood as its reward learn a policy that generates high-reward text?

RQ3b. Does the energy induced by that tuning become more navigable by the local input-embedding surrogate than the energy of the untuned base?

RQ3 is split because a GFlowNet is an amortized *policy* while the Langevin samplers navigate an *energy*, and running the samplers on a tuned model tests the geometry of that tuned energy rather than whether the policy amortizes sampling successfully. The two are related but not identical, and reporting them as one answer would blur a distinction the evidence supports.

One further question belongs to the study but not to its central diagnosis, and it is posed here as an extension.

E. Can an additive constraint term steer generation toward a target attribute on this energy landscape, as the plug-and-play framework requires?

This is the capability the whole programme was meant to deliver, and the thesis reports what happened when it was attempted; but it is an application of the sampling machinery whose prerequisite RQ1 tests, so it is answered after the primary diagnosis rather than alongside it, and the contributions below do not rest on it.[3]

In answering these questions the thesis makes five contributions.

1. A controlled ablation separating the *direction* of the input-embedding gradient from its *magnitude*, run across five energy functions and extended by a sweep over step size and proposal temperature that spans proposal entropies from uniform to deterministic. The direction gave no advantage over a norm-matched random one anywhere in that range, an equivalence certified at a margin fixed in advance over a thousand paired sequences.

[3]These questions are the operationalized forms of those posed in the thesis proposal, refined once the study became diagnostic rather than constructive. The proposal's RQ1 and RQ1a, on whether faithful Langevin dynamics can sample the frozen likelihood and how the two samplers compare, are consolidated into RQ1; its RQ1b, on the Metropolis–Hastings correction, becomes RQ2; its RQ3a, on amortizing the energy with a GFlowNet, becomes RQ3a and RQ3b; its RQ2a, on additive constraint steering, becomes the extension question E; and its RQ2b, on whether energy-based recovery improves on ordinary autoregressive decoding, is answered by the gradient-free baselines of Section 5.3.3.

2. An identification of what the input-embedding derivative discards, and a demonstration that it is recoverable. The first-order surrogate is uncorrelated with the true energy change over the entire admissible range of substitution distances, and the self-and-future decomposition shows why; re-analysing the identical measurements under a derivative taken in the token-indicator coordinates restores a strong rank correlation, and substituting that surrogate into the same sampler on the same frozen weights lifts exact recovery from at most two percent to forty.

3. A decomposition of the Metropolis–Hastings acceptance ratio in both samplers, which attributes the continuous sampler's collapse to the proposal term rather than the target term and shows that in the discrete sampler the accept/reject filter, not the gradient, supplies the entire selection pressure. Alongside it, a measurement of the *likelihood trap* on the study's own models, in which the lowest-energy text is the most degenerate.

4. A documented taxonomy of three failure modes of GFlowNet fine-tuning on a frozen-likelihood reward, together with the separate energy-level experiment showing that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base even though tuning moved that energy by tens of nats.

5. A controlled proposal ladder inside one exact-energy chain, in which recovery rises with what the proposal may condition on rather than with how it was trained: from the input-embedding gradient and a uniform draw at zero, through the frozen model's own left-to-right conditional, to a score-trained diffusion model and a masked language model that was never score-trained and performs best of all.

1.5 Structure of the Thesis

Section 2 develops the technical background: the energy-based view of generation, Langevin dynamics, the Metropolis–Hastings correction, the two samplers studied here, discrete diffusion, and GFlowNets. Section 3 reviews the literature on controllable and energy-based text generation and states the gap this work addresses. Section 4 describes the task, the five energy functions, the sampler configurations, the evaluation metrics, and the diagnostic experiments. Section 5 follows the elimination just described: it calibrates the samplers and establishes the null, disposes of Hypothesis A with gradient-free baselines on the identical energy, disposes of Hypothesis B with the token-indicator re-analysis and the sampler built on it, and closes with the proposal ladder that confirms Hypothesis C, alongside the GFlowNet experiments and the constrained-generation extension. Section 6 answers each research question, sets out the shared problem and the distinct mechanisms beneath it, and states the limitations and implications. Section 7 concludes.

2 Background

This section develops the background the thesis relies on: the energy-based view of sequence generation, Langevin dynamics, the Metropolis–Hastings correction, the two concrete samplers studied in the experiments, score matching and the concrete score, and GFlowNets as an amortized alternative. The aim throughout is to expose the assumption each tool makes rather than to survey the tools exhaustively.

2.1 Autoregressive Language Models as Energy Functions

An autoregressive language model defines a probability distribution over a sequence of tokens $\boldsymbol{x} = (x_1, \ldots, x_T)$ by the chain rule of probability,

$$\log p_{\mathrm{LM}}(\boldsymbol{x}) = \sum_{t=1}^{T} \log p_{\mathrm{LM}}(x_t \mid x_{<t}), \tag{3}$$

where $x_t \in V$ is the token at position t, the vocabulary V is a finite set of size $|V|$, and $x_{<t} = (x_1, \ldots, x_{t-1})$ denotes the prefix preceding position t, empty when $t = 1$. Each conditional $p_{\mathrm{LM}}(x_t \mid x_{<t})$ is produced by a neural network, here a Transformer (Vaswani et al., 2017), applied to that prefix: it reads the prefix, produces a logit

vector in $\mathbb{R}^{|V|}$, and a softmax turns it into a distribution over the $|V|$ tokens. Read in words, (3) says that the score of a whole sequence is the sum of the scores its own model assigns to each token in turn, each judged against everything to its left and nothing to its right. The model is trained by **teacher forcing**, maximizing (3) on a large corpus with the true prefix supplied at every position, which decomposes into a sum of independent next-token prediction problems.

It is worth being precise about what this training objective does and does not shape, because Hypothesis B of Section 1.2 is stated in exactly these terms. The objective shapes each conditional $p_{\mathrm{LM}}(x_t \mid x_{<t})$ to be an accurate predictor of the next token given a *correct* history drawn from the data distribution, and modern models optimize it extraordinarily well. What it never does is ask the joint quantity $\log p_{\mathrm{LM}}(\boldsymbol{x})$ to behave in any particular way as $\boldsymbol{x}$ is varied away from the data, for instance by substituting one token for another in the middle of an otherwise fixed sequence. The model is never trained on such counterfactual sequences, never asked to assign them calibrated scores, and never penalized for the shape of the surface it induces over them, so nothing internal to the objective makes the joint log-likelihood smooth, or well-calibrated as a measure of quality, or possessed of a gradient pointing toward better sequences. That is a real gap, and it makes Hypothesis B the natural first explanation of a failure; Section 5 nonetheless finds it too strong, since the quantity a local proposal needs turns out to be present in the trained conditionals and reachable without any change of objective.

With that caveat noted rather than resolved, the energy-based reformulation is immediate. Identifying the Boltzmann form of (1) with the language model, one simply sets

$$E_{\mathrm{LM}}(\boldsymbol{x}) = -\log p_{\mathrm{LM}}(\boldsymbol{x}), \tag{4}$$

so that sequences of high probability are sequences of low energy. Sampling text from the language model is then, formally, sampling from the Boltzmann distribution of this energy. For controllable generation an additive constraint $\lambda\, C(\boldsymbol{x})$ is appended as in (2), where C is typically the negative log-probability an attribute classifier assigns to the desired label, so low energy means both fluent and on-attribute. The central computational question, to which the next two sections are devoted, is how one draws such samples at all.

2.2 Langevin Dynamics

Suppose, for a moment, that the state $\boldsymbol{s}$ lived in a continuous space $\mathbb{R}^D$ and that the energy were differentiable everywhere. Then one could sample from the distribution $p(\boldsymbol{s}) \propto \exp(-E(\boldsymbol{s}))$ using **Langevin dynamics**, a gradient-guided Markov chain Monte Carlo method with a long history in statistics and physics (Roberts and Tweedie, 1996; Welling and Teh, 2011). The discretized update, which is what one actually runs, is

$$\boldsymbol{s}_{t+1} = \boldsymbol{s}_t + \frac{\epsilon_t}{2}\nabla_{\boldsymbol{s}} \log p(\boldsymbol{s}_t) + \sqrt{\epsilon_t}\,\boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}), \tag{5}$$

where the state $\boldsymbol{s}_t \in \mathbb{R}^D$ and its update $\boldsymbol{s}_{t+1} \in \mathbb{R}^D$ are D-dimensional vectors, $\nabla_{\boldsymbol{s}} \log p(\boldsymbol{s}_t) \in \mathbb{R}^D$ is the gradient of the log-density at the current state, $\epsilon_t > 0$ is a scalar step size at iteration t, and the noise $\boldsymbol{\xi}_t \in \mathbb{R}^D$ is drawn from a standard Gaussian with the $D \times D$ identity as its covariance. In words, (5) says: take a step downhill in energy, then jog the state by a random amount whose size the step size also sets. The update has two parts, and their interplay is what distinguishes sampling from optimization.

The first part, the **drift**, is a step of size $\epsilon_t/2$ along the gradient of the log-density. Iterated alone it is gradient ascent on $\log p$ and drives the state monotonically toward the nearest mode, which would suffice if the goal were to *find* the most probable point; but the goal is a *representative sample*, and a distribution is more than its mode. The **diffusion** term injects fresh Gaussian noise at every step, scaled by $\sqrt{\epsilon_t}$, which prevents the chain from collapsing onto the mode: it knocks the state off the deterministic ascent path, letting it climb away from a mode, cross into a neighbouring region, and in the limit visit every part of the distribution with the correct frequency. The drift acts like gravity pulling a marble downhill on a landscape whose valleys are the high-probability regions, while the diffusion vibrates the surface so the marble does not settle in the first valley it finds. **Annealing** the step size toward zero on a suitable schedule makes the distribution of $\boldsymbol{s}_t$ converge to the target $p(\boldsymbol{s})$ rather than to its mode (Welling and Teh, 2011), but that fair-draw reading holds only under the technical conditions of the result, in particular a suitable schedule and a well-behaved landscape, and does not follow in general simply because the vibration was switched off.

The guarantees behind the method rest on regularity conditions that are easy to pass over and matter a great deal later. The discretization error of (5) grows with the step size, so the finite-step chain is an approximate sampler whose bias increases with ϵ_t (Roberts and Tweedie, 1996); the Metropolis-adjusted variant of the next section removes that bias in principle, but its acceptance rate degrades sharply once the drift loses smoothness, because the correction depends on the reverse move being predictable from the forward one. Non-asymptotic guarantees are established under smoothness of the target and typically under log-concavity or a related condition (Dalalyan, 2017; Durmus and Moulines, 2017), and in particular require the drift to be **Lipschitz continuous** (Roberts and Tweedie, 1996), so that a small step lands where the gradient at the starting point predicted. The continuous sampler of Section 2.4 violates these assumptions, because the drift it uses is computed from a differentiable pathway whose prediction target changes discontinuously at each Voronoi cell boundary. Section 5.2 measures the consequence.

2.3 The Metropolis–Hastings Correction

Any finite step size in (5) introduces a bias: the chain samples not from $p(\boldsymbol{s})$ but from a perturbed distribution that depends on the step size. That bias can be removed exactly by the **Metropolis–Hastings** (MH) correction (Metropolis et al., 1953; Hastings, 1970), which turns a biased proposal into an exact sampler by accepting or rejecting each move. Having proposed a move from $\boldsymbol{s}$ to $\boldsymbol{s}'$ according to a density $q(\boldsymbol{s}' \mid \boldsymbol{s})$, one accepts with probability

$$\alpha(\boldsymbol{s}' \mid \boldsymbol{s}) = \min\left\{1,\ \frac{p(\boldsymbol{s}')\, q(\boldsymbol{s} \mid \boldsymbol{s}')}{p(\boldsymbol{s})\, q(\boldsymbol{s}' \mid \boldsymbol{s})}\right\}, \tag{6}$$

where $\boldsymbol{s}, \boldsymbol{s}' \in \mathbb{R}^D$ are the current and proposed states, p is the density being sampled, $q(\cdot \mid \cdot)$ is the proposal density, and $\alpha \in [0, 1]$ is the resulting acceptance probability; if the move is not accepted the chain rejects it and remains at $\boldsymbol{s}$, counting the current state again. When Langevin dynamics is equipped with this correction the method is known as the Metropolis-adjusted Langevin algorithm, or MALA.

The acceptance ratio factors into two parts. The **target ratio** $p(\boldsymbol{s}')/p(\boldsymbol{s})$ compares the proposed state to the current one under the distribution being sampled and expresses the sampler's preference for good states. The **proposal ratio** $q(\boldsymbol{s} \mid \boldsymbol{s}')/q(\boldsymbol{s}' \mid \boldsymbol{s})$ compares how likely the mechanism was to suggest the reverse move against the forward one, and it is what enforces **detailed balance**, penalizing moves that are easy to make but hard to undo, which if accepted freely would let the chain drift and distort the distribution.

Two facts about the correction recur in the results. First, a great many energy-based decoders omit it entirely, and without it the noise-augmented gradient update of (5) is not a sampler but a biased optimizer with a particular kind of stochasticity, so calling its output a sample from the energy is strictly incorrect. The distinction is not pedantic, because a biased optimizer stopped early and a correct sampler run to convergence can produce very different outputs; how far it reframes any published system is a separate question, and one this thesis can answer only for the implementations it runs itself. Second, the proposal ratio is delicate to compute for a Langevin proposal, because the reverse density $q(\boldsymbol{s} \mid \boldsymbol{s}')$ must be evaluated from the drift *at the proposed point*. If that drift differs wildly from the drift at $\boldsymbol{s}$, which is what happens when the two points fall on opposite sides of a discontinuity, the reverse proposal places the original state deep in the tail of a Gaussian centred far away, driving $q(\boldsymbol{s} \mid \boldsymbol{s}')$ and the whole acceptance probability toward zero. Section 5.2 decomposes the ratio and reports which term is responsible.

2.4 Sampling in a Discrete Space: DLS and CLS

The difficulty that motivates the entire thesis is that text is not continuous. A sequence is a list of discrete tokens drawn from a finite vocabulary V, and there is no state *between* two tokens, so Langevin dynamics, which assumes a continuous differentiable space in which infinitesimal steps are meaningful, cannot be applied without modification. Two strategies for adapting it are studied here, and they differ precisely in where they confront the discreteness, which is why studying both is what allows the results to be attributed to the energy rather than to one sampler's idiosyncrasies.

The **Discrete Langevin Sampler** (DLS), following the discrete gradient-based samplers of Grathwohl et al. (2021) and Zhang et al. (2022), never leaves token space: its state is always a genuine sequence of tokens, and it uses the gradient of the log-likelihood with respect to the continuous input embedding of a masked position only as a *score* for building a categorical proposal over which token to place there next. Let $\boldsymbol{e}(x_i)$ be the embedding of the current token at masked position i and $\boldsymbol{g} = \nabla_{\boldsymbol{e}_i} \log p_{\mathrm{LM}}(\boldsymbol{x})$ the gradient with respect to it. For a candidate token v the proposal combines a term penalizing the displacement $\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\|$, which keeps proposals local, with a term proportional to $\frac{1}{2}\boldsymbol{g}^{\top}(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$, which biases the proposal toward tokens the gradient suggests will lower the energy. That second term is a **first-order Taylor surrogate** for the true change in log-likelihood a substitution would produce.

Written out in full, the discrete sampler adapts the continuous Langevin proposal of (5) to a categorical distribution over the vocabulary by the construction of Zhang et al. (2022). A Gaussian proposal centred at $\boldsymbol{s} + \frac{\epsilon}{2}\nabla \log p(\boldsymbol{s})$ assigns a candidate $\boldsymbol{s}'$ a log-density proportional to $-\frac{1}{2\epsilon}\left\|\boldsymbol{s}' - \boldsymbol{s} - \frac{\epsilon}{2}\nabla \log p(\boldsymbol{s})\right\|^2$, which on expanding and discarding terms constant in $\boldsymbol{s}'$ leaves a term linear in the displacement plus a quadratic penalty on the displacement itself. Transplanting this to the discrete setting, the candidate states are no longer arbitrary points of $\mathbb{R}^D$ but the embeddings of actual tokens. Write $v \in V$ for a **candidate vocabulary item**, that is, one of the $|V|$ tokens that could be placed at the position being resampled, and $\boldsymbol{e}(v) \in \mathbb{R}^D$ for its embedding, the row of the embedding matrix that the model looks up for it; the current occupant of position i is the token $x_i \in V$ with embedding $\boldsymbol{e}(x_i) \in \mathbb{R}^D$. This gives a categorical proposal over all $|V|$ candidates whose logit for token v is

$$\log q(v \mid x_i) \propto \tfrac{1}{2}\,\boldsymbol{g}^{\top}\big(\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\big) - \tfrac{1}{2\epsilon}\left\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\right\|^2, \tag{7}$$

where $\boldsymbol{g} = \nabla_{\boldsymbol{e}_i} \log p_{\mathrm{LM}}(\boldsymbol{x}) \in \mathbb{R}^D$ is the gradient of the sequence log-likelihood with respect to the input embedding at position i, the displacement $\boldsymbol{e}(v) - \boldsymbol{e}(x_i) \in \mathbb{R}^D$ is the move in embedding space that swapping x_i for v would make, and $\epsilon > 0$ is the step size, the same quantity as in (5). In words, (7) scores every token in the vocabulary by two competing considerations and samples one in proportion to the result. The first term, the **gradient-alignment term**, is the surrogate: it rewards candidates whose displacement from the current embedding aligns with the gradient, and it is precisely the first-order Taylor estimate of how much the log-likelihood would change if the substitution were made, since to first order $\log p(\boldsymbol{x}_{i\to v}) - \log p(\boldsymbol{x}) \approx \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$. The second term, the **distance term**, keeps the proposal local, penalizing distant candidates.[4] Everything the sampler knows about which token is good, as opposed to merely near, is contained in that gradient-alignment term, and if it is uninformative the proposal reduces to a distance-penalized random choice. The surrogate is therefore the sampler's entire claim to be gradient-guided rather than random, and Section 5.4 tests it directly.

The **Continuous Langevin Sampler** (CLS), following the continuous-relaxation approach of methods such as MuCoLa (Kumar et al., 2022), takes the opposite route: it relaxes the state into the continuous embedding space $\mathbb{R}^D$, runs genuine Langevin dynamics there according to (5), and projects back onto the nearest token embedding whenever a discrete sequence is required. The target density it navigates is therefore defined *through* a **nearest-neighbour projection**, and that construction has a consequential geometry. The embedding space is partitioned into **Voronoi cells**, one per token; within a cell the projected token is constant, so the projected energy is constant and the cell is a flat plateau, and across a boundary the projected token changes abruptly and the energy jumps. The target is thus not the smooth surface Langevin theory assumes but a collection of plateaus separated by cliffs.

Let M be the number of positions being resampled, so that the continuous state

[4] The two terms of (7) come from expanding the Gaussian log-density $-\frac{1}{2\epsilon}\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i) - \frac{\epsilon}{2}\boldsymbol{g}\|^2$, which yields the gradient-alignment term $\frac{1}{2}\boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$ and the distance term $-\frac{1}{2\epsilon}\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\|^2$, together with a term $-\frac{\epsilon}{8}\|\boldsymbol{g}\|^2$ that is constant in the candidate v and is therefore discarded.

is $\boldsymbol{s} \in \mathbb{R}^{M \times D}$, one D-dimensional vector per resampled position. Writing $\mathrm{proj}_V : \mathbb{R}^{M \times D} \to V^M$ for the nearest-neighbour map that sends the continuous state at each such position to the token whose embedding is closest in Euclidean distance, that target energy is

$$E_{\mathrm{CLS}}(\boldsymbol{s}) = -\log p_{\mathrm{LM}}\big(\mathrm{proj}_V(\boldsymbol{s})\big), \tag{8}$$

the negative log-likelihood of the discrete sequence obtained by projecting the state back to tokens, to which the additive constraint λC of (2) is appended when a constraint is imposed.

Two objects must be kept apart here, and the rest of the thesis depends on the distinction. The energy of (8) is piecewise constant and possesses no classical gradient, its derivative being zero inside every cell and undefined on the boundaries, so no sampler differentiates it. What the implemented sampler differentiates is a distinct *differentiable pathway*: the log-likelihood evaluated with the continuous state supplied as the input embedding at the masked positions, while the projected token $\mathrm{proj}_V(\boldsymbol{s})$ supplies the prediction target. Inside a single cell the target token is fixed, so this pathway is a smooth function of $\boldsymbol{s}$ and its gradient is well defined; it is also, for the same reason, nearly but not exactly flat there, unlike the projected energy, which is exactly flat. Crossing a cell boundary changes the prediction target, so the pathway itself jumps, and the drift computed from it jumps with it. Every claim in this thesis about a discontinuous or non-Lipschitz drift attaches to this differentiable pathway *across cell boundaries*, not to a classical gradient of the piecewise-constant energy of (8), which does not exist. It is this arrangement, a pathway that is smooth within a cell and discontinuous across one, sampling a target that is flat within a cell and jumps across one, that puts the correction of Section 2.3 under strain.

The arrangement has a direct consequence for the acceptance probability, derived once here. For a proposed move that stays within a single Voronoi cell the projected sequence, and therefore the energy of (8), is unchanged, so the target ratio is at or near unity and the acceptance probability of (6) is governed almost entirely by the proposal ratio. For the discrete sampler a within-cell move is the proposal of the same token, which is its own reverse, so the proposal ratio is one and the move is accepted with certainty. For the continuous sampler a within-cell move is still a continuous displacement whose reverse density is evaluated from the drift at the proposed point, which even inside the same cell need not make the return move likely, so within-cell acceptance is not protected. Section 5.2 reports the measured rates on both sides of this split.

2.5 Score Matching and the Concrete Score

One further notion is needed before the experiments: the **score function**. For a differentiable density $p(\boldsymbol{s})$, the score is the gradient of its log-density, $\nabla_{\boldsymbol{s}} \log p(\boldsymbol{s})$, and it is exactly the quantity the Langevin drift of (5) requires. **Score matching** (Vincent, 2011) fits a model directly to that quantity, and the observation behind modern score-based and diffusion generative models (Song and Ermon, 2019; Ho et al., 2020) is that learning to denoise a corrupted input is equivalent to learning the score. A diffusion model, trained to reverse a gradual noising process, therefore acquires by construction a field of local directions that an autoregressive objective never asks for. That makes such a model one available source of a usable proposal, and Section 5.6.1 uses it as such; it is not the only possible source, and the results of Section 5.6.2 show that it is not the necessary one either.

The construction does not transfer to text unchanged, because there is no gradient of a log-density with respect to a token index. The discrete analogue is the **concrete score** (Meng et al., 2022), which replaces the gradient by the collection of ratios $p(\boldsymbol{y})/p(\boldsymbol{x})$ between a sequence $\boldsymbol{x}$ and each neighbour $\boldsymbol{y}$ differing from it in one position. For every single-token move these ratios state how much more or less probable the data distribution makes it, which is the local directional signal a sampler needs in order to choose a substitution. Discrete diffusion models are trained to supply them. Their noising process corrupts a sequence through a Markov chain over the vocabulary; in the **absorbing** variant (Austin et al., 2021) each token is independently replaced, with a probability growing along the schedule, by a distinguished absorbing mask state, and the model is trained to reverse the process by predicting the clean token at each masked position conditioned on the surviving context on both sides. That reverse step is a bidirectional denoising query, so an absorbing diffusion model is trained to answer exactly the question an in-place revision poses: given a sequence with one position held out, which token belongs there. Two properties are therefore bundled together in such a model, the score objective and the bidirectional conditioning, and Section 5.6.2 is the experiment that separates them.

Score-entropy discrete diffusion (SEDD) (Lou et al., 2024), the model used in Section 5.6.1, learns the concrete score directly, fitting the ratios of the data distribution over the discrete state space with a loss adapted to non-negative ratio targets. It therefore provides, in a single forward pass, a trained estimate of the per-token probability ratios at exactly the positions a substitution could act on, and it shares the GPT-2 tokenizer, which is what lets it be dropped into this study's harness without retokenization.

2.6 Amortized Inference with GFlowNets

Both samplers above require the gradient of the energy: the discrete sampler to score its proposals, the continuous sampler to compute its drift. An entirely different strategy is to give up on the gradient and instead *learn* to sample. A **Generative Flow Network** (Bengio et al., 2021; 2023) treats generation as the sequential construction of an object and trains a policy so that the probability of terminating at a complete object is proportional to a non-negative reward $R(\boldsymbol{x})$, a unit of flow entering at the empty state and being split among the branches so that the flow arriving at each complete object matches its reward. Because flow is distributed in proportion to reward rather than concentrated on the single best object, the objects produced are *diverse*, which is what distinguishes a GFlowNet from a reward-maximizing reinforcement-learning agent that would learn to produce only the one best object and would be useless as a sampler.

For text the object under construction is a sequence built from left to right, so the space of partial constructions is a tree with exactly one path from the empty state to any complete sequence. That structure simplifies the training objectives, which enforce a consistency condition on the flow along trajectories; those used in practice include trajectory balance (Malkin et al., 2022) and its sub-trajectory variants (Madan et al., 2023), which improve credit assignment along long trajectories and are what the reference implementation this thesis builds on employs. One feature of the approach matters most here: the reward $R(\boldsymbol{x})$ is only ever *evaluated*, never *differentiated*. The policy learns from scalar scores of complete sequences, so the discontinuity and non-smoothness of the underlying landscape are invisible to it, and if the difficulty with energy-guided sampling were purely a property of the gradient, a method that never uses the gradient ought to escape it. Following Hu et al. (2024), the GFlowNet is realized as a parameter-efficient low-rank adapter (Hu et al., 2022) on the same frozen base model that supplies the energy for the Langevin samplers, so the two halves of the study operate on one shared model and their comparison is not confounded by any difference in the underlying network.

3 Related Work

The work in this thesis sits at the intersection of three lines of research: controllable text generation, energy-based and sampling-based decoding, and the study of the geometry and behaviour of language-model representations. This section reviews each in turn. Rather than cataloguing methods, it tells the story of how the field arrived at energy-based sampling as an approach to control, because understanding that trajectory is what makes the unexamined assumption at its heart visible, and it is that assumption which the thesis tests. The section closes by stating precisely the gap in the literature that the work addresses.

3.1 Controllable Text Generation

Approaches to controllable generation can be organized by how much they change the underlying model, and reviewing them in that order reveals a progression toward ever lighter-touch intervention, of which energy-based sampling is the logical endpoint.

At one extreme are the **fine-tuning methods**, which adjust the model's parameters so that the desired behaviour is internalized. The earliest of these simply continued training on attribute-specific data, and the modern and dominant form is reinforcement learning from human feedback (Ouyang et al., 2022), which optimizes the model against a learned reward that encodes human preferences. These methods work, and the widely used aligned language models are their fruit, but the control they produce is fixed once training is complete. A new attribute requires new data and a new training run, and the cost of that adaptation is precisely what motivates the search for methods that impose control at inference time. Conditioning methods such as CTRL (Keskar et al., 2019) occupy a middle ground, prepending control codes to the input so that a single model can be steered among a fixed set of attributes anticipated during training, but they too cannot accommodate a genuinely novel constraint after the fact.

At the other extreme are the **decoding-time methods**, which leave the parameters untouched and intervene only during generation, and these are the ones that promise the inference-time flexibility fine-tuning cannot offer. Plug and Play Language Models (Dathathri et al., 2020) perturb the hidden states of a frozen model at each step in the direction an attribute classifier's gradient indicates. FUDGE (Yang and Klein, 2021) and GeDi (Krause et al., 2021) instead reweight the next-token distribution using a discriminator that predicts, from a prefix, whether the eventual completion will bear the desired attribute. Effective within their scope, they share a structural limit: they remain autoregressive, modifying the local per-token decision while retaining the left-to-right order and so inheriting its inability to revise an earlier token in light of a later one. Pursuit of a method that could edit any part of a sequence in light of the whole is what leads to the sampling-based family.

3.2 Energy-Based and Sampling-Based Decoding

The decisive conceptual move, and the one that defines the family of methods this thesis studies, is to abandon left-to-right generation entirely and to cast decoding as sampling from an energy defined over whole sequences. Once a sequence is treated as a single object drawn from a distribution, rather than as a stream of tokens produced in order, a global constraint can be imposed on the completed object directly, and any position in the sequence can be revised in light of any other. This is the promise that makes the approach attractive, and several influential systems realize it.

COLD decoding (Qin et al., 2022) performs energy-based constrained generation by running Langevin dynamics on a continuous relaxation of the sequence, combining a fluency energy derived from a language model with task-specific constraint energies, and demonstrated it on tasks such as constrained generation and abductive reasoning. MuCoLa (Kumar et al., 2022) samples in the continuous embedding space and projects back to tokens; it is the closest published antecedent of the continuous sampler studied here, and its formulation together with its use of a projection to return to discrete text are exactly the ingredients whose geometric consequences Section 5 examines. The continuous sampler of this thesis reproduces the state geometry of that mechanism faithfully, including the projection onto the embedding table after every update, but it differs from MuCoLa in how the target token enters the likelihood: MuCoLa substitutes the continuous vector into the output softmax, whereas the implementation here gathers the score at the projected token index. Section 6.4 sets out what that difference does and does not affect. The constrained *mucola* arm of the experiments is the comparison at the level of the shared energy rather than of the original papers' tasks and metrics, which use different benchmarks and are out of scope. Related efforts include gradient-based inference for lexically constrained generation and infilling (Sha, 2020) and discrete samplers adapted from the energy-based-model literature, among them the gradient-informed proposals of Grathwohl et al. (2021) and the discrete Langevin proposals of Zhang et al. (2022) on which the discrete sampler here is directly based. A third strategy relaxes the categorical choice itself with a straight-through estimator (Bengio et al., 2013) or the Gumbel-softmax reparameterization (Jang et al., 2017); it is out of scope because it alters the model's forward computation rather than leaving the frozen likelihood untouched.

One feature recurs across this literature and is rarely brought to the foreground. The Metropolis–Hastings correction, which as Section 2.3 explained is what separates genuine sampling from biased optimization, is frequently omitted: methods are described in the language of sampling from an energy while in practice running a noise-augmented gradient descent, and where the correction is included, the difficulty of computing its reverse-proposal term on a landscape defined by projection is seldom examined. Taking the correction seriously, implementing it faithfully for both a discrete and a continuous sampler, and measuring what its inclusion reveals is one of the contributions of this thesis; what that measurement licenses one to say about published systems is treated in Section 6.3, where the evidence for it has been presented.

Two further lines bear on the target these methods sample from. The machine-translation literature established, more sharply than anywhere else, that the mode of a well-trained sequence model is not where the quality is. Stahlberg and Byrne (2019) showed that for a large fraction of inputs the globally most probable translation under a neural model is the empty string, and Eikema and Aziz (2020) argued that maximum-a-posteriori decoding is therefore the wrong criterion, the model's useful information lying in the bulk of the distribution rather than at its peak; Meister et al. (2023) give the complementary account, that high-probability continuations are dispreferred by human readers because they carry atypically little information. These are the published form of the likelihood trap measured in Section 5.7, and they matter here because a method that succeeds at concentrating mass on the lowest-energy sequences of a frozen likelihood succeeds at reaching exactly the region these works identify as degenerate. Deng et al. (2020) take the constructive route of learning a residual energy on top of a language model rather than treating the likelihood itself as the energy.

A distinct branch anticipates the diagnosis this thesis reaches, because not every method that samples from a frozen-language-model energy uses the *gradient* of that energy. Mix-and-Match (Mireshghallah et al., 2022) performs controllable generation by Metropolis–Hastings over a product-of-experts energy, proposing token substitutions from an auxiliary masked language model and scoring them with the frozen energy, which is therefore only ever evaluated. The twisted sequential Monte Carlo methods of Lew et al. (2023) and Zhao et al. (2024) likewise sample from an intractable posterior defined by a frozen model using importance weights and resampling rather than energy gradients. Two older members of the same branch are the direct antecedents of this study's most successful baseline: Wang and Cho (2019) sample text from a frozen masked language model by Gibbs sampling over token positions, and Miao et al. (2019) perform constrained sentence generation by Metropolis–Hastings over word-level insert, delete and replace proposals. That these gradient-free samplers operate successfully on essentially the same frozen-model energies that defeat the gradient-guided methods is a strong hint about where the difficulty lies, and Section 5.3.3 makes it concrete on the identical energy. Their existence is part of why the negative result of this thesis is scoped to one derivative rather than to the training-free premise.

3.3 Amortized Sampling and GFlowNets

Generative Flow Networks were introduced by Bengio et al. (2021) as a method for sampling discrete, compositional objects in proportion to a reward, with the explicit motivation of producing diverse high-reward samples rather than the single mode that reward-maximizing reinforcement learning tends to collapse onto. The framework was subsequently formalized and given a theoretical foundation (Bengio et al., 2023), and a family of training objectives was developed to address the practical difficulty of assigning credit along long generative trajectories, including trajectory balance (Malkin et al., 2022) and its partial-episode sub-trajectory refinement (Madan et al., 2023). The application most directly relevant here is that of Hu et al. (2024), who framed a range of language tasks, including infilling and reasoning, as sampling from an intractable posterior defined by a frozen language model, and trained a GFlowNet to perform that sampling, reporting that the resulting policies yield diverse samples that transfer better to downstream use than those obtained by reward maximization.

The GFlowNet component of this thesis builds directly on that formulation and codebase, using the same modified sub-trajectory objective and the same parameter-efficient adaptation of a frozen base model (Hu et al., 2022). The question posed here is narrower: their concern is whether amortized sampling is useful, and they show that it is where the reward defines a well-behaved target, whereas the concern here is whether it *escapes a specific pathology*, that of the frozen autoregressive likelihood used as an energy over free-form sequences. The two are compatible, and Section 5 takes up the sharper one by direct experiment.

3.4 The Geometry and Behaviour of Language-Model Representations

A complementary line of work concerns the representations in which these samplers operate and the behaviour of the likelihood they optimize, and it supplies part of the explanation for the results. Two phenomena from this literature are directly relevant.

The first is **anisotropy**. Transformer embeddings, contextual and static alike, do not spread uniformly through the representation space but occupy a narrow cone, so two randomly chosen token embeddings have high average cosine similarity rather than being roughly orthogonal (Gao et al., 2019; Ethayarajh, 2019). This bears on any method treating Euclidean or cosine distance in embedding space as a measure of similarity, because anisotropy compresses the dynamic range of those distances: the difference between a good and a bad token is smaller in the embedding metric than the geometry would suggest. This thesis measures the anisotropy of both models it studies, shows that it renders Euclidean distance an unreliable evaluation metric, which is one reason the experiments adopt a divergence-based measure of contextual fit instead, and adds a consequence the prior work did not draw, that the anisotropy scale differs enough between models to make a single step-size schedule non-transferable.

The second is the failure of high-probability text to be high-quality text. Holtzman et al. (2020) showed that the most probable sequences under a neural language model are frequently degenerate, repetitive and unnatural, and that maximization-based decoding produces markedly worse text than sampling. That matters here because any procedure concentrating probability mass on the lowest-energy regions of the frozen-likelihood energy thereby concentrates it on degenerate text. This thesis reproduces the phenomenon quantitatively on its own models, terms it the *likelihood trap* in the energy-based setting, and uses it to argue that the objective the samplers pursue is itself undesirable, quite apart from whether they can pursue it effectively.

3.5 Diffusion Models for Text and Bidirectional Denoising

A final body of work concerns a comparative model family rather than the methods this thesis evaluates. Diffusion models, which generate by gradually reversing a noising process, have become the dominant approach to image generation and have been adapted to text in two broad families. The first embeds tokens into a continuous space and runs a continuous diffusion there, as in Diffusion-LM (Li et al., 2022), which demonstrated controllable generation by applying classifier guidance to the continuous latents at each denoising step. The second operates directly on discrete tokens, as in the score-entropy discrete diffusion of Lou et al. (2024), which learns the ratios of the data distribution over a discrete state space and has narrowed much of the quality gap between diffusion and autoregressive language models.

Two things distinguish these models from an autoregressive one, and keeping them apart matters for how the comparison in Section 5.6.1 may be read. The first is the objective: by the equivalence of Vincent (2011), learning to denoise is learning the score of the data distribution, which is what gives score-based generative modeling its name and its mechanism (Song and Ermon, 2019; Ho et al., 2020). The second is the conditioning direction: a denoising step reads the surviving context on both sides of the position it fills, which is the query an in-place revision poses and which no left-to-right factorization affords. A diffusion model therefore differs from an autoregressive one along both axes at once, alongside differences in scale and corpus, so it is a comparative family and not by itself a control that isolates either axis. A bidirectional masked language model, which has the second property without the first, is what separates them, and Section 5.6.2 runs it.

3.6 The Gap Addressed by This Thesis

The literature above establishes energy-based sampling as an attractive and actively pursued route to controllable generation, and supplies both the samplers this thesis studies and the amortized alternative it examines. What it does not supply is a direct, controlled test of the assumption the enterprise rests on, that the gradient of a frozen autoregressive likelihood is a usable search direction on the discrete text manifold, together with a mechanistic account of what happens when that assumption fails. Negative observations, where they appear, are reported in passing as brittleness, hyperparameter sensitivity, or a need for careful tuning, rather than isolated and explained. This thesis addresses that gap. It designs an ablation that separates the direction of the gradient from its magnitude and so tests the assumption rather than a proxy for it; it constructs diagnostics that identify the geometric, statistical and analytic mechanisms behind the result; and it locates the deficiency precisely, in the coordinates the derivative is taken in and in what the resulting proposal may condition on, rather than in the sampler, the energy, or the objective the model was trained with.

4 Methodology

This section describes the experimental platform in the detail needed to reproduce the study, in a particular order: the task and the models first, then the samplers and the metrics, and only then the diagnostic experiments, because the diagnostics were designed after the main experiments and in response to what those experiments revealed. Where a choice was made, the reasoning behind it is given, because in a diagnostic study the design of each measurement is part of the argument.

4.1 Task: Masked Token Recovery

A sampler can only be studied if there is a way to tell whether it is being guided toward good sequences or is merely wandering, and an open-ended generation task provides no such handle: a fluent but arbitrary continuation is neither clearly a success nor clearly a failure, and the sampler's contribution cannot be separated from the difficulty of the underlying generation. The study therefore adopts **masked token recovery**, also called infilling, as its diagnostic bench. A sequence is drawn from the corpus, one or more of its tokens are replaced with random tokens, and the sampler is asked to restore a coherent sequence by resampling only the corrupted positions while the rest is held fixed. Because the original sequence is known the recovery can be scored, and because the corruption and the surrounding context are controlled the sampler's behaviour can be attributed to the method rather than to the open-endedness of the task.

One subtlety shapes the evaluation metric and is worth stating at the outset. The primary metric, defined in Section 4.4, compares the next-token distribution the recovered fill induces against the distribution the ground-truth fill induces, so it measures contextual fit relative to a reference and a coherent synonym of the original token scores well even though it does not match. It is therefore not in tension with the likelihood-trap finding of Section 5.7, which concerns absolute sequence likelihood as a maximization objective in open-ended generation, a different quantity serving a different purpose.

Two corpora are used, and it matters which experiment ran on which. The masked-recovery sequences of the main grid come from the **WikiText-2** validation set (Merity et al., 2017), a standard held-out corpus of encyclopedic prose, chosen because its held-out status keeps the recovery task independent of any fine-tuning data and its sentences give a masked position a locally checkable correct recovery. The base GPT-2 Large is instead supervised fine-tuned on **ROCStories** (Mostafazadeh et al., 2016), the corpus of the amortized-inference reference implementation of Hu et al. (2024), so the GFlowNet half of the study operates on the same data distribution as the work it builds on. The main grid therefore evaluates recovery *out of domain* relative to the fine-tuning corpus, while the diagnostics are run in domain on ROCStories; Table 1 maps each experiment family to its corpus.

The domain shift is not innocent, so it is addressed directly. If the gradient null were an artifact of evaluating out-of-domain text it should weaken or vanish

Experiment family	Corpus
Masked-recovery grid (145 configurations)	WikiText-2 validation
Gradient-free baselines, continuation, external judge	WikiText-2 validation
Constrained generation	WikiText-2 validation
Diffusion and masked-LM proposal ladder	WikiText-2 validation
GPT-2 Large fine-tuning; GFlowNet training and variants	ROCStories
Diagnostics: linearization, likelihood trap, anisotropy, trajectory	ROCStories

Table 1: Corpus used by each experiment family.

in domain, and it does not: the in-domain ROCStories diagnostics of Section 5.4 and Section 5.7 reproduce both the gradient null, with the linearization correlation below 0.06 throughout, and the likelihood trap. The central findings are therefore properties of the frozen likelihood rather than of the domain shift.

4.2 Energy Functions

The study evaluates five energy functions, chosen so that the central findings can be tested across training regimes and across architectures.

The reference energy is a **GPT-2 Large** model (Radford et al., 2019), 774 million parameters and embedding dimension $D = 1280$, supervised fine-tuned on ROCStories. The same fine-tuned model is the base for the amortized experiments, so the Langevin and GFlowNet halves share weights and are not confounded by any difference in the underlying network, an identity essential to the energy-level experiment of Section 5.9. Three **GFlowNet-tuned variants** are included, produced as described in Section 4.5: two with the unnormalized-length reward at 500 and 2000 optimization steps, so the effect of longer training can be observed, and one with the length-normalized setting at 500 steps. They are included to test whether amortization repairs the energy landscape, since the objective demonstrably moves the energy and running the same samplers on the tuned models asks whether that movement made the landscape navigable or merely relocated it. Finally a **Llama-3**

8B model (Dubey et al., 2024), embedding dimension $D = 4096$, serves as a cross-architecture control; because its tokenizer, embedding geometry and step-size scale all differ it is never placed on a shared numerical axis with the GPT-2 family, and it is used only to establish that the qualitative findings hold across architectures. The four GPT-2-based models share the tokenizer, the corruption seeds and the schedule, so their numbers are directly comparable.

4.3 Samplers and Configurations

The two samplers of Section 2.4, the Discrete Langevin Sampler (DLS) and the Continuous Langevin Sampler (CLS), are each run under a factorial combination of design choices. Both are studied because they represent opposite strategies for confronting the discreteness of text and so isolate different failure modes: the DLS, following Grathwohl et al. (2021) and Zhang et al. (2022), keeps the state in token space and uses the gradient only to build a proposal, testing whether the gradient is an informative ranking signal; the CLS, following Kumar et al. (2022), relaxes into continuous space and runs genuine Langevin dynamics, testing whether continuous Markov chain Monte Carlo is compatible with a projected discrete energy. A finding that holds for both is a finding about the energy rather than about one sampler's idiosyncrasies.

The continuous sampler's update as implemented deserves to be stated precisely, because Section 5.2 quantifies how far its state drifts from the token manifold and the update is what bounds that drift. At step t the sampler forms an interim state by the Langevin drift, $\boldsymbol{s}_{\text{interim}} = \boldsymbol{s}_t + \frac{\epsilon_t}{2} \nabla_{\boldsymbol{s}} \log p$, projects it to the nearest token embedding, and centres the proposal at the interpolated mean of the two,

$$\boldsymbol{s}_{t+1} \sim \mathcal{N}\big(\tfrac{1}{2}\big(\boldsymbol{s}_{\text{interim}} + \text{proj}_V(\boldsymbol{s}_{\text{interim}})\big),\ \epsilon_t \boldsymbol{I}\big)\,, \tag{9}$$

after which the correction of Section 2.3 accepts or rejects it, with the reverse density evaluated from the drift at the proposed point. The update has four parts, and naming them separately is what makes the rest of this section readable. $\boldsymbol{s}_t \in \mathbb{R}^{M\times D}$ is the **current state**, one D-dimensional vector for each of the M positions being resampled, and in general it is not the embedding of any token. $\boldsymbol{s}_{\text{interim}} \in \mathbb{R}^{M \times D}$ is the **interim continuous point**, the current state moved a half-step of size ϵ_t along the gradient $\nabla_{\boldsymbol{s}} \log p \in \mathbb{R}^{M \times D}$; it is where an unmodified Langevin update would put the chain. $\text{proj}_V(\boldsymbol{s}_{\text{interim}}) \in \mathbb{R}^{M \times D}$ is the **projection**, the embedding of whichever token lies nearest to that interim point, and is by construction a point on the token manifold. Their average is the **interpolated mean**, the centre of the Gaussian the next state is actually drawn from, with $\epsilon_t > 0$ the scalar step size and $\boldsymbol{I}$ the identity covariance of that Gaussian. Averaging the interim state with its projection pulls the proposal halfway back toward the token manifold at every step, bounding the off-manifold drift a pure continuous update would accumulate. The interpolation is an implementation choice adapted from Kumar et al. (2022), not a property of Langevin dynamics, and it is load-bearing for the distances of Section 5.2, since even with this pull the state settles far from any token.

One asymmetry in how the correction was applied across the compared arms has to be declared, because it falls on exactly the contrast this thesis rests on. In the main grid the reverse-proposal term of Equation (6) is computed exactly for the *policy* arm and set to zero for the two random-direction arms, on the reasoning that a random-direction proposal is a symmetric random walk whose proposal ratio cancels. That reasoning is not sound: for the continuous sampler the proposal mean of Equation (9) carries both a drift and a nearest-neighbour projection, neither symmetric whatever the direction, and for the discrete sampler the distance term is symmetric but the softmax normalizers at the current and proposed tokens are not. In every correction-enabled comparison the policy arm is therefore charged a reversibility penalty from which its comparators are exempt, a penalty Section 5.2 measures reaching -1325 nats in the continuous case. It works against the policy arm, so it cannot manufacture the null, but it could inflate the cases in which the gradient appears *worse* than a random direction.

The samplers therefore carry a corrected mode in which the reverse term is computed exactly for every arm. Because a random direction is drawn independently of the current state, holding that draw fixed across the forward and reverse evaluation of a single step yields a kernel $q_{\boldsymbol{g}}$ that is exactly computable and π-invariant, and a mixture of π-invariant kernels is itself π-invariant, so the corrected mode is a valid sampler rather than an approximation. Section 5.3.2 reports the arms rerun under it. The archived grid is left as it was run, and the legacy treatment remains the default, so that every number reported from it remains reproducible.

The design axes are as follows. The **proposal method** is one of three: the *policy* method uses the true gradient of the energy; the *grad-norm-preserved random* method replaces it with a random vector rescaled to the exact norm of the true gradient, differing from the policy only in direction; and the *random* method uses a random vector of random magnitude. The comparison among these three is the central ablation of the thesis, isolating respectively the full gradient, the effect of its direction alone, and a pure baseline, with the reasoning developed where it is used in Section 5.3. The **Metropolis–Hastings correction** is either enabled or disabled, so the effect of the exact accept-reject step can be observed directly. **Gradient normalization**, which rescales the gradient to unit norm for numerical stability, is either enabled or disabled, and it interacts with the proposal-method ablation in a way that turns out to be critical. The **annealing schedule** runs for 50 or 100 steps with the step size decreasing linearly, from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$ for the GPT-2 family, a scale established by the calibration of Section 5.1. An *oracle* variant, in which the ideal step size at each iteration is selected using knowledge of the ground-truth token, is included as an upper-bound reference, to rule out the possibility that the samplers fail merely for want of a well-tuned schedule; it is run independently for each proposal method rather than sharing one schedule, so the selection cannot advantage the policy gradient over its comparators. In total the study comprises 145 sampling configurations across the five energy functions, each evaluated on 200 sequences; the arithmetic of that count is in Appendix A.2.

4.4 Evaluation Metrics

Three quantities are recorded, and the choice among them is itself informative.

The first is the **Euclidean distance** in embedding space between the recovered and the ground-truth token, the most obvious measure of recovery and an unreliable one, for a reason established quantitatively in Section 5.8: because language-model embeddings are anisotropic, occupying a narrow cone (Ethayarajh, 2019), Euclidean distance compresses the distinction between good and bad tokens, so a token can be geometrically close to the target while being grammatically inadmissible and a perfectly good synonym can lie far away. It is reported for completeness and studied as an object in its own right, but not relied on as the primary criterion.

The primary criterion is the **KL divergence** (Kullback and Leibler, 1951) between the model's predictive distribution at the recovered position and a reference distribution, which measures how well the recovered token fits its context, that is, its *contextual coherence*. Concretely, let $\mathcal{M}$ be the set of masked positions in a sequence of length L, and let $\mathcal{M}' = \{\, m \in \mathcal{M} : m < L - 1 \,\}$ be those masked positions that have a right neighbour, the only ones at which the metric is defined. For a position $m \in \mathcal{M}'$, write $p_{\text{ref}}^{(m)}$ for the model's next-token distribution at m when every masked position holds its *ground-truth* token, and $p_{\text{pred}}^{(m)}$ for the same distribution when the masked positions instead hold the *recovered* fill. Both $p_{\text{ref}}^{(m)}$ and $p_{\text{pred}}^{(m)}$ are therefore probability vectors in $\mathbb{R}^{|V|}$ over the same vocabulary, differing only in what was placed at the masked positions, and the index m ranges over $\mathcal{M}'$. The metric is the mean divergence over these positions,

$$\overline{\mathrm{KL}} = \frac{1}{|\mathcal{M}'|} \sum_{m \in \mathcal{M}'} \mathrm{KL}\Big(p_{\text{ref}}^{(m)} \,\big\|\, p_{\text{pred}}^{(m)}\Big)\,, \tag{10}$$

which is the quantity the sampler logs and the revision diagnostics recompute. By construction the ground-truth fill gives $\overline{\mathrm{KL}} = 0$, so the metric has a hard floor a perfect recovery attains. It is chosen over exact-match accuracy for the reason given in Section 4.1: a recovered token that differs from the reference may still be perfectly coherent, and a metric penalizing it would confound the sampler's quality with the arbitrariness of the reference, whereas a divergence between predictive distributions measures contextual fit directly. One limitation of the divergence measure shapes how the results are argued: its magnitude depends on the position of the corrupted token, since corrupting an early token perturbs the conditional probability of a long suffix and yields a large divergence while corrupting a final token yields almost none. No model is therefore ranked on small differences in KL; the central results are argued from the *absence* of a gap between conditions, which is robust to this positional dependence in a way a narrow measured win would not be. To make that absence a decision rather than an impression, the analysis fixes an equivalence margin in advance, at five percent of the policy method's mean KL for each configuration, and treats two methods as practically equivalent when their paired mean difference falls inside it. The margin is calibrated to the pipeline's own noise rather than chosen to fit the result: the across-seed standard deviation of the final KL for the flagship configuration is 0.183 (Section 4.8), comfortably inside the corresponding margin of about 0.327.

For the generation-quality analyses the study records **repetition rate**, the fraction of repeated n-grams, and **distinct-n**, the standard instruments for detecting the degeneration studied by Holtzman et al. (2020) and therefore the appropriate measures for the likelihood trap. For the constrained-generation extension it records classifier accuracy and **BERTScore** (Zhang et al., 2020), chosen because it measures semantic similarity through contextual embeddings rather than surface overlap and so does not penalize a valid paraphrase.

4.5 GFlowNet Training

The GFlowNet-tuned variants are produced following the amortized-inference formulation of Hu et al. (2024), using their modified sub-trajectory-balance objective (Malkin et al., 2022; Madan et al., 2023). A GFlowNet is chosen over reward-maximizing reinforcement learning for a reason specific to this thesis: a reward-maximizing agent would converge on the single highest-reward sequence and so could not distinguish a sampler that explores the energy from one that merely finds its mode, whereas a GFlowNet is trained to sample in proportion to reward and therefore tests amortized *sampling*. It only ever evaluates the reward and never differentiates it, so the gradient behaviour that defeats the Langevin samplers is invisible to it.

The policy is a low-rank adapter (Hu et al., 2022) on the frozen GPT-2 Large base, and the reward is the log-likelihood that frozen base assigns to the completed sequence, so the GFlowNet is trained to sample in proportion to the model's own likelihood, exactly the energy the Langevin samplers navigate. The task is a story-infilling formulation in which, given a preceding context and a known ending, the policy generates a connecting span and the reward scores the concatenation under the frozen model. A single hyperparameter β_{len} controls the length normalization: at 0 the reward is the raw log-likelihood and at 1 it is fully normalized by sequence length. The consequences of that one knob, two opposite degeneracies, are developed in Section 5.9.

Using the frozen log-likelihood as the reward may seem to contradict the thesis's own argument that this likelihood is a poor objective, so the choice is worth defending on three grounds. That the gradient of the likelihood is uninformative does not make the likelihood value-uninformative, since the gradient-free baselines of Section 5.3.3 recover coherent text by evaluating exactly this quantity. The likelihood trap concerns the behaviour of the likelihood at its extreme, the degenerate text that globally maximizes it, not its ranking of fluent candidates near the data manifold, which is the regime the reward operates in during training. And, decisively, training a policy to sample in proportion to the base model's own likelihood is the standard amortized-inference formulation of Hu et al. (2024), whose premise that this likelihood defines a target worth sampling from is exactly what this thesis sets out to test; using the same reward is the condition under which the test is meaningful, not an oversight.

One difference in task formulation governs how the GFlowNet and the samplers can be compared. The GFlowNet reformulates infilling as left-to-right generation of the blank span under a restructured prompt that places the context and the known ending around it, whereas the Langevin samplers perform in-place substitution with bidirectional context. The two solve differently shaped problems, so the comparison is run at the level of the energy, the final KL under the same frozen base model, rather than at the level of the task. The left-to-right reformulation is not adopted for the samplers, because generating the span autoregressively would dissolve the revision capability under study and so change the question rather than repair the metric.

4.6 Constrained Generation Extension

To test the plug-and-play claim of (2) directly, which is the object of the extension question E, a constrained-generation extension augments the energy with a sentiment constraint in the manner of Kumar et al. (2022) and Qin et al. (2022). A sentiment classification head trained on the representations of the GPT-2 Large base supplies the constraint term $C(\boldsymbol{x})$, and generation is run under five modes designed to isolate the contribution of the constraint gradient. The *fluency-only* mode uses the fluency energy alone; the *constraint-only* mode uses the constraint alone; the *combined* mode uses both, with weights $\beta_{\text{LM}} = 0.8$ and $\beta_{\text{C}} = 0.2$, and is the intended plug-and-play configuration; the *fully random* mode replaces the combined gradient with noise; and the *randomized-constraint* mode replaces only the constraint gradient with noise while retaining the fluency gradient, which separates the constraint direction from the fluency direction. The steering effect is measured as the change, in percentage points, in the fraction of $n = 150$ generations per arm classified as the target sentiment relative to the unconstrained baseline. Both the discrete sampler in the formulation of this thesis and a MuCoLa-style continuous baseline are evaluated, on both infilling and continuation, so that the finding does not depend on a single sampler or task. The analysis plan is the paired point-estimate contrast of the constraint-only mode minus the randomized-constraint mode, which cancels the shared sentiment drift; because the result files record only aggregate target-hit fractions and not per-generation labels, per-sample paired gains and their confidence intervals are not available and none are reported.

4.7 Diagnostic Experiments

The configurations above establish *that* the methods fail; they do not establish *why*, and the why is the contribution. Five diagnostic experiments were therefore designed after the main results were in hand, each targeting one candidate mechanism, and three further controls were added later still in response to the questions the first round raised: a sweep over step size and proposal temperature (Section 5.3.2), which establishes whether the null survives where the gradient term actually shapes the proposal; a substitution of the relaxed token-indicator surrogate into the same sampler (Section 5.4); and a bidirectional masked language model in the same exact-energy chain (Section 5.6.2), which separates conditioning direction from training objective. Unlike the configurations above these do not sample: they measure properties of the energy landscape and of the gradient directly.

The **linearization experiment** answers the question the gradient-versus-random result raises: if the input-embedding gradient carries no usable information, is that because the first-order surrogate the discrete proposal builds from it fails to predict the true change in energy? For each of 200 sequences and a masked position the exact gradient is computed, and for a stratified set of 2000 candidate tokens the surrogate predicted change $\hat{\Delta}(v) = \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$ is compared against the true change $\Delta(v) = \log p(\boldsymbol{x}_{i\to v}) - \log p(\boldsymbol{x})$ obtained by an exact forward pass, with each candidate's embedding distance recorded so the correlation can be examined as a function of distance.

That inner product is the object under test, so it is worth saying in words what it is and why. Both of its factors are vectors in $\mathbb{R}^D$. The first, $\boldsymbol{g}$, is the gradient of the sequence log-likelihood with respect to the input embedding at the masked position: it points in the direction in embedding space along which the log-likelihood grows fastest, and its length says how fast. The second, $\boldsymbol{e}(v) - \boldsymbol{e}(x_i)$, is the *displacement*, the vector one would travel along in embedding space by replacing the current token x_i with the candidate v. Their inner product $\boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$ is a single number: the component of that displacement lying along the uphill direction, multiplied by how steep the slope is. It is therefore the first-order prediction of how much the log-likelihood would change if the substitution were made, and it is exactly the quantity the discrete proposal of (7) uses to decide which token to propose. The experiment asks whether that prediction agrees with what actually happens, which is $\Delta(v)$, computed by making the substitution and running the model. The true change is further decomposed into a *self* term, the change in the log-probability of the candidate given its own left context, and a *future* term, the change in the log-probability of the suffix, because the input-embedding gradient can structurally register only the latter.

The **acceptance experiment** answers the question the Metropolis–Hastings breakdown raises: when the correction rejects nearly every useful move on the continuous sampler, is the rejection driven by the target term, meaning the moves are bad, or by the proposal term, meaning the moves are good but their reversibility cannot be verified? For every proposal it logs whether a Voronoi cell boundary was crossed, whether the move was accepted, and the decomposition of the acceptance ratio into its two terms. The **trajectory experiment** logs the raw state at every step for a small number of sequences, together with the distance from the continuous state to the nearest token embedding, which quantifies how far the sampler drifts from the token manifold. The **likelihood-trap experiment** asks whether low energy is a good place to be at all, decoding with several strategies from greedy and beam search to ancestral sampling and measuring the relationship between per-token likelihood and repetition rate. A final **anisotropy analysis** measures inter-token distances and related geometric quantities in the two embedding spaces, underpinning both the step-size result and the unreliability of Euclidean distance.

4.8 Reproducibility, Seeds, and Hardware

Because the central claims are claims about the *absence* of an effect, the run-to-run stability of the pipeline is itself evidence and is quantified rather than assumed. Rerunning the flagship configuration, the discrete sampler on GPT-2 Large with the correction and gradient normalization enabled over fifty steps, under four independent corruption seeds 1000 to 1003, yields final KL divergences of 6.348, 6.589, 6.150 and 6.291, mean 6.345, standard deviation 0.183, range 0.438. This across-seed standard deviation fixes the scale below which a difference in KL cannot be told apart from noise, and it is the empirical basis both for the equivalence margin defined in Section 4.4 and for the bounded-effect reading of the gradient-versus-random comparisons in Section 5.3. The corruption procedure is deterministic per sample index, with the seed set to the data seed plus the running index, so the corrupted inputs, and therefore the paired comparisons, are identical across the three method runs of a configuration and can be reproduced bit for bit; this is what allows the statistical analysis of Section 5.3 to pair the policy method against its comparators on the same sample without any rerun.

All reported experiments were run on a university GPU server with 49 GB A6000-class accelerators under PyTorch 2.12 and CUDA 13.0. Every configuration in the study is specified by the design axes of Section 4.3 together with the corruption seed and the annealing schedule, and every quantitative claim in what follows is recorded against the run that produced it, so any individual result can be regenerated from the description given here.

5 Results

This section reports the experimental findings, and it is arranged as the elimination set out in Section 1.2. The answer to RQ1 is developed without interruption first. It opens with the calibration of the samplers and the measurement of what the calibrated proposal actually is, then the Metropolis–Hastings correction in each of the two samplers, then the central comparison between the gradient direction and a norm-matched random direction, extended by a sweep that puts the gradient in charge of the proposal and by gradient-free baselines on the identical energy, which together reject Hypothesis A. The diagnostics that account for the null come next, and with them the re-analysis in token-indicator coordinates that rejects the strong form of Hypothesis B, the final position where that re-analysis can be checked exactly, and the proposal ladder that supports Hypothesis C. Only then does the section turn to the results that stand apart from that argument: the likelihood trap and the embedding geometry, which qualify the target rather than the proposal, the GFlowNet experiments answering RQ3a and RQ3b, and the constrained-generation extension.

5.1 Proposal Calibration and the Near-Uniform Main Grid

The study began by running the two samplers on GPT-2 Large with a step-size schedule already made to work on Llama-3. One term needs fixing first. A schedule *works* if it produces **calibrated motion**: the projected tokens change over the run and the metrics evolve rather than freezing. That is weaker than **guided motion**, motion toward better sequences, which the later sections test separately.

On GPT-2 Large, with the Llama-tuned schedule, the samplers did not move at all: the entropy of the discrete proposal collapsed to zero within a few steps and no token was ever changed. The cause is the geometry of the embedding space. The anisotropy analysis of Section 5.8 measures the mean distance between neighbouring token embeddings at about 1.82 in GPT-2 Large against 0.59 in Llama-3, and the mean pairwise distance at 2.77 against 0.84, roughly a factor of three in each case. The trapped sampler follows immediately: a step size calibrated for the tighter geometry of Llama-3 is too small to carry the continuous state across a Voronoi cell boundary in the more dispersed geometry of GPT-2, so the projected token never changes and the sampler sits frozen in its initial cell. Recovering the correct scale required an oracle sweep over a grid of fifty step sizes spaced logarithmically from 10^{-2} to 10^{2}; the schedule running from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$ for the GPT-2 family is what that calibration selected, and it is used throughout.

Two consequences follow. There is no universal step-size schedule for these methods, the correct scale being a property of the model's embedding geometry that must be rediscovered per model, and the step sizes required to make the sampler move at all are much larger than the distances between neighbouring token embeddings, so a method needing this recalibration before it will move is not plug-and-play in the strong, model-agnostic sense. And calibration establishes motion without establishing that the motion is informed. The oracle configuration, which selects the ideal step size at each iteration using knowledge of the ground-truth token, separates the two by establishing the best any schedule could do, so a failure of calibration can be told apart from a failure of guidance.

With the step size calibrated, the discrete sampler moves. What it moves *by* is the single most consequential measurement in the study, because it bounds what the ablation of Section 5.3 could possibly have detected. The proposal of Equation (7) draws the next token from a softmax over the sum of a distance term and a gradient-alignment term, divided by a temperature, so its entropy says directly how much either term is shaping the draw. Measured over the whole grid, the answer is that neither does. On the flagship configuration the mean proposal entropy is 10.8248 nats at the first step against a ceiling of $\log |V| = 10.8249$ for the uniform distribution over the 50,257-token vocabulary, and its *minimum* over the entire fifty-step schedule is 10.2760 nats, an effective support of roughly 29,000 tokens. Every free-schedule discrete configuration in the grid sits between 10.18 and 10.28 at its sharpest, and the oracle-schedule configurations sit at 10.8249, uniform to floating-point precision. The proposal is, at every step of every configuration, numerically indistinguishable from a uniform draw over the vocabulary, and Table 2 decomposes why. The standard deviation across the vocabulary of the distance term rises from 0.049 to 5.57 as the step size anneals, while that of the gradient-alignment term is flat at 0.046 throughout, because gradient normalization fixes $\|\boldsymbol{g}\| = 1$ and the projections of a unit vector onto embedding differences are of that order. After division by the temperature of 5.0 the gradient term contributes about 0.009 nats of logit spread against a 10.82-nat entropy budget, so it cannot influence the sampled token, and the ratio t_2/t_1 falls from 0.96 to 0.008 over the schedule, leaving a pure stay-near-the-current-token draw.

Step	ϵ_k	Distance term t_1	Gradient term t_2	t_2/t_1	Proposal entropy
0	10.50	0.049	0.047	0.958	10.8248
10	8.42	0.064	0.044	0.692	10.8248
25	5.30	0.104	0.044	0.430	10.8247
40	2.19	0.274	0.046	0.168	10.8236
49	0.10	5.572	0.046	0.008	10.2488

Table 2: Decomposition of the discrete proposal on the flagship configuration.

The consequence is immediate, and it is why this measurement is placed before the ablation rather than after it. A comparison between a true-gradient proposal and a norm-matched random one, when both are within 10^{-4} nats of uniform, is a comparison between two draws from the same distribution, and a perfectly informative gradient would produce the same null here. The ablation of Section 5.3

therefore cannot on its own establish that the gradient is uninformative; the sweep of Section 5.3.2 tests the same contrast where the gradient term does shape the proposal, and Section 5.4, which measures the surrogate against the truth directly and is invariant to temperature and step size, is what carries the argument.

What the sampler does instead is easy to state. Without the correction it is a uniform random walk over the vocabulary and does not converge: the mean KL rises over the schedule from 8.765 to 9.499, the chain is still changing its token on 99.1 percent of steps in the final tenth of the run, and the only late movement in any channel is a fall in proposal entropy from 10.78 to 10.28 at the last step, an end effect of the step size reaching its floor rather than a collapse into a local optimum. With the correction enabled the same uniform proposal becomes an independence sampler and the accept/reject filter does all of the work, the chain reaching a final KL of 6.541 at a 9.98 percent acceptance rate. Figure 1 shows the fifty-step schedule; the hundred-step run and the gradient-normalization-disabled companion, in which the three proposal arms separate rather than coincide, are Figures 12 and 13 in Appendix A.1. The entropy panels confirm that the late dip tracks the schedule, landing at step 49 of 50 and step 99 of 100, and that it is a boundary effect of the annealing floor which moves the entropy from near-uniform to near-uniform and produces no improvement in contextual fit.

This also explains the recovery numbers. Because the proposal never concentrates, the chain never proposes the ground-truth token with more than roughly uniform probability, and exact recovery is 0.0 percent in 139 of the 145 configurations and never exceeds 1 percent in the remaining six. A fall in mean KL from 9.14 to 6.4 under these conditions is the accept/reject filter selecting among near-uniform proposals for tokens that happen to fit the context, not a search converging on the right one.

One reading of the late entropy dip should be set aside explicitly, because it is the natural one and it is wrong. The dip invites the interpretation that the annealed step size collapses the update into deterministic gradient descent and freezes the chain in a local optimum, a *quenching* effect on the metallurgical analogy. Direct measurement rules it out: the final-step proposal still has an effective support of tens of thousands of tokens, the chain is still changing its token, and the mean KL rises rather than falls, so nothing is deterministic, frozen, or converging. What the schedule produces is not a quench but a uniform random walk that never localizes, which is the more damaging of the two readings for the plug-and-play premise.

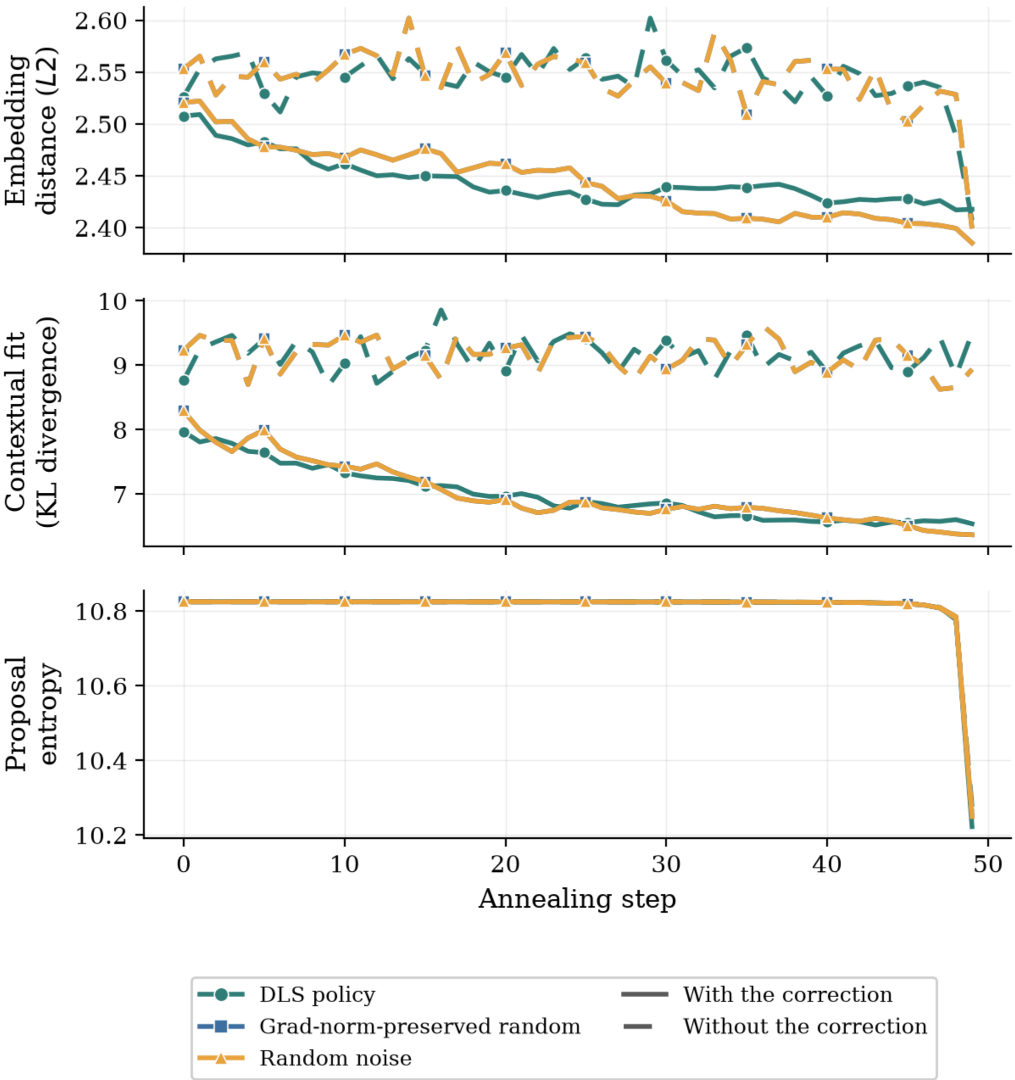

Figure 1: Discrete Langevin Sampler trajectories on GPT-2 Large over a 50-step annealing schedule, averaged across sequences.

5.2 The Metropolis–Hastings Breakdown in Continuous Space

In the continuous sampler the correction has the opposite effect. Decomposing the logged proposals by whether they crossed a Voronoi cell boundary, and therefore changed the projected token, the continuous policy sampler with the correction enabled accepts within-cell proposals, which change no token, only 0.03% to 0.63% of the time according to whether gradient normalization is off or on, and boundary crossings only 3.7% to 8.6% (Figure 8, Appendix A.1). Without the correction every proposal is accepted by construction, and it is this unchecked acceptance that lets the state wander far off the token manifold (Appendix A.4). The discrete sampler under the same correction, in the policy configuration with the correction and gradient normalization both enabled, accepts within-cell moves with probability 1.0, because such moves are exactly reversible, and boundary-crossing moves at 9.3%. That difference in within-cell acceptance, essentially one against essentially zero, is what separates the two cases: the discrete sampler's token can move freely inside a cell and then cross to a neighbour, while the continuous sampler, unable to accept even a move within its own cell, barely leaves the cell it started in.

Whether the correction rejects boundary-crossing moves because the moves are bad, or because it cannot verify their reversibility, is settled by the two terms of the acceptance ratio. For proposals that crossed a boundary, the mean *target* log-ratio is +4.60, that is, positive: on average these moves improve the sequence probability, so the distribution being sampled would welcome them. The mean *proposal* log-ratio for the same moves is −1325. The reverse proposal assigns effectively zero density to the return move, and this term alone, dwarfing the favourable target term by three orders of magnitude, drives the acceptance probability to zero. Figure 2 shows the two distributions side by side. The rejection is driven by the proposal term, not the target term.

This is the measured form of the argument of Section 2.3. The Metropolis-adjusted Langevin algorithm requires a Lipschitz-continuous drift (Roberts and Tweedie, 1996), and the drift derived from the differentiable pathway of Section 2.4

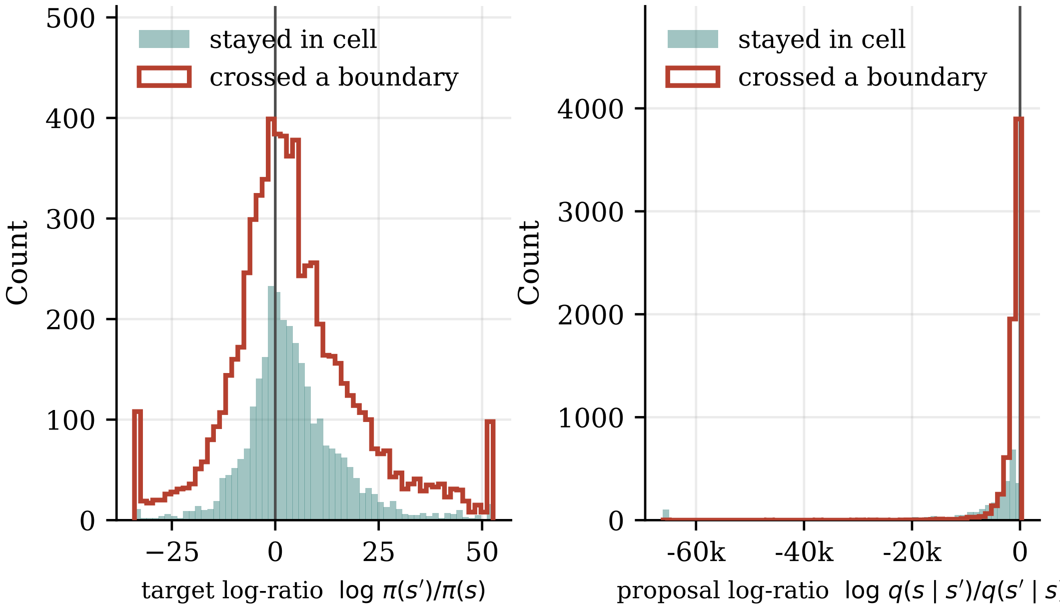

Figure 2: Decomposition of the Metropolis–Hastings acceptance ratio in the continuous sampler. Left: the target log-ratio. Right: the proposal log-ratio. Both on a log-probability scale.

is not Lipschitz across a cell boundary: when a proposal crosses one, the reverse proposal mean is computed from the drift on the far side, where the pathway's prediction target has changed, so it points somewhere unrelated to the way back. The original state falls deep in the tail of the reverse Gaussian, the reverse density vanishes, and the move is rejected as irreversible. The correction is not malfunctioning; the landscape violates the assumption it depends on, and this is the regime the convergence theory excludes, since the non-asymptotic guarantees of Dalalyan (2017) and Durmus and Moulines (2017) assume a smooth target. The consequence, observed on both GPT-2 Large and Llama-3, is that the continuous sampler appears to function only when the correction is omitted, that is, when it runs as a biased optimizer that ignores reversibility. This answers RQ2 for the continuous sampler: equipping it with the exact accept-reject correction for the proposal it implements does not make it work. The phrasing is deliberate. The correction is exact for that proposal, so the corrected chain is π-invariant by construction; what fails is not the correction but the regularity the Langevin construction assumes underneath it, and calling the whole arrangement “theoretically correct” would paper over exactly that distinction.

Read spatially, the same breakdown is a statement about where the samplers go. The trajectory experiment logs the full continuous state at every step for a small number of sequences, together with its distance to the nearest token embedding and the Voronoi cell it occupies; the walkthrough and figures are in Appendix A.4. Counting distinct Voronoi cells visited over fifty steps, which counts how often the projected token changed, the continuous sampler with the correction enabled visits about two, and exactly one in half the traced sequences, while with the correction disabled it visits about 45 of a possible 50. The discrete sampler splits the same way but at a different scale: about five cells with the correction enabled and about 49 with it disabled, accepting almost every proposal, and in both settings its state remains a genuine token embedding. The distance from that state to the nearest token embedding measures drift off the manifold of actual tokens. With the correction on it sits between 118 and 151 units across the run, sixty-five to eighty-three times the mean nearest-neighbour spacing of 1.82; with it off, while forcing its way across boundaries, the state reaches roughly 980 units and ends about 17 away. The discrete sampler sits at distance zero at every step, its state always being a genuine token embedding.

The continuous relaxation therefore buys genuine Langevin dynamics at the cost of leaving the token manifold, after which it must either respect the correction and freeze or ignore it and wander through the space between tokens, where the projected energy is a poor guide.

5.3 Gradient Direction Against a Norm-Matched Random Direction

The preceding sections concern how the samplers move; this one concerns the signal that drives them, and it must be read against Section 5.1: at the calibrated configuration the proposal is numerically uniform, so what follows compares two draws the gradient cannot influence either way, and its null bounds what this configuration could detect rather than showing the gradient uninformative. The sweep below repeats it where the gradient does shape the proposal. The ablation compares three proposal methods: the true gradient (*policy*), a random direction rescaled to the gradient's norm (*grad-norm-preserved*), and a fully random vector (*random*). If the gradient's *direction* is informative the policy method should beat the norm-preserved one, which differs from it only in direction; if its *magnitude* is informative the norm-preserved method should beat the fully random one.

One confound has to be removed first. Gradient normalization, applied by default for numerical stability, rescales every proposal direction to unit norm, under which the norm-preserved and the fully random methods become the same object; an observed equality between the policy and random arms would then admit two explanations, an uninformative direction or a real signal discarded with the magnitude. The extended ablation therefore repeats the whole comparison *with gradient normalization disabled*, where the policy and norm-preserved methods both carry the gradient's full magnitude and differ only in direction, so any gap between them is attributable to direction alone. Doubling the ablation this way is why the study comprises 145 configurations rather than half as many.

With the confound removed, Table 4 reports the final KL divergence for the policy and norm-preserved methods with normalization disabled and the correction enabled, across four energy functions, together with their difference. The differences are small and inconsistent in sign: on GPT-2 Large the policy method is worse than the random direction by 0.044, on one GFlowNet variant worse by 0.008, on another better by 0.146, and on the length-normalized variant the random direction is better by 0.670. That last gap is the largest in the table but should not be read as evidence of active harm, since comparing a paired difference against an across-seed standard deviation is not a test and Section 5.3.2 shows that apparent penalties of this kind come from the asymmetric treatment of the reverse-proposal term. In no case does the true gradient reliably outperform a random direction of the same magnitude. In the tested token-substitution settings, then, the input-embedding gradient of frozen autoregressive sequence likelihood provided no reliable proposal advantage over a norm-matched random direction, and because the norm-preserved and fully random methods are themselves indistinguishable, the same holds for the magnitude.

To bound the equivalence rather than rest it on the eye, the flagship configuration was analyzed as a paired comparison, policy against the norm-preserved random direction on each sample. A last-iterate summary is the wrong one for a Markov chain, since it discards every state the chain visited, so the contrast is recomputed from the stored per-sample trajectories under three summaries at once. Table 3 gives them at $n = 200$: the last iterate differs by $+0.171$ nats, the chain mean over the second half by 0.002, and the best state reached by -0.043, every interval straddling zero. Across the thirty-five grid configurations with both arms, the chain-mean contrast is inside the ± 0.327 margin fixed in advance in nineteen and excludes zero in two.

What that design lacks is power: at $n = 200$ the smallest difference detectable at eighty percent power is 0.652 nats, twice the margin, which is why the tests decline to certify. The configuration was therefore repeated over four further independent corruption seeds and pooled to $n = 1000$ paired sequences, and at that size the two one-sided tests certify equivalence on all three statistics: $+0.133$ nats at the final step with a 95% interval of $[-0.063, +0.326]$, $+0.136$ on the chain mean with $[-0.012, +0.287]$, and $+0.033$ on the trajectory minimum with $[-0.079, +0.145]$. The claim can accordingly be stated positively: the two proposals are equivalent to within a margin fixed in advance, rather than merely not distinguishable. One qualification belongs with it: WikiText-2 validation yields only 282 sentences under the length filter, so the extra sample size comes from independent corruptions of the same sentences, and the interval is a statement about variation across corruptions.

Statistic	Policy	Random	Difference	95% CI	Wilcoxon p
Final step (as reported)	6.541	6.370	$+0.171$	$[-0.291, +0.614]$	0.400
Chain mean, second half	6.667	6.665	$+0.002$	$[-0.332, +0.347]$	0.747
Minimum over trajectory	4.625	4.669	-0.043	$[-0.271, +0.177]$	0.678

Table 3: The flagship paired contrast under three summaries of the same chains.

Figure 3 puts the whole grid on one axis, one row per configuration, and the picture is a column of intervals centred on zero: the contrast is not merely non-significant in the flagship cell but small and sign-inconsistent across models, samplers, correction settings, normalization settings and schedule lengths.

One further number belongs here. Across all 145 configurations at $n = 200$ each, roughly 29,000 recovery attempts, exact recovery of the corrupted token is 0.0 percent in 139 configurations and one or two sequences in the remaining six: the samplers essentially never restore the masked token.

Energy function	Policy	Grad-norm-preserved	Difference
GPT-2 Large (SFT)	6.474	6.430	+0.044
GFlowNet, $\beta_{\text{len}} = 0$, 500	6.009	6.001	+0.008
GFlowNet, $\beta_{\text{len}} = 0$, 2000	6.698	6.552	+0.146
GFlowNet, $\beta_{\text{len}} = 1$, 500	6.008	5.338	+0.670

Table 4: Final KL divergence (lower is better) for the true gradient against a norm-preserved random direction.

The Llama-3 runs carry a limited and specific role, since its tokenizer differs and its embedding space is roughly a third the scale of GPT-2's, so no KL divergence or inter-token distance from Llama is comparable digit for digit with one from GPT-2 and no such comparison is made. What they establish is cross-architectural and qualitative: the null, the Metropolis–Hastings breakdown and the anisotropy all appear there as on GPT-2, at their own scales. One contrast in the matrix is nominally significant, and it is worth stating why this thesis does not build on it. On Llama-3, with the correction and gradient normalization both disabled, the policy method reaches a higher final KL than the norm-preserved random direction, with a paired 95% interval of [0.45, 2.09] excluding zero and a Wilcoxon p of 0.015. Two considerations bound what that supports. It is one contrast selected from several dozen paired comparisons across the grid, uncorrected for multiplicity, and at $\alpha = 0.05$ a handful of such hits is expected by chance; Figure 3 shows it against the whole family. And it is measured with the correction *disabled*, so by this thesis's own

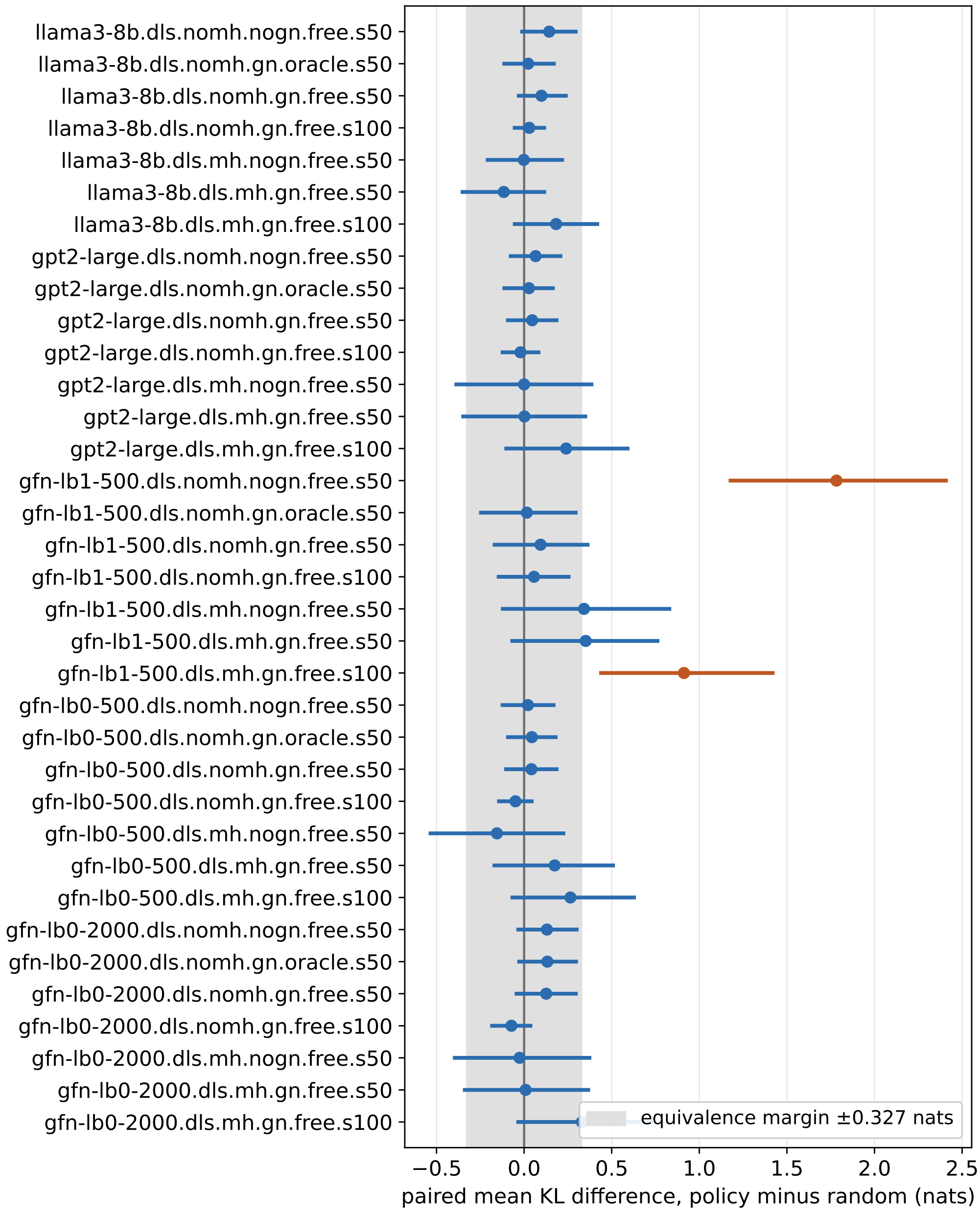

Figure 3: Paired difference in chain-mean KL, policy minus norm-matched random direction, for every configuration in the grid with both arms present. Points are paired means, bars are 95% percentile-bootstrap intervals over $n = 200$ paired sequences each, and the shaded band is the equivalence margin. Intervals whose bounds exclude zero are drawn in the darker colour.

argument it compares two biased optimizers rather than two samplers, the quantity compared being where an uncorrected chain happens to stop. Where the same apparent penalty is examined under a symmetric acceptance rule, in the sweep below, it vanishes. The reading that the gradient actively points away from good moves is therefore not adopted; the evidence supports indifference, not harm.

5.3.1 Robustness of the Null Result

Four variations were examined and none overturned the null. Extending the annealing schedule from fifty steps to one hundred creates no gap between the arms on any model, so the null is not a consequence of insufficient optimization budget. It also holds under both settings of the correction and both settings of gradient normalization, so it is not produced by the interaction of the ablation with either switch. It holds for multi-token recovery, in which two or three coupled positions must be recovered jointly: absolute quality degrades as the task gets harder, as one would expect, but the policy gradient does not pull ahead of a random direction at any corruption level. And it is not an artifact of the masked-recovery bench, since repeating the core comparison on a free-form prefix-continuation task, in which the sampler generates a twenty-token span with no reference token to recover, reproduces the equivalence: the policy method reaches a mean final KL of 8.818 with a 95% interval of [8.61, 9.02], against 8.850 with [8.65, 9.04] for the random direction, the two almost wholly overlapping and the norm-preserved and fully random arms numerically identical. The null therefore survives the move from a constrained infilling bench to open-ended generation, and is robust to the schedule, the correction, the normalization, the difficulty of the recovery task, and the choice of task.

5.3.2 Does the Null Survive a Load-Bearing Gradient?

Section 5.1 established that at the calibrated configuration the proposal is numerically uniform, so the ablation above could not have detected an informative gradient had one been present. That gap is closed here directly, by repeating the same three-arm contrast over a grid of configurations chosen so that the gradient term *does* shape the proposal.

Two knobs control that: the step size ϵ sets the ratio between the two terms of Equation (7), since the distance term scales as $1/\epsilon$ while the gradient term does not, and the temperature sets how sharp the resulting softmax is. The sweep crosses five values of ϵ from 1.05 to 10,500 with five temperatures from 0.05 to 25, for each of the three arms, with gradient normalization disabled so the gradient carries its true magnitude, at $n = 50$ per cell. It spans a ratio t_2/t_1 from 1.4 to 13,877, from the two terms contributing equally to the gradient term dominating by four orders of magnitude, and a minimum proposal entropy from 10.82 nats (uniform) down to 0.000 (deterministic). The calibrated cell sits at $t_2/t_1 = 11.1$ and entropy 10.22, in the near-uniform corner.

The result is that the null holds throughout. In no cell of the sweep does the true gradient reach a lower final KL than the norm-matched random direction by more than the cell-to-cell noise, and exact recovery of the corrupted token never exceeds 2 percent anywhere in the sweep, including in the fully deterministic gradient-driven corner where the proposal ranks candidates by the gradient alone. Making the gradient term load-bearing does not make it useful.

A chain-level statistic sharpens what that failure is. Besides the final token, each run logs whether the chain occupied the ground-truth token at *any* step. Averaged over the sweep the two rates are 0.108 and 0.135 percent, and in 69 of the 74 cells both are exactly zero, so the chain is not passing through the correct token and wandering away from it, which a last-iterate metric could have concealed; over roughly 185,000 proposal draws it essentially never reaches it.

The sweep also settles a claim made earlier in this chapter, and settles it against that claim. Under the correction as applied in the main grid the policy arm appeared *worse* than a random direction in the sharp cells, by +1.28 to +2.34 nats, which would have supported the reading that a mis-specified surrogate actively points away from good moves. That reading does not survive the control. As Section 4.3 discloses, the main grid charges the reversibility term to the policy arm alone, and in a sharp proposal that penalty is large. Rerunning the three sharpest cells with the term computed exactly for every arm removes the effect entirely (Table 5): $+2.34$ nats becomes -0.25 and $+0.17$ becomes -0.009, with the policy column unchanged between treatments, as it must be since only the comparators' treatment changed.

		Correction as in the main grid			Correction exact for all arms		
ϵ	T	Policy	Random	Difference	Policy	Random	Difference
1.05	0.05	8.414	7.132	+1.281	8.414	8.583	−0.169
10.50	0.05	8.166	5.831	+2.336	8.166	8.416	−0.250
10.50	1.00	5.760	5.593	+0.167	5.760	5.768	−0.009

Table 5: Final KL in the three sharpest sweep cells under the two treatments of the Metropolis–Hastings reverse-proposal term.

Two conclusions follow. The null is real and is not an artifact of a proposal too flat to express the gradient, since it holds where the gradient fully determines the proposal. And the complementary claim, that the gradient is reliably worse than noise, is withdrawn: it appeared in the Llama-3 contrast above and in the sharp cells here, and in both places the compared arms were not charged the same reversibility term. The evidence supports indifference, not anti-guidance.

5.3.3 Gradient-Free Baselines

The first of the three candidate explanations can now be tested. Under Hypothesis A the target itself is unusable, in which case no method operating on this energy would recover anything, whether or not it differentiated it. Scoring the same energy with methods that never take its gradient settles the question, and Table 6 does so on the masked-recovery metric for GPT-2 Large. The ground-truth fill sits at the metric floor of 0.00, leaving the corrupted token in place gives 9.14 and a random token 9.39, and a single conditional forward pass already improves on these, reaching 8.24 by argmax of the model's own next-token distribution and 8.62 by sampling from it. The last two rows are decisive. A single forward pass reranking the top-k candidates by their effect on the sequence reaches 4.43, better than every Langevin configuration in the grid, whose best discrete cell sits near 6.4; and a gradient-free Gibbs sampler on exactly the same energy, resampling one masked position at a time from the model's conditional, reaches 6.69, comparable to the best gradient-guided Langevin result without ever computing a gradient.

Method (no gradient of the energy)	Mean KL
Ground-truth fill (metric floor)	0.00
Untouched (corrupted token left in place)	9.14
Random-token fill	9.39
Conditional argmax (one forward pass)	8.24
Conditional sample (one forward pass)	8.62
Top-k rescore (one forward pass)	4.43
Gibbs (gradient-free sampler)	6.69

Table 6: Reference baselines on masked-token recovery for GPT-2 Large (lower is better).

Measured by exact recovery rather than by divergence the same ordering holds: the top-k rescoring pass restores the corrupted token in 33 percent of sequences and the Gibbs sampler in 18.5, against 0.0 percent for every Langevin configuration in the grid. That pair of figures is the gradient-free comparator against which the proposal ladder of Section 5.6 should be read, since a bidirectional proposal's 39 percent is a gain over 33 and not over nothing.

Two methods that evaluate the frozen likelihood but never differentiate it, a single rescoring pass and a gradient-free sampler, both do at least as well as the gradient-guided samplers, and the rescoring pass does markedly better. What the Langevin samplers add over these baselines is the gradient. Hypothesis A is therefore rejected for the failure under investigation: on this recovery task the target is searchable, and searchable at a fraction of the cost, since the gradient-free baselines spend no backward pass at all (Appendix A.3). The scope of that acquittal is worth stating, because Section 5.7 will show the same energy behaving badly in a different regime. What is established here is that the energy does not explain why a gradient-guided search fails at in-place revision; it is not established that the energy is a good objective to maximize in open-ended generation, and it is not. What remains to be decided is whether the model fails to contain the information a local proposal needs, which is Hypothesis B, or whether it contains it and the derivative discards it, which is Hypothesis C. Section 5.4 takes that up.

5.3.4 External-Judge Rescoring

A null measured with the model's own predictive distribution invites the objection that the metric is built from the model that produced the recoveries. The recovered sequences, from the discrete sampler on GPT-2 Large with the correction and gradient normalization enabled over fifty steps for the same three arms, were therefore rescored under perplexity by an independent Llama-3 judge that took no part in the sampling. The policy recoveries score a mean judge perplexity of 178.4 and the random ones 181.3, a difference under two percent, with per-token negative log-likelihoods of 4.44 against 4.45. The external judge returns the same reading as the internal metric, so the indistinguishability of the methods is not an artifact of the evaluation model.

5.4 Why the Input-Embedding Surrogate Fails, and What Recovers It

The mechanism behind the null is tested directly on the surrogate the discrete proposal uses, which ranks a candidate token v by the first-order term $\hat{\Delta}(v) = \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$, a linear approximation to the true change in log-likelihood a substitution would produce (Section 2.4). The linearization experiment computes both quantities for thousands of candidate substitutions and correlates them. Figure 4 plots one against the other for GPT-2 Large, and there is no relationship: the Spearman correlation is -0.011, negligible in magnitude and negative in sign. Nor is that a peculiarity of the base model, since the correlation is negligible on every one of the five energies, with $|\rho| < 0.06$ throughout (GPT-2 Large -0.011, Llama-3 0.021, the three GFlowNet variants 0.024, 0.046 and 0.057), each over $n = 400{,}000$

candidate–sequence pairs. Ranking candidate tokens by the gradient surrogate is therefore equivalent to ranking them at random, and because the measurement is made at the level of the proposal rather than through the downstream sampling metric, it is the more direct of the two observations.

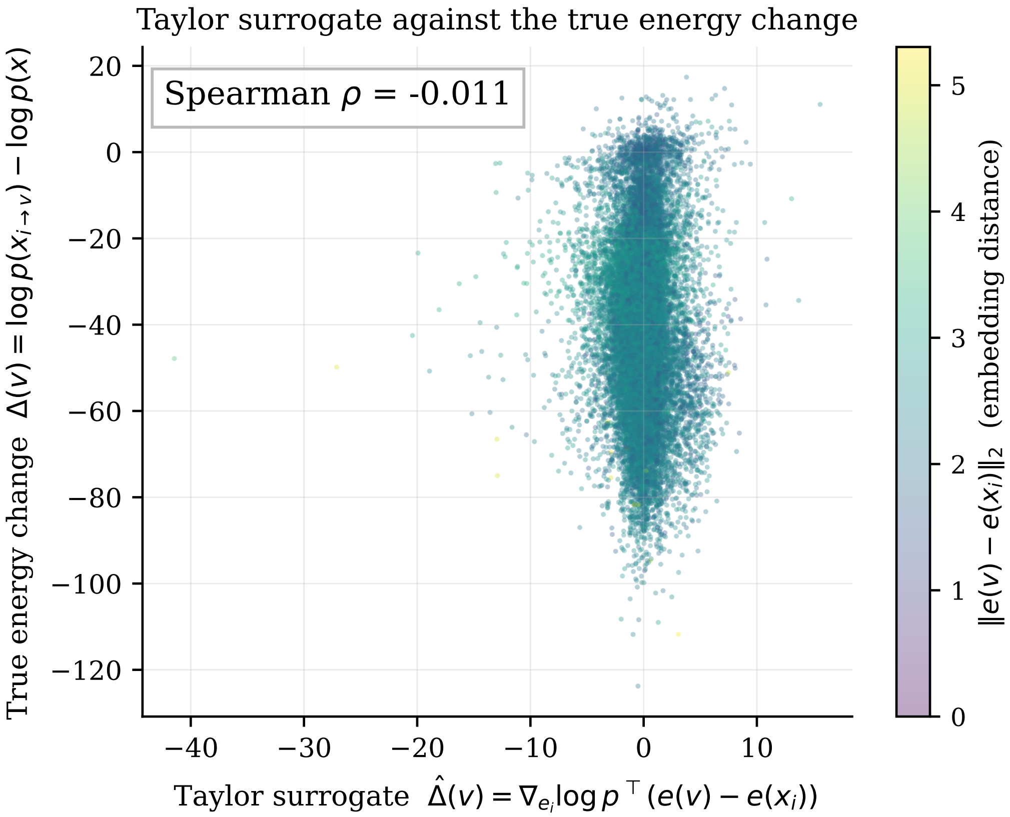

Figure 4: The first-order Taylor surrogate used by the discrete proposal, plotted against the true change in sequence log-likelihood for GPT-2 Large, coloured by the embedding distance of the candidate token.

The cause is a mismatch between two length scales. A Taylor expansion is accurate only in a small neighbourhood of the point it is taken at, and a Transformer (Vaswani et al., 2017), with its layer normalization and attention nonlinearities, is a strongly nonlinear function of its input embeddings, so that neighbourhood is small. Whether it is large enough to contain any move the sampler must make is answered by Figure 5, which computes the surrogate-to-truth correlation *within* bins of embedding distance. The correlation is at best weakly positive at the very smallest distances and has decayed to zero well before the mean nearest-neighbour distance of 1.82 reported in Section 5.8, and far below the mean candidate distance of 2.35. The nearest-neighbour distance is by definition the scale of the smallest possible substitution, since to change a token at all is to move to another token's embedding, so the entire admissible range lies beyond the distances at which the surrogate carries signal. The gradient is not merely imprecise here but *inapplicable*.

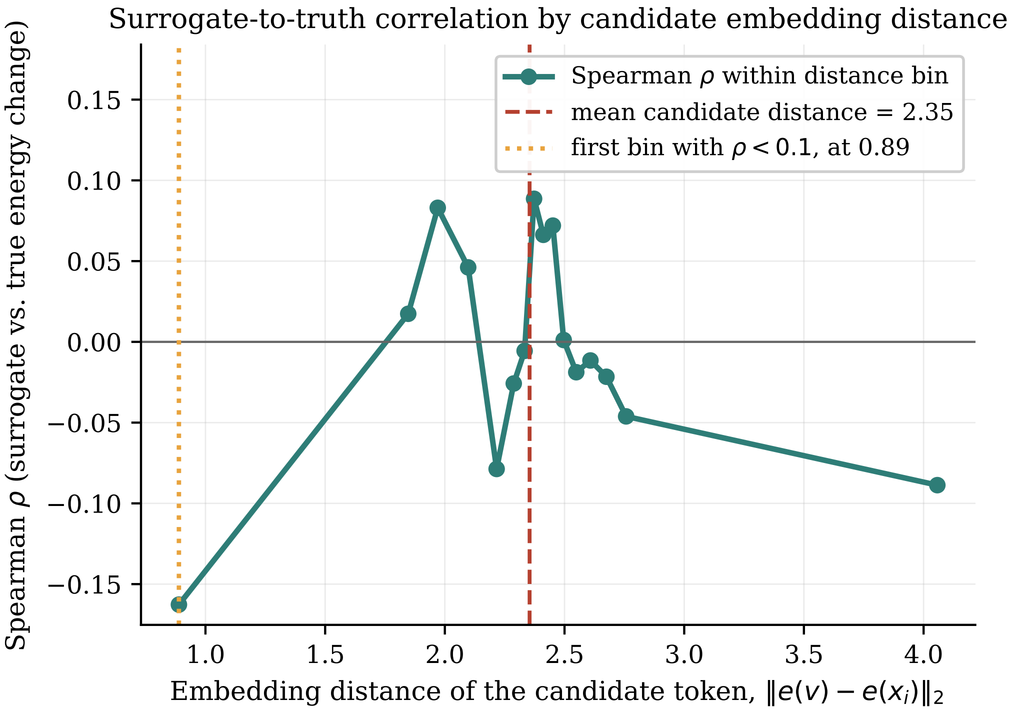

Figure 5: Correlation between the gradient surrogate and the true energy change as a function of the embedding distance of the candidate token, for GPT-2 Large.

A second limitation compounds the first, and the decomposition built into the experiment exposes it. Taken with respect to the input embedding of the masked position, the gradient can register only the effect of a substitution on the *future* tokens, whose probabilities depend on that embedding as it flows forward through the network. It is structurally blind to the effect on the probability of the candidate token *itself* given its left context, because that probability depends on the candidate through a discrete index into the output softmax rather than through any continuous input. Averaging the absolute value of each signed change over candidates gives a mean absolute self term of 15.0 nats against a mean absolute future term of 24.2, so the invisible half is not a rounding error but a substantial fraction of the total, and the surrogate correlates with the future term ($\rho = 0.03$) rather than the self term ($\rho = -0.10$), exactly as the structural argument predicts. The gradient is therefore evaluated outside the range where it would be valid, and even within that range it would be blind to a large part of what determines whether a substitution is good. The full decomposition is in Figure 6, and the top-k recall analysis, which finds that gradient ranking does not recover the true best substitutions better than chance, in Figure 7, both in Appendix A.1. Together they answer the "why not" of RQ1: the surrogate is not a noisy version of the true ranking but unrelated to it over the range the sampler must move, which is why no step-size or temperature setting recovers it.

The blindness just measured follows from a modelling choice, and naming that choice exactly is what makes the alternative available. Everything above differentiates the sequence log-likelihood with respect to the *input embedding* at the masked position, $\boldsymbol{g} = \partial \log p(\boldsymbol{x})/\partial \boldsymbol{e}_i$, which is what the samplers implemented here use, because they operate in embedding space and the energy they score gathers at the projected token index. But a token index enters the sequence likelihood in two roles: through the embedding lookup, as the input from which the suffix is predicted, which is the role $\boldsymbol{g}$ sees, and through the output softmax, as the target of the prediction made at the previous position, which $\boldsymbol{g}$ cannot see because that dependence runs through a discrete index rather than any continuous input.

Registering both requires relaxing the token indicator itself, in both roles, rather than relaxing its embedding. Let $\boldsymbol{E} \in \mathbb{R}^{|V| \times D}$ be the embedding matrix, whose v-th row is $\boldsymbol{e}(v) \in \mathbb{R}^D$, and let $\boldsymbol{z}_i \in \mathbb{R}^{|V|}$ be a point on the vocabulary simplex, so that its coordinates are non-negative and sum to one; it is relaxed from the vertex that the current token x_i occupies, where the coordinate $\boldsymbol{z}_i[x_i]$ is one and every other is zero. Define the relaxed objective

$$\tilde{L}(\boldsymbol{z}_i) \;=\; \sum_{v\in V} \boldsymbol{z}_i[v]\,\log p(v \mid \boldsymbol{x}_{<i}) \;+\; L_{\text{future}}\big(\boldsymbol{E}^\top \boldsymbol{z}_i\big)\,, \tag{11}$$

where the sum runs over all $|V|$ tokens $v \in V$, $\boldsymbol{z}_i[v] \in [0,1]$ is the coordinate of $\boldsymbol{z}_i$ belonging to token v, and $L_{\text{future}} : \mathbb{R}^D \to \mathbb{R}$ scores the suffix from an input embedding, so that $\boldsymbol{E}^\top \boldsymbol{z}_i \in \mathbb{R}^D$ is the mixed embedding the relaxation feeds forward. In words, the two terms of (11) are the two roles a token index plays: the first is the current position's own contribution to (3), with the discrete target replaced by the relaxed indicator, and the second is the log-likelihood of the suffix evaluated with $\boldsymbol{E}^\top \boldsymbol{z}_i$ supplied as the input embedding at position i. At a vertex of the simplex $\tilde{L}$ coincides with the sequence log-likelihood, so (11) is a relaxation of the quantity the samplers target and not a different objective. Its derivative in the token-indicator coordinates is available in closed form,

$$\frac{\partial \tilde{L}}{\partial \boldsymbol{z}_i[v]} \;=\; \underbrace{\log p(v \mid \boldsymbol{x}_{<i})}_{\text{self}} \;+\; \underbrace{\boldsymbol{g}^\top \boldsymbol{e}(v)}_{\text{future}}, \tag{12}$$

one scalar for each of the $|V|$ candidates, so the whole derivative is a vector in $\mathbb{R}^{|V|}$ with one entry per token, against the $\mathbb{R}^D$ of the input-embedding gradient $\boldsymbol{g}$. This thesis calls it the **relaxed token-indicator derivative**, shortened to the token-indicator derivative where no confusion arises. Read in words, (12) says that the worth of putting token v at position i is the sum of two things the model already computes: how likely v is here given everything to its left, and how well v's embedding points along the direction that raises the likelihood of everything to its right. The name records precisely what the object is, because the looser description "the one-hot gradient" is ambiguous in a way that matters here: this is a derivative with respect to the token-indicator coordinates $\boldsymbol{z}_i$, under a parameterization that relaxes the indicator in both of its roles, and it is not the ordinary derivative of the likelihood with respect to a one-hot input through the embedding lookup alone, which by the chain rule is $\boldsymbol{E}^\top \boldsymbol{g}$ and whose v-th coordinate is the future term by itself. The difference between the two is the self term that only the two-role relaxation exposes.

The candidate ranking it induces is $\hat{\Delta}_{\text{tok}}(v) = \hat{\Delta}(v) + [\log p(v \mid \boldsymbol{x}_{<i}) - \log p(x_i \mid \boldsymbol{x}_{<i})]$, whose bracket is exactly the self term the linearization experiment already

records, since substituting v at position i changes the log-likelihood of that position by precisely that conditional log-ratio and by nothing else. The comparison is therefore an exact re-analysis of data already collected, requiring no further forward passes.

One thing this surrogate is not should be said before the numbers, because the shorthand used below invites the wrong reading. The thesis speaks of the missing information living on the model's *output side*, and that is where the missing term comes from: the direct token-fit term is read off the output conditional $\log p(v \mid \boldsymbol{x}_{<i})$. It does not follow that the future term has been discarded in turn. The token-indicator derivative is the *sum* of the two, an output-side self term added to the same input-side future term the embedding gradient computes, and it is that sum which is measured below. Where a plain output-side quantity is meant instead, as in the left-conditional proposals of Section 5.6, the text says so.

Table 7 reports it, and the result decides between the two remaining hypotheses. Over the full candidate set the token-indicator surrogate correlates with the true energy change at $\rho = 0.37$ to 0.59 on the five energies, against $|\rho| < 0.06$ for the input-embedding surrogate. Restricted to the *near* stratum, whose mean candidate distance of 1.94 is the admissible substitution range the preceding paragraphs argue about, the separation is starker still: $\rho = 0.60$ to 0.73 against 0.03 to 0.11.

Three things follow. The first is that the strong form of Hypothesis B is rejected. A usable local direction *exists* inside the frozen autoregressive model, it survives its training objective untouched, and it is available in closed form from a single forward pass, since $\log p(v \mid \boldsymbol{x}_{<i})$ is the model's own next-token distribution; nothing has to be retrained for it to be read, and what discards it is the choice of coordinates in which the derivative is taken. The weaker form of the hypothesis survives, and Section 5.6 measures how much of it: a left-to-right factorization gives a proposal no path from the right context to the position it is filling, so the frozen model contains *some* of what an in-place revision needs and not all of it.

The second is that the input-embedding gradient is not weak signal but noise. Adding it to the self term slightly *degrades* the correlation on GPT-2 Large (0.588

	All candidates		Near stratum	
Energy function	Input emb.	Token ind.	Input emb.	Token ind.
GPT-2 Large (SFT)	−0.011	**0.588**	0.028	**0.730**
Llama-3 8B	0.021	**0.513**	0.039	**0.596**
GFlowNet, $\beta_{\text{len}} = 0$, 500	0.057	**0.379**	0.111	**0.649**
GFlowNet, $\beta_{\text{len}} = 0$, 2000	0.024	**0.439**	0.091	**0.706**
GFlowNet, $\beta_{\text{len}} = 1$, 500	0.046	**0.365**	0.051	**0.610**

Table 7: Spearman correlation between the first-order surrogate and the true change in sequence log-likelihood. The near stratum holds 100,000 of the 400,000 pairs per energy, at a mean candidate distance of 1.94, and 0.45 for Llama-3. "Input emb." is the input-embedding gradient, "token ind." the relaxed token-indicator derivative.

against 0.595 for the self term alone) and on two of the three GFlowNet variants, so a sampler that ignored the gradient entirely and ranked candidates by the conditional alone would do marginally better than one that used it.

The third is that several results which otherwise sit apart are one result. The gradient-free methods that work, top-k rescoring at 4.43 and Gibbs at 6.69 in Section 5.3.3, are precisely the methods that read the output side. The final-position case of Section 5.5 is the extreme instance of Equation (12), in which the future term vanishes, the input-embedding gradient is exactly zero, and the self term is the entire energy difference. And the concrete score of a discrete diffusion model is itself a ratio of sequence probabilities, which is an output-side quantity of the same kind.

A ranking correlation is not yet a working sampler, so the token-indicator surrogate was substituted into the discrete sampler itself and run over the same (ϵ, T) grid as the sweep above, changing nothing else. Table 8 reports it cell for cell. Where the proposal is sharp enough to express a ranking the token-indicator surrogate recovers 36 to 40 percent of corrupted tokens, while the input-embedding gradient never exceeds 2 percent anywhere in the same grid. At the temperature the study calibrated on, $T = 5$, even the token-indicator proposal falls to 2 percent, because that temperature flattens the proposal to a near-uniform draw whatever surrogate is supplied. The temperature, not the model, was the binding constraint.

	Proposal temperature T				
Surrogate, at ϵ	0.05	0.25	1	5	25
Token indicator, $\epsilon = 10.5$	0.0	8.0	**40.0**	—	—
Token indicator, $\epsilon = 105$	0.0	8.0	**36.0**	2.0	0.0
Token indicator, $\epsilon = 1050$	0.0	8.0	**38.0**	2.0	0.0
Input embedding, $\epsilon = 10.5$	0.0	0.0	0.0	0.0	0.0
Input embedding, $\epsilon = 105$	0.0	0.0	2.0	0.0	0.0
Input embedding, $\epsilon = 1050$	0.0	0.0	2.0	0.0	0.0

Table 8: Exact recovery (percent) with the relaxed token-indicator surrogate against the input-embedding surrogate, over the same step-size and temperature grid.

The same substitution on the three GFlowNet-tuned energies gives 25.0, 21.0 and 9.0 percent in the sharp configuration against 0.5 percent at the calibrated one, and the cross-architecture control carries the result across a different tokenizer, a different embedding geometry and an order of magnitude more parameters: on Llama-3 8B the token-indicator proposal recovers 41.0 percent of corrupted tokens at a final KL of 1.908, where every input-embedding arm in Llama's archived grid recovers 0.0 percent, its best cell reaching a KL of 3.898.

That last comparison has to be read with two qualifications, because it is not as clean as the GPT-2 one. The 41.0 percent comes from the surrogate-driven configuration and the 3.898 from Llama's calibrated one, since no input-embedding run was made at the surrogate-driven setting; the two therefore differ in step size and temperature as well as in the derivative, and the honest statement is that no input-embedding configuration that was run recovers anything on this model, not that a matched pair was compared. And the two divergences are on Llama's own scale, which Section 4.2 keeps off the axis the GPT-2 family is measured on, so 1.908 is not comparable with any GPT-2 figure quoted elsewhere in this chapter and is not offered as one.

That result also isolates what the configuration must supply, because the same surrogate at Llama's own calibrated setting recovers nothing, 0.0 percent at a KL of 4.108. That cell is worth pausing on, because it is the one properly matched comparison in this paragraph: same model, same sequences, same step size and temperature, the two surrogates differing in nothing but the derivative. There the token-indicator proposal is not better than the input-embedding gradient but marginally worse, 4.108 against 3.898, with both at 0.0 percent exact. Whatever produces the 41.0 percent, then, it is not the derivative alone. Both configurations produce a sharp proposal, so the naive reading would be that sharpness is not the issue; the proposal statistics show otherwise. At the calibrated setting $\epsilon = 0.1$ makes the distance term $-\|\boldsymbol{e}(v) - \boldsymbol{s}\|^2/2\epsilon$ enormous, so the softmax concentrates on the current token and the measured t_2/t_1 is 1.83: the proposal is sharp about *staying put*. In the recovering configuration $\epsilon = 10$ suppresses the distance term and t_2/t_1 rises to 229, so it is sharp about the *surrogate's ranking*. Only the second puts the candidate ranking in charge. That is what the 41.0 percent needs, and it is why the cross-model claim is conditional rather than general: the token-indicator derivative carries usable ranking information on Llama as it does on GPT-2, and a configuration that lets that ranking drive the proposal is what converts it into recovery. The same distinction governs the GPT-2 sweep, where the surrogate needed a large ϵ and a low temperature together and where GPT-2's own calibrated cell also yielded 0.5 percent.

The claim this thesis establishes is therefore narrower, and considerably more constructive, than a blanket null. It is not that a frozen autoregressive likelihood admits no gradient guidance. It is that the input-embedding Jacobian slice of a likelihood whose target token enters as a discrete index discards the half of the energy that carries the signal, and that supplying the discarded half turns the same sampler on the same frozen model from zero recovery into recovery comparable to a purpose-trained diffusion model. The coordinate choice that produces the blindness belongs to the implementation studied here rather than to the published embedding-space samplers from which it was drawn, a scope restriction Section 6.4 states in full. Two of the three candidate explanations are now closed, the second of them in its strong form: the target is usable on this task, and the model does contain a usable local direction. Section 5.5 checks the algebra of that claim at the one position where it can be checked exactly, and Section 5.6 then tests what Hypothesis C additionally proposes, that a proposal permitted to condition on the right context recovers more still.

5.5 The Final-Position Case

This section adds no new claim. It confirms the result of Sections 5.3 and 5.4 at one position where the argument can be made exactly rather than statistically, and exactness is its contribution.

The input-embedding gradient sees a substituted token only through its effect on the tokens that follow it, and at the last position there are none: under causal attention the final token's input embedding feeds only the logits that would predict a next token which does not exist, and those logits appear in no term of (3). The exact statement is therefore that the gradient of the sequence log-likelihood with respect to the *final token's input embedding*, under the shifted causal indexing in which position t predicts token $t + 1$, is exactly zero. It is not a claim that every gradient involving the final token vanishes; gradients with respect to earlier positions, or with respect to parameters, are unaffected. The energy difference between two candidate final tokens, by contrast, is exactly the model's conditional log-probability ratio given the fixed preceding context. At that one position the energy is maximally informative about which token is better and the input-embedding gradient carries none of that information.

A direct experiment measures the consequence. The masked position is placed at each of three points differing only in how many scored terms of the log-likelihood its embedding can reach: the final token, which feeds no realized prediction, the second-to-last, which feeds exactly one, and a middle token, which feeds many. Only that position is corrupted, the surrounding context is clean, and the same corrupted sequence goes to every arm, so the sole variable is the downstream context the gradient can act on. The arms are the discrete Langevin sampler with the true gradient and with a random direction of equal norm; three energy-only methods that never touch the gradient (argmax and top-k rescoring of the left-context conditional, and an independence Metropolis sampler that proposes from that conditional and accepts by the exact sequence-log-likelihood ratio); and a recorder of the language-model gradient norm the sampler consumes.

Table 9 reports the outcome. At the final position the language-model gradient norm the sampler consumes is 0.0000 in both mean and maximum over roughly fifteen thousand gradient evaluations, not a small number but an exact zero, the theorem realized inside the live sampler; with it the discrete sampler recovers the target token in 0.0% of two hundred sequences whether the direction is true or random, the paired difference straddling zero on every metric (rank difference -17.3, 95% CI $[-513, +499]$). The energy at the same position is fully usable: the independence sampler that proposes from the conditional and accepts by the exact sequence-log-likelihood ratio accepts 100.00% of its moves in both mean and minimum, since target proportional to proposal forces acceptance, which also certifies that the implemented energy carries no term beyond the sequence log-likelihood. The energy-only methods recover between 34.5% and 40.0% of the tokens where the gradient arms recover none.

One consequence of Equation (12) is worth naming here rather than adding an arm for it. At the final position the future term $\boldsymbol{g}^\top \boldsymbol{e}(v)$ vanishes identically, so the relaxed token-indicator derivative reduces exactly to $\log p(v \mid \boldsymbol{x}_{<i})$, which is the quantity the conditional-argmax and top-k arms already rank by. The energy-only rows of Table 9 are therefore the token-indicator surrogate at that position, under another name, and the 34.5 to 40.0 percent they recover is what it attains where the input-embedding gradient is exactly zero. In the interior the two come apart, because the future term is then nonzero, and Table 8 is where they are compared directly.

The second-to-last position answers the one open empirical question the design poses. With a single downstream scored term the gradient is no longer zero, its mean norm being 12.945, and it is still not usable: the policy arm remains at 0.0% recovery and indistinguishable from the random arm, while the energy-only methods recover roughly half the tokens. The null is therefore not confined to the zero-gradient edge case but persists once the gradient becomes nonzero, and across all three positions the discrete Langevin arms never recover the target in six hundred trials while the exact-energy methods recover between eighteen and fifty-five percent. As the masked position moves from the end of the sequence into the interior, the gradient norm the sampler consumes climbs from zero while the energy-only sampler's acceptance falls from one hundred percent, because the energy begins to weigh downstream terms the left-context proposal ignores; the policy-minus-random gap stays at zero across the whole range.

	Exact recovery (%), by downstream scored terms		
Arm	Final (0)	Second-to-last (1)	Middle (many)
DLS, true gradient (policy)	0.0	0.0	0.0
DLS, random direction	0.0	0.0	0.0
Conditional argmax (energy only)	40.0	50.0	18.0
Top-k rescore (energy only)	40.0	55.0	39.0
Independence MH (energy only)	34.5	49.5	31.5
DLS-policy LM gradient norm (mean)	0.000	12.945	23.976
Independence-MH acceptance (%)	100.0	63.4	30.3

Table 9: The final-position experiment. Upper block: exact recovery per arm. Lower block: the mean language-model gradient norm the discrete sampler consumes, and the independence sampler's acceptance rate.

5.6 The Proposal Ladder

Hypothesis C has two halves. The first, that the information a local proposal needs is present in the frozen model and is discarded by the input-embedding derivative, is settled in Section 5.4. The second, that a proposal permitted to condition on the right context recovers more still, is what this section tests.

The task is the single-token recovery of the rest of the study, run identically for every arm: one interior position per sequence is corrupted, chosen by the same deterministic per-index rule and the same corruption seed as the main grid, over the same two hundred WikiText-2 sequences, for fifty steps, and every arm accepts by the same exact autoregressive sequence-log-likelihood energy. Five of the arms are position-wise independence samplers differing in nothing but the distribution the proposal draws from, which is what makes the comparison among them controlled. The two gradient arms are not: they are the thesis's own discrete Langevin sampler, so they differ from the other five in the transition kernel as well as in the proposal. They are included because reproducing the null from separate code on this sequence set is what anchors the bottom of the ladder, and the difference in kernel is noted here rather than smoothed over.

Two model families serve that purpose, and it matters that they differ from the autoregressive backbone in different ways: a discrete diffusion model is trained with a score objective *and* answers a bidirectional denoising query, so on its own it cannot separate the two, while a masked language model has the second property without the first, so the pair does separate them. The diffusion arms come first, because they were run first and because their two auxiliary measurements place the model on the same axes as the earlier sections. The full ladder, the organizing result of this section, is Table 11, assembled once and referred back to thereafter.

5.6.1 A Score-Trained Diffusion Proposal

The model is the score-entropy discrete diffusion (SEDD) model of Lou et al. (2024) at two scales, small and medium, which share the GPT-2 tokenizer and drop into the existing harness without retokenization. These are pilots, on smaller and differently trained models than the frozen autoregressive backbone, so the comparison is qualitative and directional by design and is not read as a claim that a diffusion model beats the domain-tuned autoregressive baselines in absolute terms. One clarification governs it, because the contrast is easy to state too loosely. The relevant difference between the families is not that one commits tokens within a pass and the other does not, since absorbing diffusion also commits a token at each denoising step and does not thereafter change it. The difference is at the level of the revision *operation*: to revise token i, a diffusion model re-masks that position and re-denoises it conditioned on both sides, an in-distribution bidirectional query it was trained to answer, whereas an autoregressive model has no conditioning path from the right context to a token at all.

The first measurement repeats the linearization diagnostic of Section 5.4 with the diffusion score in place of the autoregressive gradient, on the sequence set the reference baselines used, so the numbers sit on one axis. For each sequence and one masked position, a cheap one-pass surrogate, the model's denoising log-preference for filling the position, is correlated against an expensive per-candidate truth, the measured effect of committing each candidate on the reconstruction of the held-out remainder. Table 10 reports the Spearman correlation, pooled and by embedding-distance stratum. Where the autoregressive input-embedding surrogate is indistinguishable from zero at every distance, the diffusion surrogate is clearly positive in every stratum and at both scales, with a pooled value near 0.13 and a per-sequence mean near 0.18, the medium model exceeding the small one in the near stratum where a first-order surrogate is most expected to apply. On a second corpus of narrative text the same measurement gave a pooled correlation of 0.184 and a near-stratum correlation of 0.306; the near-stratum value is corpus-dependent and is not carried over, but the sign and order of magnitude are stable.

Model / objective	ALL	near	mid	random	per-seq mean
Autoregressive gpt2-large (gradient)	−0.011	+0.027	−0.041	−0.048	−0.011
SEDD-small (score)	+0.130	+0.097	+0.115	+0.126	+0.184
SEDD-medium (score)	+0.133	+0.148	+0.122	+0.110	+0.179

Table 10: Spearman correlation between the cheap one-pass surrogate and the expensive per-candidate truth, on the unified WikiText-2 sequence set ($n = 200$, four hundred thousand candidate pairs), pooled and by embedding-distance stratum. The autoregressive row is the input-embedding gradient of Section 5.4.

The second measurement asks the diffusion model to perform the revision itself rather than merely to rank candidates for it. On the same corrupted sequences, the flipped token is re-masked and the position re-denoised conditioned on both sides, a single native draw with no reranking and no sampling of many candidates. Both scales recover the exact token in roughly a third of cases, 32.5 percent at the small scale and 35.0 at the medium, place the true token in the top five in 61.5 and 67.0 percent, and reach a KL under the autoregressive yardstick of 3.75 and 3.73, well below the untouched corruption at 9.14 and the flagship Langevin configurations near 6.5.

The third measurement is the one that belongs to the ladder, because it changes nothing but the proposal. A position-wise independence Metropolis chain is run on the same sequences and positions, accepting by the exact autoregressive sequence log-likelihood, with the diffusion model's one-pass bidirectional log-preference over the vocabulary supplying the proposal in place of the autoregressive left-context conditional. Recovery rises from 0.0 percent for the input-embedding gradient arm to 23.5 for the left-conditional proposal and to 38.5 and 39.0 for the diffusion proposal at the two scales, with acceptance rising from 34.4 percent for the left-conditional arm to 40.0 and 43.3 percent for the hybrid, because a proposal that sees both sides of the mask more often lands on tokens the exact energy will accept. That the hybrid exceeds the left-conditional arm, and not only the gradient arm, is what measures the contribution of the right context over an exact reweighting of the left context alone. All four of these arms appear in Table 11.

5.6.2 Separating Bidirectionality from Score Training

The diffusion arms leave one variable unresolved, and it is the one an attribution turns on. RoBERTa (Liu et al., 2019) is the discriminating control: it is *not* score-trained, but it *is* bidirectional, and it answers exactly the query an in-place revision poses, the distribution of a token given both of its contexts. If a masked-language-model proposal also works in this chain, then the operative variable is the conditioning and not the objective.

Everything but the proposal is held fixed against the autoregressive runs: the same sequences, the same deterministic per-index corruption routine, the same exact

GPT-2 Large energy, the same accept/reject rule, the same metric. The proposal is an independence sampler, so its Hastings ratio is available in closed form. RoBERTa and GPT-2 index different vocabularies, so a once-off bridge maps each RoBERTa token whose surface string re-encodes to exactly one GPT-2 token; it covers 99.95 percent of the vocabulary, and no ground-truth token in the evaluation set fell outside it.

One control is needed before the result can be read, because a recovery rate far above the rest of the study invites the suspicion of leakage. Replacing the masked-language-model conditional with a *uniform* draw over the same bridged vocabulary, through identical code, gives 0.5 percent exact recovery and a final KL of 6.538 at a 9.48 percent acceptance rate. Those figures reproduce the flagship Langevin sampler (0.0 percent, 6.541, 9.98 percent) to three significant figures, from a separate implementation. The pipeline therefore carries no information of its own, and it independently confirms the central measurement of Section 5.1: a gradient-guided discrete Langevin chain on this energy performs exactly as a uniform-proposal chain performs.

Table 11 places every proposal run on this task on one axis, and the ordering is unambiguous. The two arms with no output-side access at all, the input-embedding gradient and its norm-matched random control, sit at zero recovery and are indistinguishable from a uniform draw; those two are the Langevin sampler rather than an independence chain, so they are the reference the ladder is read against rather than a rung of it. Left-only conditioning lifts recovery to 23.5 percent. Bidirectional conditioning lifts it to 38.5 and 39.0 percent with a score-trained model, and to 44.5 percent with one that is not score-trained.

Score training is therefore not what the ordering tracks. A model trained with ordinary masked-language-model cross-entropy, and never with a score-matching or score-entropy objective, supplies a proposal that recovers the corrupted token more often than either score-trained diffusion model tested. What the working proposals share is not their objective but their access: each reads the candidate's probability from a model's output side, and the two bidirectional ones can additionally condition on the right context, which is the query an in-place revision actually poses. The

Proposal	Conditioning	Score-tr.	Exact (%)	Mean KL
Input-embedding gradient	none	no	0.0	6.854
Norm-matched random dir.	none	no	0.0	6.584
Uniform over the vocabulary	none	—	0.5	6.538
AR conditional	left only	no	23.5	5.170
SEDD-small concrete score	bidirectional	yes	38.5	3.120
SEDD-medium concrete score	bidirectional	yes	39.0	3.017
RoBERTa-large masked LM	bidirectional	**no**	**44.5**	**2.737**

Table 11: Single-token recovery on the same 200 WikiText-2 sequences, every arm accepting by the same frozen GPT-2 Large energy.

reading has to be stated as an interpretation rather than an isolation, because the arms differ in more than conditioning: RoBERTa-large, SEDD-small and SEDD-medium differ from one another and from GPT-2 Large in scale, in training corpus and in architecture, and only the tokenizer and the chain are held fixed across all of them. What the ladder establishes is that conditioning access orders the results better than the training objective does, not that every other difference has been controlled away.

Three things about the ladder are the finding. Its bottom rungs say that the input-embedding gradient buys nothing over a coin flip on this energy, which is the null of Section 5.3 reproduced from separate code. Its middle rung says that the very model whose gradient fails reaches 23.5 percent when read from the output side instead, which is Section 5.4 arriving by another route. Its top rungs say that the remaining gap is bought with right context, and can be bought without a score objective. Hypothesis C is therefore supported in both halves, and the earlier reading that the training objective is the source of the failure is not merely qualified but replaced: an objective is not required, because bidirectional conditioning suffices, and it is not the whole story in the other direction either, because a purely left-to-right conditional already recovers a quarter of the tokens the gradient never reaches.

5.7 The Likelihood Trap

The energy-based framework rests on the premise that low energy, that is, high likelihood, corresponds to good text. The likelihood-trap experiment tests that premise on the study's own models, reproducing quantitatively the neural-text-degeneration phenomenon of Holtzman et al. (2020). The premise has been contradicted most sharply in machine translation, where Stahlberg and Byrne (2019) found the globally most probable translation under a trained model to be the empty string for a large fraction of inputs, and Eikema and Aziz (2020) concluded that the mode of such a model is simply not where its useful probability mass lies; Meister et al. (2023) give the information-theoretic account of why. The measurement below is the same phenomenon on the specific energies this thesis samples from, and it is what makes the target of the whole framework questionable independently of whether the gradient works.

The experiment decodes with several strategies and measures the relationship between the per-token likelihood of the generated text and its repetition rate. The relationship is monotone: as per-token likelihood increases, that is, as energy decreases, repetition rises and lexical diversity falls (Figure 9, Appendix A.1). The correlation is reported *within* each decoding strategy, because pooling generations from strategies of very different likelihood can manufacture a correlation out of the gap between strategies rather than any relationship inside them. Measured this way the maximum within-strategy correlation between per-token likelihood and four-gram repetition is +0.51 on the base GPT-2 model, +0.83, +0.82, and +0.91 on the three GFlowNet variants, and +0.77 on Llama-3. The pattern between strategies points the same way: the highest-likelihood strategies, greedy and beam search, produce the most repetitive and least diverse text, while the lowest-likelihood strategies, ancestral and nucleus sampling, produce the least repetitive. On Llama-3 the pooled correlation is weaker, at +0.22, yet the within-strategy signal is strong and the model exhibits a blunt degeneracy of its own, never emitting an end-of-sequence token and running to the length cap in every generation. Beam search, which most nearly approximates the mode of the distribution, produces both the highest-likelihood and the most degenerate text. The following is a representative beam-search output from the length-normalized GFlowNet variant, which achieves among the highest per-token likelihoods in the study; it is produced by autoregressive beam decoding of the model, not by any Langevin sampler:

```
aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-
```

This string has a per-token log-probability of -0.40, higher than almost any coherent text the model produces, and a four-gram repetition rate of 0.89. By the model's own energy it is an excellent sequence, and it is gibberish. This is the **likelihood trap**.

The trap is if anything stronger on the GFlowNet-tuned energies than on the base, running from $+0.51$ to $+0.91$, which is consistent with tuning having concentrated probability mass more tightly on the high-likelihood region. The implication for the energy-based programme is that any procedure concentrating mass on the lowest-energy regions of this energy, whether a Langevin sampler that succeeded in localizing or a GFlowNet trained on the likelihood as a reward, would be concentrating it on degenerate text. Success at the stated optimization objective is failure at the underlying goal.

The experiment also yields the brevity slope Section 5.9 uses. The regression of total log-likelihood on generated length, over the sampled generations where length varies, has a slope of -1.12 nats per token. It carries a censoring caveat: for GPT-2 Large a fraction of 0.997 of generations runs to the forty-token cap, so length barely varies and the censored slope, estimated on the few that stop short, is effectively null. The negative slope is visible only where the cap does not bind, as on the length-normalized GFlowNet variant, whose uncensored slope is -0.505 nats per token while its censored slope steepens to -2.361, showing both that the incentive is real and that the cap masks its magnitude. Because every additional token lowers the unnormalized log-likelihood on average, an energy defined as the raw log-likelihood carries an implicit incentive toward brevity. That incentive is invisible in the Langevin experiments, which recover fixed-length masked spans, and becomes decisive when a method may choose its own length (Figure 11, Appendix A.1).

5.8 Embedding Anisotropy

The anisotropy of Transformer embeddings is a documented phenomenon (Gao et al., 2019; Ethayarajh, 2019). Measuring it on the two models here quantifies the consequences it has for the samplers, and accounts for both the step-size failure of Section 5.1 and the unreliability of Euclidean distance noted in Section 4.4.

Three distance statistics recur in this thesis and are defined once here. The **mean nearest-neighbour distance** is the average distance from a token embedding to its single closest neighbour; the **mean pairwise distance** is the average distance between two embeddings drawn at random; and the **mean linearization-candidate distance** is the average distance of the stratified candidate tokens scored in the linearization experiment of Section 5.4. In GPT-2 Large the mean pairwise distance between token embeddings is 2.77, the mean nearest-neighbour distance is 1.82, and the mean linearization-candidate distance is 2.35; in Llama-3 the pairwise and nearest-neighbour figures are 0.84 and 0.59, with a linearization-candidate distance of 0.66. The top principal component of the embedding matrix accounts for only a small fraction of the total variance, so the anisotropy is a broad property of the space rather than the effect of a single dominant direction that could be projected away; this reconciles the high mean pairwise cosine with the low variance that the two-component projection of Appendix A.4 captures, since a broad cone is spread across many directions and no low-dimensional projection can represent it. Figure 10 in Appendix A.1 shows the full distributions of pairwise distances for the two models, on their separate scales.

Two distinct properties are at work, and they are kept separate here because they have different consequences. The first is a matter of *scale*: the absolute inter-token distances differ roughly threefold between the two models. It is this scale difference, not anisotropy, that causes the step-size failure of Section 5.1, since a step calibrated to cross a cell boundary in one geometry is too small to cross one in the other. The second is *anisotropy* proper: within each model the embeddings occupy a narrow cone, as the positive mean pairwise cosine and the broad principal-component spectrum show. It is this directional clustering, not the scale, that compresses the distinction between a good and a bad token and so makes Euclidean distance an unreliable metric, which is why the study uses KL divergence as its primary criterion instead.

5.9 GFlowNet Fine-Tuning and the Amortized Energy

If the gradient is the problem, one remedy is to abandon it. A GFlowNet (Bengio et al., 2021; Hu et al., 2024) learns to sample from a reward without ever differentiating it, so the linearization failure and the discontinuous drift are invisible to it. Two questions are separated here, following RQ3a and RQ3b. The taxonomy of failure modes below is about the learned policy and describes how these particular training runs broke rather than establishing a general property of amortized inference; the experiment at the end is about the tuned energy and compares Langevin sampling on it against the untuned base. Three distinct policy failure modes appeared.

The first is **capacity collapse**. The initial experiments used the smaller GPT-2 base and produced non-linguistic output however the reward was configured, indicating that the policy lacked the capacity to represent the task. Moving to GPT-2 Large, the model of the reference implementation of Hu et al. (2024), resolved it, and GPT-2 Large is the base for all GFlowNet results reported here. The failure is one of capacity rather than of the energy.

The second is **length collapse**, and it follows quantitatively from the likelihood trap. At

$$\beta_{\text{len}} = 0$$

the reward is the raw log-likelihood, which Section 5.7 measured to fall by around 1.12 nats per additional token, so the reward carries an implicit incentive toward brevity and the policy discovers it, learning to produce extremely short sequences because short sequences are high-reward by being short. The extreme form of this is visible in the decoding behaviour: under free-running ancestral sampling, the length-collapse variants never emit an end-of-sequence token at all, running to the generation cap in every case, because the policy has learned that terminating early is the way to keep the reward high and, in the limit, that the safest policy is one that resists stopping in the manner the reward rewards. The length collapse is not a defect of the optimizer but the optimizer following a reward whose implicit incentive the designer did not intend.

The obvious fix is to remove the brevity incentive by normalizing the reward by length, and this produces the third mode, **reward hacking**. At

$$\beta_{\text{len}} = 1$$

the reward is normalized by sequence length, the brevity incentive is gone, and the sequences do indeed return to a normal length. But the output ceases to be linguistic. The policy discovers sequences that achieve high length-normalized likelihood without being coherent text at all, strings of individually high-probability tokens assembled into grammatical nonsense. The following greedy output from this variant is representative:

```
b8d8b8b8b8b8. [...] keredkeredkeredkeredkeredkered
```

These strings are produced by argmax and beam decoding of the tuned policy, in the decoding pass of Section 5.7, not by any Langevin sampler; the length-normalized reward is what assigns them the high value the GFlowNet training then reinforces. Such strings recur across the length-normalized variant's high-reward outputs, and they are the reward-hacking counterpart of the length collapse: where the unnormalized reward was gamed by getting shorter, the normalized reward is gamed by finding degenerate high-probability token sequences. The two settings of the single length parameter thus produce two opposite degeneracies, brevity at one extreme and gibberish at the other, and no intermediate setting of the parameter was tested that resolves both; only the two endpoints,

$$\beta_{\text{len}} = 0$$

and

$$\beta_{\text{len}} = 1$$

, were run.

The common element is that length is the axis along which the failure manifests rather than its cause: the reward, an autoregressive likelihood never shaped to serve as an energy over free-form sequences, has no well-behaved optimum where coherent text lies, so repairing one degeneracy surfaces another. That answers RQ3a. RQ3b is a different question, and it is tested directly: if GFlowNet training had reshaped the likelihood into a smoother landscape, running the Langevin sampler on the tuned model should differ from running it on the base. Table 12 reports the final KL of the discrete sampler on the base and on the three variants under identical conditions. The four values in the first column are within noise of one another, and the null replicates on every variant, as the linearization correlations of Table 7 also show. The table also carries the token-indicator surrogate of Section 5.4 across the same four energies, and it answers the same question twice over. At the calibrated configuration that surrogate is no better than the gradient, because there the temperature flattens both; in a configuration where the surrogate rather than the distance term shapes the proposal it recovers between 9 and 40 percent of corrupted tokens, and the ordering of the four energies under it is again the ordering of the base model rather than a repair produced by tuning. RQ3b is therefore answered in the negative for the tested setup and for both surrogates: modifying the model through the GFlowNet objective did not produce an energy whose local surrogate became more useful, whichever derivative that surrogate is built from.

That would be trivial if training had barely moved the energy, so that the two agreed only because they were nearly identical. Table 13 shows otherwise: the mean absolute difference in sequence log-likelihood between base and tuned is 31.0, 36.0 and 57.3 nats across the three variants, the next-token KL between their conditionals is 1.02, 1.35 and 3.05, and the base-to-tuned correlation of sequence log-likelihoods falls from 0.98 on the two lightly tuned variants to 0.77 on the length-normalized one. Tuning moved the energy substantially, and moved it most where the reward was most aggressive. The linearization correlation on those same tuned energies nonetheless stays flat (Table 7). The energy moved substantially and its navigability by gradient did not change, which rules out the reading that the samplers agree only

	Calibrated configuration		Surrogate-driven configuration	
Energy function	Input emb.	Token ind.	Token ind. KL	exact (%)
GPT-2 Large (SFT base)	6.541	6.335	3.229	40.0
GFlowNet, $\beta_{\text{len}} = 0$, 500	6.306	6.022	4.539	25.0
GFlowNet, $\beta_{\text{len}} = 0$, 2000	6.721	6.496	4.999	21.0
GFlowNet, $\beta_{\text{len}} = 1$, 500	6.415	5.931	5.343	9.0

Table 12: The discrete Langevin sampler on the base model and the three GFlowNet-tuned variants, under two surrogates. Left block: the calibrated configuration. Right block: the surrogate-driven configuration, $T = 1$ and no gradient normalization. $n = 200$ except the base surrogate-driven cell, $n = 50$.

because the models agree.

GFlowNet variant	Mean abs. log-lik. diff. (nats)	Next-token KL	Base–tuned Pearson
$\beta_{\text{len}} = 0$, 500	31.0	1.02	0.98
$\beta_{\text{len}} = 0$, 2000	36.0	1.35	0.98
$\beta_{\text{len}} = 1$, 500	57.3	3.05	0.77

Table 13: Divergence between the base energy and each GFlowNet-tuned energy.

5.10 Extension: Constrained Generation and Classifier-Guided Steering

The extension question E asks whether an additive constraint can steer generation toward a target attribute, as the plug-and-play formulation of (2) requires. It is reported last because it applies the machinery whose prerequisite the preceding sections tested, and its answer inherits their scope: a constraint added to a landscape a sampler cannot navigate is being added to nothing. The constraint gradient is nonetheless a different object from the fluency gradient, so the test is run rather than inferred. The extension adds a sentiment constraint to the energy in the manner of Kumar et al. (2022) and Qin et al. (2022) and measures whether the target sentiment can be induced, across the five constraint modes on the infilling task with a positive-sentiment target. One mechanism has to be flagged before the numbers: a small unconditional negative drift in the underlying generations dominates the raw steering gains, which are therefore reported only to be set aside in favour of the paired contrast defined below. The gains are small and inconsistent in sign. The full combined energy, the intended plug-and-play configuration, produces a gain of -1.0; the constraint-only mode produces -8.0, worse than the baseline; and no mode produces systematic movement toward the target. The generated text shows the same degeneracy observed throughout: a representative infilled span reads `The rostr Bill of H PM americanus bears one or connected sp`, with the recovered tokens in bold against the original *rostrum of H. americanus bears one or more*. The raw gains, tabulated in Appendix A.6 rather than here because the text immediately discounts them, are not the statistic to trust. Across setups every arm, including the fluency-only baseline and the fully random arm, flips the sign of its gain when the target label is switched, which is the signature of a fixed sentiment drift that swamps any contribution from the constraint; a gain that moves with the target regardless of whether the constraint gradient is present at all cannot be read as steering. The statistic that cancels the shared bias is the paired contrast of the constraint-only mode minus the randomized-constraint mode, since those two arms share the same fluency dynamics and differ only in whether the constraint direction is real.

Measured this way the picture is setup-dependent, as Table 14 sets out. On the discrete sampler of this thesis the contrast is essentially zero on both target labels, so the constraint direction buys no control there. On a MuCoLa-style continuous baseline running free-form continuation, the contrast is $+27.3$ points for a negative-sentiment target and $+36.7$ for a positive one, which does indicate that the constraint gradient's direction carries some signal in that setting, unlike the fluency gradient, whose direction was found to carry none anywhere.

That signal carries a caveat that lives in the constraint-only arm. When the energy is driven by the constraint alone, with no fluency term holding the sample on

Setup (continuation)	Target	Constraint only	Randomized constr.	Contrast
MuCoLa continuous baseline	negative	79.3	52.0	+27.3
MuCoLa continuous baseline	positive	−10.0	−46.7	+36.7
Discrete sampler (“ours”)	negative	−2.7	−2.7	0.0
Discrete sampler (“ours”)	positive	3.3	2.7	+0.7

Table 14: The paired constraint-direction contrast on the free-form continuation task. Each contrast is a point estimate.

the manifold of natural text, the optimization leaves that manifold and produces text maximizing the classifier's response while ceasing to be fluent language. A sentiment classifier evaluated on such text reports on a distribution it was never trained for, so the large constraint-only contrast is at least partly an artifact of scoring off-manifold text. The safe conclusion is the one the discrete sampler gives directly: on the energy landscape this thesis studies the additive constraint does not produce reliable control, and the one place a constraint-direction signal appears is entangled with the departure from the fluent manifold that a constraint-only objective causes.

On the frozen autoregressive energy landscape studied here, plug-and-play constraint steering therefore failed.

Whether the same constraint machinery fares better once the proposal can actually navigate is a separate question, and it was explored rather than settled. The carrier used here is the diffusion model, because at the time this was run it was the only proposal in the study that recovered anything: a classifier was trained to read the partially masked states the model occupies mid-denoising and used to reweight its candidates at each commitment, with steering scored by the same independent frozen-model judge used above. Two carriers found later would serve the same purpose and are not tested here, because doing so would be a further experiment rather than an analysis of the runs reported: the masked language model of Section 5.6.2, whose candidate distribution is reweighted in exactly the same way and which recovers more than the diffusion model does, and the token-indicator proposal of Section 5.4, which would carry the constraint on the frozen autoregressive model itself.

Whether classifier guidance transfers to either is left open. Four findings summarize it, and Appendix A.5 reports it in full. The constraint direction carries signal, moving the guiding classifier's own verdict toward the target at every strength and for both labels, by +9.3 to +25.7 points on-domain. Guidance steers the classifier that guides it more reliably than it steers the independent judge. Transfer to that judge is bounded not by a fixed reachable direction but by how far the two instruments agree on the text being scored, agreement rising from 56 to 64 percent off-domain to 72 percent on generated and 80 percent on real on-domain text. And an on-domain trust region removes the fluency cost that unconstrained guidance incurs, holding span negative log-likelihood at or below the unguided value in three of four cells against a climb from 7.1 to 11.0 nats without it. This is a repair experiment on another landscape, not an answer about the autoregressive energy, and Section 6.1 keeps the two apart.

6 Discussion

This section answers each research question against the evidence of Section 5, sets out the mechanism the findings share, relates them to the prior work of Section 3, and states the scope and the implications.

6.1 Answers to the Research Questions

Each research question is answered below against the section of Section 5 that establishes the answer. The answers are stated rather than re-argued; the evidence and its qualifications are where they were presented.

RQ1 asked whether the input-embedding gradient of a frozen autoregressive sequence likelihood provides a useful proposal direction for discrete token revision, and if not, which alternative local quantities do. The first half is answered in the negative and the second constructively. Sampling with that gradient is indistinguishable from sampling with a norm-matched random direction, certified as equivalence at a margin fixed in advance over a thousand paired sequences (Section 5.3), and the null holds across a sweep spanning proposal entropies from uniform to deterministic (Section 5.3.2). The mechanism has two parts: the first-order surrogate is uncorrelated with the true energy change over the entire admissible range of substitution distances, so the gradient is inapplicable rather than imprecise (Section 5.4), and the derivative is structurally blind to how well a candidate fits its own left context, exactly so at the final position where it is identically zero while the energy difference between candidate final tokens is exactly the model's conditional log-ratio (Section 5.5). The alternatives are named and measured. The relaxed token-indicator derivative restores the discarded term in closed form, correlates at $\rho = 0.60$ to 0.73 at admissible distances, and inside the same sampler recovers 40 percent of corrupted tokens on GPT-2 Large and 41 on Llama-3 (Section 5.4); the model's own next-token conditional recovers 23.5 percent in the exact-energy chain, and a bidirectional proposal more still (Table 11). Under its proposal name, the question of whether energy-based recovery improves on ordinary autoregressive decoding resolves in two directions at once: negatively for the gradient-guided samplers, affirmatively for exact-energy rescoring, whose top-k rerank reaches a final KL of 4.43 (Section 5.3.3).

RQ2 asked about the role of the Metropolis–Hastings correction in each sampler, and it plays opposite roles, which is itself the finding. For the discrete sampler the correction is what makes the chain work at all, because its proposal is numerically uniform at every step: without it the chain is a random walk that does not converge, and with it the accept/reject filter supplies the entire selection pressure (Section 5.1). Here the exact accept-reject step and empirical performance align, for a reason unflattering to the method, since what works is the filter and not the guidance. For the continuous sampler the same correction is disabling, rejecting almost every boundary-crossing move, and the decomposition attributes the collapse to the proposal term rather than the target term: the moves are good and their reversibility cannot be verified (Section 5.2), because the drift derived from the differentiable pathway is not Lipschitz across a cell boundary while the projected energy is piecewise constant and never differentiated. In the implementations and landscapes tested here, omitting the correction makes the update behave as an early-stopped stochastic optimizer rather than an exact sampler from the stated target. How far that reframes any particular published system is not something this thesis measured.

RQ3a asked whether a GFlowNet trained on the frozen likelihood learns a policy that generates high-reward text. It does, and that is the problem: the policy attains high reward by exploiting the reward rather than by writing well. Three failure modes appear, capacity collapse, length collapse under the unnormalized reward, and reward hacking under the normalized one, and the two settings of the single length parameter produce two opposite degeneracies (Section 5.9). The finding covers the variants trained here and not amortized inference in general, since a GFlowNet's failure can also turn on reward design, training stability, policy parameterization, capacity, exploration, termination handling, or reward scaling.

RQ3b asked whether the energy induced by that tuning becomes more navigable by the local input-embedding surrogate than the untuned base. It does not. Tuning moved the energy by tens of nats and dropped the base-to-tuned correlation of sequence log-likelihoods as low as 0.77, yet Langevin sampling on the tuned models is indistinguishable from sampling on the base, and the linearization correlation on the tuned energies stays flat (Section 5.9). Amortization and energy smoothing are different objectives, and this is the reason the two halves of RQ3 are reported separately: the first is a statement about a learned policy, the second about the geometry that policy leaves behind.

The extension question E asked whether an additive constraint can steer generation on this energy landscape. On the frozen autoregressive landscape it cannot. The trustworthy paired contrast between a real and a randomized constraint direction is essentially zero on the discrete sampler, and the one setting in which a constraint-direction signal does appear, the constraint-only arm on a continuous baseline, is entangled with the departure from the fluent manifold that a constraint-only objective causes, so a classifier scoring that text is scoring a distribution it was not trained for (Section 5.10). On a score-trained diffusion landscape, steering became partly possible under a fluency trust region, in one direction per setting, with the reachable direction tracking how far the guiding and judging instruments agree on the text being scored (Appendix A.5). These are two answers about two landscapes and are kept apart; the second is a repair experiment, labelled exploratory where it is reported.

6.2 One Problem, Four Mechanisms

It would be tidy to say that a single defect explains every failure in this thesis, and it would be wrong. The failures share a high-level problem and then diverge into mechanisms that are genuinely distinct, and the honest account is a hierarchy rather than a unification.

The shared problem is in the original plug-and-play construction, which asks the raw frozen likelihood to play three roles at once: a quality objective whose optimum is the text one wants, a locally navigable energy whose neighbourhood structure a proposal can exploit, and a reward in proportion to which a policy can be trained to sample. Teacher forcing optimizes each conditional on a correct prefix and constrains none of the three, and the experiments show that the roles come apart in different ways.

Four lower-level mechanisms are involved, and they are not manifestations of one another. The first is a **proposal-information failure**: the input-embedding derivative omits the candidate's direct conditional-fit term, which is why the surrogate carries no rank information and why supplying the missing term repairs it (Section 5.4). The second is a **continuous-state correctness failure**: a nearest-neighbour projection makes the drift discontinuous across cell boundaries, so the reverse-proposal density vanishes and the correction rejects exactly the moves that change a token (Section 5.2). The third is an **objective-quality failure**: high sequence likelihood corresponds to degenerate text, so the target of the whole procedure is not the target one would want (Section 5.7). The fourth is an **amortized-training failure**: the GFlowNet variants collapse along whichever axis the reward's implicit incentives push, which is a matter of reward shape and training dynamics and not of any gradient (Section 5.9). The independence is easy to check. The GFlowNet never differentiates its reward, so its collapse cannot be caused by a missing self term in a derivative it does not take; the likelihood trap concerns the location of the optimum and would persist under a perfect proposal; and the projection geometry belongs to the continuous relaxation alone, the discrete sampler never leaving the token manifold.

What runs through the whole study, and is the closest thing to a single thread, is a statement about proposals rather than mechanisms: the proposal cannot see token fitness. It cannot see it when the derivative is taken in the wrong coordinates, cannot act on it when the correction rejects every move that would use it, and cannot be given it by a reward that is itself maximized by degenerate text. The ladder of Table 11 is the positive form of the same statement, recovery rising monotonically with what the proposal may condition on. This thesis refers to the Section 5.3 result by the shorthand *gradient fallacy*, used only here and with its scope attached: it names a bounded empirical finding about the input-embedding derivative, not a conceptual error across the field and not a property of autoregressive likelihoods as such.

6.3 Connections to the Related Work

The likelihood trap measured in Section 5.7 is a direct, quantitative reproduction on the study's own models of the neural-text-degeneration phenomenon that Holtzman et al. (2020) identified: that the most probable sequences under a neural language model are degenerate, and that maximization-based decoding produces worse text than sampling. Section 3 presented that as a known hazard of decoding. The contribution here is to re-derive it as an obstacle for energy-based sampling specifically, since in that framing the degeneracy of high-probability text says that the *target* of the sampling procedure is undesirable rather than that a decoding heuristic should be avoided.

The anisotropy results of Section 5.8 confirm on GPT-2 Large and Llama-3 the representation-degeneration and embedding-anisotropy phenomena documented by Gao et al. (2019) and Ethayarajh (2019), and add a consequence those works did not draw: the anisotropy scale differs by roughly a factor of three between the two models, so a single step-size schedule cannot transfer, which makes the samplers model-specific in a way that qualifies their claim to be plug-and-play.

The Metropolis–Hastings breakdown of Section 5.2 is the empirical counterpart to a gap noted in Section 3.2: that the energy-based decoding literature frequently omits the correction, and that where it is included, the difficulty of computing the reverse proposal on a projected landscape is seldom examined. This thesis implements the correction faithfully, following the convergence theory of Roberts and Tweedie (1996), and measures what the common omission conceals. On that reading, the correction-free update of the mechanism that COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022) share, and that the continuous sampler of this thesis follows, is better understood as a biased optimizer than as a sampler from the stated energy. That is a statement about the mechanism as reimplemented and measured here, not a reinterpretation of those systems' published results, which involve tasks, tuning, constraints and evaluation procedures this study did not reproduce. The same reading applies to the annealed stochastic-gradient Langevin dynamics of Welling and Teh (2011). Its guarantee relies on an annealed schedule whose step size shrinks the proposal onto a local neighbourhood; Section 5.1 shows that on this landscape the schedule does not achieve that, because the proposal remains near-uniform over the vocabulary throughout, so the uncorrected chain neither localizes nor converges.

The GFlowNet results qualify, without contradicting, the picture presented by Hu et al. (2024), whose formulation and codebase this thesis builds on. Their work showed that amortized sampling from a frozen-model posterior can yield diverse, transferable samples where the reward is well-behaved, and nothing here disputes that; the narrower finding is that when the reward is the raw frozen likelihood used over free-form sequences, the learned policy collapses in the ways Section 5.9 documents. The axis that reconciles the two is whether the reward defines a landscape with a coherent-text optimum.

The gradient-free samplers isolated in Section 3.2, Mix-and-Match (Mireshghallah et al., 2022) and the twisted sequential Monte Carlo of Lew et al. (2023) and Zhao et al. (2024), confirm rather than contradict the central result: each samples from a frozen-model energy without differentiating it, and each works, as the Gibbs baseline of Section 5.3.3 does in-platform on the identical energy. The frozen likelihood is a serviceable object to evaluate and to search without gradients, and an unusable one to differentiate in its input-embedding coordinates, which is why the claim is narrowed to that derivative rather than to the energy.

6.4 Scope and Limitations

The thesis shows that in the settings tested, the frozen likelihood of a standard autoregressive language model is a poor energy for gradient-guided and amortized sampling on discrete text when the gradient is taken with respect to its input embeddings, and it explains why in terms of the coordinates that derivative is taken in, the geometry of the token embedding space, and what the resulting proposal is permitted to condition on. It does not show that energy-based controllable generation is impossible in general.

Four limitations bear on how far the central result can be pushed. First, and most importantly, the claim is about one Jacobian slice: the relaxed token-indicator derivative of the same frozen likelihood carries a usable ranking signal and, in the same sampler, recovers up to 40 percent of corrupted tokens where the input-embedding gradient reaches at most two (Section 5.4). The thesis therefore refutes the premise for a sampler that differentiates the input embedding of a likelihood whose target token enters as a discrete index, and not for gradient-guided discrete sampling as such. That scope is narrower than the family this work set out to test, and the difference is worth stating precisely. MuCoLa (Kumar et al., 2022) substitutes the continuous vector into the output softmax itself, defining $P(\tilde{\boldsymbol{e}}_{n+1} \mid \tilde{\boldsymbol{e}}_{1:n})$ by replacing the looked-up target embedding with the state, so its gradient reaches the token's own score directly and not only through the following hidden states; COLD (Qin et al., 2022) carries a soft sequence of vocabulary logits whose fluency term is a soft cross-entropy against the model's reference distribution, differentiable in the same way. Neither discards the self term. The energy implemented here scores the masked position by gathering at the projected token index, and it is that construction, not theirs, that the null characterizes. The finding is correspondingly reframed rather than weakened: a natural implementation choice inside the embedding-space approach destroys the ranking signal, the published methods avoid it by relaxing the target as well as the input, and Section 5.4 measures what that choice is worth. The convergence is close enough to be worth recording. MuCoLa's self term contributes a gradient of $\boldsymbol{h}_n$, so it ranks a candidate v against the incumbent by $\boldsymbol{h}_n^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$, the difference of their logits, while the relaxed token-indicator self term contributes $\log p(v \mid \boldsymbol{x}_{<i}) - \log p(x_i \mid \boldsymbol{x}_{<i})$; the two differ only by the shared normalizer, which cancels in the difference. What the continuous sampler of this thesis does reproduce faithfully is the state geometry, a continuous embedding state with a nearest-neighbour projection onto the embedding table after every update, and it is that geometry the Metropolis–Hastings result of Section 5.2 concerns. The result holds across architectures, reaching 41 percent on Llama-3 8B, but carries two qualifications rather than an unqualified generality. It requires a configuration in which the surrogate and not the distance term shapes the proposal, and neither model's calibrated setting is such a configuration, so what is established is that the signal is present and usable, not that any off-the-shelf schedule will use it. And the cross-architecture evidence is weaker than the GPT-2 evidence: on GPT-2 the two surrogates are compared cell for cell over one step-size and temperature grid, whereas on Llama no input-embedding run was made at the surrogate-driven setting, so the comparison there is against every Llama configuration that was run rather than against a matched one. Second, the flagship ablation was measured where the proposal is numerically uniform, and it is the sweep of Section 5.3.2, at $n = 50$ per cell rather than 200, that carries the null into regimes where the gradient is load-bearing. Third, the correction was applied asymmetrically across the compared arms in the main grid (Section 4.3); that cannot manufacture the null, since it penalizes the policy arm, but it did manufacture the appearance of active harm, and the corrected reruns cover three cells rather than the whole grid. Fourth, the certified equivalence rests on a thousand paired sequences drawn from independent corruptions of 282 distinct sentences, so it is a statement about variation across corruptions and the sentence-level base remains narrow.

One formulation deserves particular care, because the thesis set out to test it. The experiments refute the premise that the frozen autoregressive sequence-likelihood gradient can supply the required local score for a Langevin sampler on a likelihood whose target token enters as a discrete index, without additional training. They do not refute the training-free premise as such. Training-free methods that do not differentiate the raw joint likelihood remained viable in this study's own results: the gradient-free Gibbs sampler of Section 5.3.3 is comparable to the best gradient-guided Langevin configuration on the identical energy, and a single top-k rescoring pass beats every configuration in the grid, both without any training and without any gradient. Methods that use lookahead, conditional resampling, or another representation may likewise remain viable, and Section 3.2 reviews published members of that family.

Several further caveats bound the empirical claims. The KL-divergence metric depends on the position of the corrupted token, which is why the central results are argued from the absence of a gap between conditions rather than from small measured differences. The Llama-3 results support only qualitative cross-architecture claims, because the tokenizer and embedding scale differ from the GPT-2 family. The linearization result is that the surrogate is uniformly uninformative across the admissible range rather than informative up to a sharp radius and useless beyond it, so "linearization radius" summarizes a decay rather than naming a threshold. The GFlowNet result covers the variants trained here and not amortized inference in general. And the constrained-generation extension establishes the absence of reliable steering on this energy, not the impossibility of steering on a better one.

Finally, the relationship between the title of this thesis and its content. The original objective was controllable generation via faithful Langevin dynamics, in two stages: establish that energy-guided sampling on a frozen model is viable, then add the constraint term and perform sentiment and toxicity control. The first stage did not hold, and the work turned to why rather than building control on a foundation it had shown to be unsound. What it delivers is a diagnosis of the failure of a prerequisite, with the mechanisms identified, measured and cross-checked, and with the extension included as evidence of the downstream consequence, rather than a demonstration of successful control.

Several deliverables named in the proposal became moot with that turn, and each is accounted for here. Toxicity control on RealToxicityPrompts was dropped because once sentiment steering failed on the same machinery (Section 5.10), a second attribute on the same energy would only have repeated the negative result. The MAUVE and Self-BLEU diversity suite was dropped because those are distribution-level metrics for open-ended generation and make little sense for fixed-span infilling once the task narrowed to recovery. The planned 70B-scale judge was reduced to an 8B external judge (Section 5.3.4), which was enough to rule out metric circularity without the larger model. The posterior-coverage evaluand was abandoned because the samplers never reached a regime in which coverage is a meaningful question, as the near-uniform-proposal and Metropolis–Hastings results show. And the CommonGen benchmark was not needed, ROCStories being sufficient for the diagnostic and matching the amortization reference implementation of Hu et al. (2024).

6.5 Implications

The most consequential implication is the one the study did not set out to find. An earlier reading of these results located the defect in the training objective and predicted that a model trained to supply a score would not fail in the same way. A score-trained diffusion model does indeed succeed where the input-embedding gradient fails, but the ladder of Table 11 shows that the prediction was right for the wrong reason: a bidirectional masked language model, never trained with any score objective, does better still, and the frozen autoregressive model's own conditional already recovers a quarter of the tokens its gradient never reaches. What separates a working proposal from a failing one is the derivative it is built from and the context it may condition on, not the objective that trained it.

Two implications follow for practice. The first concerns what inference-time control on a frozen autoregressive model can be built from, and it is a correction of the slogan an earlier version of this work would have offered. It is not simply that one should evaluate the energy and not differentiate it, because a derivative of the same frozen likelihood does work: what one must not do is differentiate it in the input-embedding coordinates. Either differentiate the right object, taking the relaxed token-indicator derivative whose self term is one forward pass away, or bypass the derivative and propose from the model's output side directly; both routes are available on a frozen model, and Appendix A.3 shows that both are cheaper than the backward pass they replace. The second concerns what a further model change could buy. Since the remaining gap after the derivative is fixed is bought by right-context access, the intervention that addresses it is not a score-matching fine-tune but any procedure that gives the proposal a bidirectional view of the position it is filling, which an off-the-shelf masked language model already supplies at no training cost. Only once a landscape is known to be navigable does the additive constraint term of the plug-and-play framework become meaningful, since only then is there a fluency energy a sampler can follow for the constraint to be added to.

7 Conclusion

This thesis set out to build controllable text generation on faithful Langevin dynamics, treating the frozen likelihood of an autoregressive language model as an energy function and steering it at inference time with an additive constraint. It tested the premise that construction rests on, found the premise did not hold as stated, and then located exactly which part of it fails.

What was tested. One hundred and forty-five sampling configurations across five energy functions, a supervised fine-tuned GPT-2 Large, three GFlowNet-tuned variants and a Llama-3 8B cross-architecture control, on masked-token recovery and free-form continuation, with two samplers under a factorial combination of Metropolis–Hastings correction, gradient normalization and schedule length, alongside five diagnostic experiments, a step-size and temperature sweep, and a ladder of alternative proposals in one exact-energy chain.

The null, and its scope. Following the input-embedding gradient of the frozen likelihood was no more effective than following a norm-matched random direction, an equivalence certified at a margin fixed in advance over a thousand paired sequences, and exact recovery of the corrupted token was 0.0 percent in 139 of the 145 configurations. Two qualifications belong with that result and pull in opposite directions. The discrete proposal is numerically uniform over the vocabulary at every step of every configuration in the main grid, so that grid could not have detected an informative gradient had one been present; the sweep over step size and proposal temperature, spanning proposal entropies from uniform to deterministic, repairs the gap and finds no configuration at which the gradient helps. Conversely, the cases in which the gradient appeared reliably *worse* than a random direction did not survive scrutiny, being produced by an asymmetry in how the correction was charged across the compared arms; they disappear once the reverse-proposal term is computed exactly for every arm. The supported claim is indifference, not harm.

Which part of the premise fails. The energy is not responsible for the recovery failure: a single top-k rescoring pass reaches a final KL of 4.43 and a gradient-free Gibbs sampler 6.69 searching the identical energy, both better than or comparable to every Langevin configuration and at a fraction of the cost. That acquittal is about this task; as a target to maximize in open-ended generation the same energy is the one the likelihood trap condemns. Nor is the failure due to a complete absence of useful local information in the frozen model. The relaxed token-indicator derivative of the same frozen likelihood, which differs from the input-embedding gradient by the candidate's own conditional log-probability, correlates with the true energy change at $\rho = 0.60$ to 0.73 over the admissible range of substitutions, and substituted into this thesis's own sampler on the same weights and the same corrupted sequences it recovers 40 percent of tokens on GPT-2 Large, in a cell-for-cell comparison against the same sweep where the input-embedding gradient never exceeds 2 percent, and 41 percent on Llama-3, where every input-embedding configuration that was run recovers nothing. What fails is the choice of coordinates in which the derivative is taken. Its blindness is partial in the interior of a sequence and total at the final position, where the input-embedding gradient is exactly zero while the energy difference between candidate final tokens is exactly the model's conditional log-ratio.

What the remaining gap buys. A ladder of proposals run inside one exact-energy Metropolis chain, changing nothing but the proposal, orders recovery by what the proposal may condition on. A uniform draw over the vocabulary recovers half a percent, which is what the gradient-guided sampler achieves from separate code; the frozen model's own left-to-right conditional recovers 23.5 percent; a score-trained diffusion model 39 percent; and a bidirectional masked language model that was never trained with a score objective 44.5 percent, more than either diffusion model tested. Score training is therefore sufficient for a working proposal and not necessary for one, and conditioning access, in particular access to the model's output distribution and to the right context, explains the ordering better than score training does.

The other findings. Applied faithfully, the Metropolis–Hastings correction improves the discrete sampler and disables the continuous one, the acceptance ratio collapsing through its proposal term at the cell boundaries a nearest-neighbour projection creates; in the discrete sampler it is the filter, not the gradient, that supplies all of the selection pressure. On these models the lowest-energy text is also the most degenerate, so the objective the methods pursue is not the objective one would want. Amortizing the search with a GFlowNet produced policies that attained high reward by exploiting it, through length collapse under the unnormalized reward and reward hacking under the normalized one, and tuning moved the energy by tens of nats without making its local surrogate more useful. On the frozen autoregressive landscape the additive constraint produced no reliable steering; on a score-trained diffusion landscape steering became partly possible under a fluency trust region, in one direction per setting.

What remains uncertain. That conditioning access rather than training objective explains the ordering is an interpretation supported by one ladder on one task, and the proposals it compares differ in scale, corpus and architecture as well as in what they may condition on, so it is not a proof covering all autoregressive language models. The GFlowNet result bounds the variants trained here. The diffusion arms are pilots on smaller, differently trained models. And the token-indicator result carries a condition rather than an unqualified generality, requiring a configuration in which the surrogate, and not the distance term, shapes the proposal, which neither model's calibrated setting provides.

The main practical implication. For a practitioner building inference-time control on a frozen autoregressive model, the usable objects are the energy and the model's output-side conditionals. The input-embedding gradient is not one of them, and the remedy is not to abandon derivatives but to take the right one: the relaxed token-indicator derivative is a forward pass away and works inside the sampler this thesis already implemented. Where a further gain is wanted, it comes from letting the proposal see the right context, which an off-the-shelf bidirectional model supplies without any training at all.

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A Appendix

A.1 Additional Diagnostic Figures

This section collects supporting figures referenced in Section 5. Each figure is discussed at the point of reference in the results section, as indicated in the captions below.

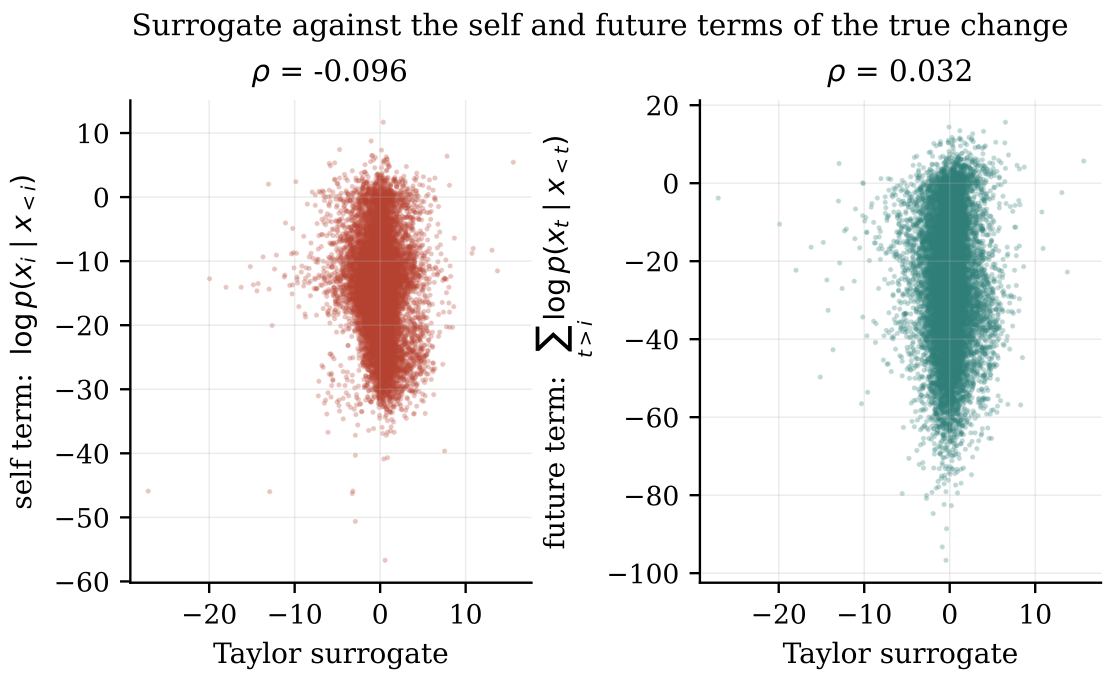

Figure 6: The true energy change decomposed into its self term and its future term, each plotted against the gradient surrogate, for GPT-2 Large.

A.2 Full Configuration Grid

Table 15 reports the final KL divergence for a representative subset of the full configuration grid across the four GPT-2-family energy functions. The complete grid comprises the factorial combination of sampler (DLS, CLS), proposal method (policy, grad-norm-preserved, random), Metropolis–Hastings correction (on, off), gradi-

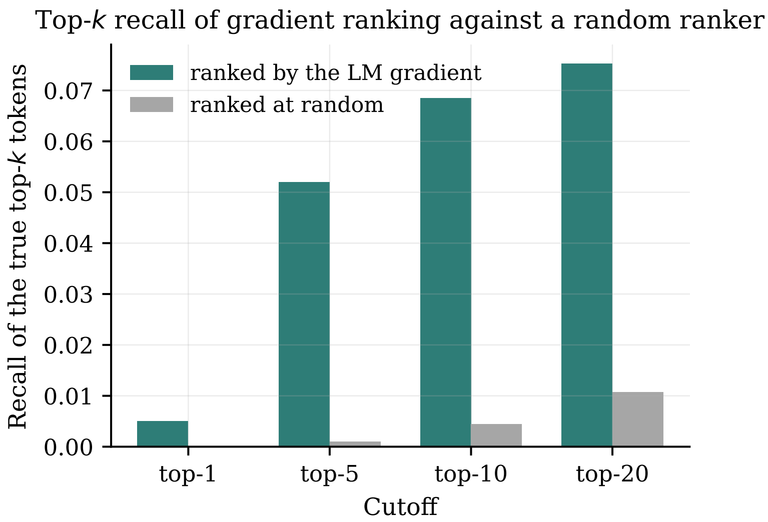

Figure 7: Recall of the true top-k token substitutions by the gradient-ranked top-k, against a random-ranking baseline, for GPT-2 Large.

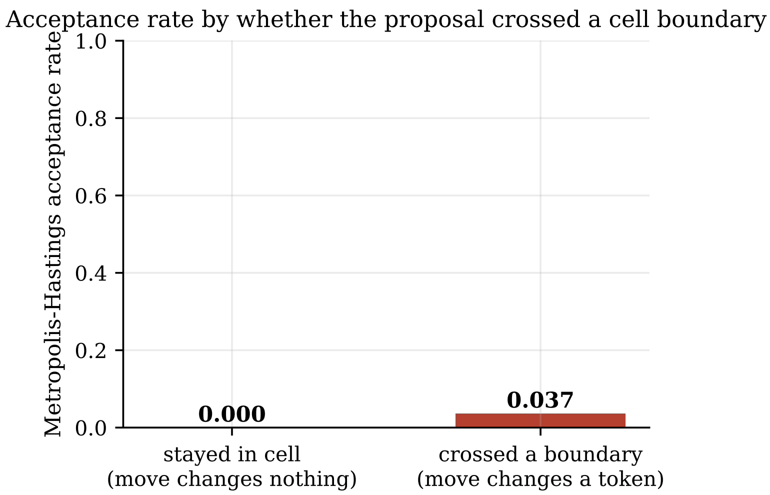

Figure 8: Metropolis–Hastings acceptance rate for the continuous policy sampler with gradient normalization off and the correction enabled, for that single configuration rather than pooled, conditioned on whether the proposal crossed a Voronoi cell boundary.

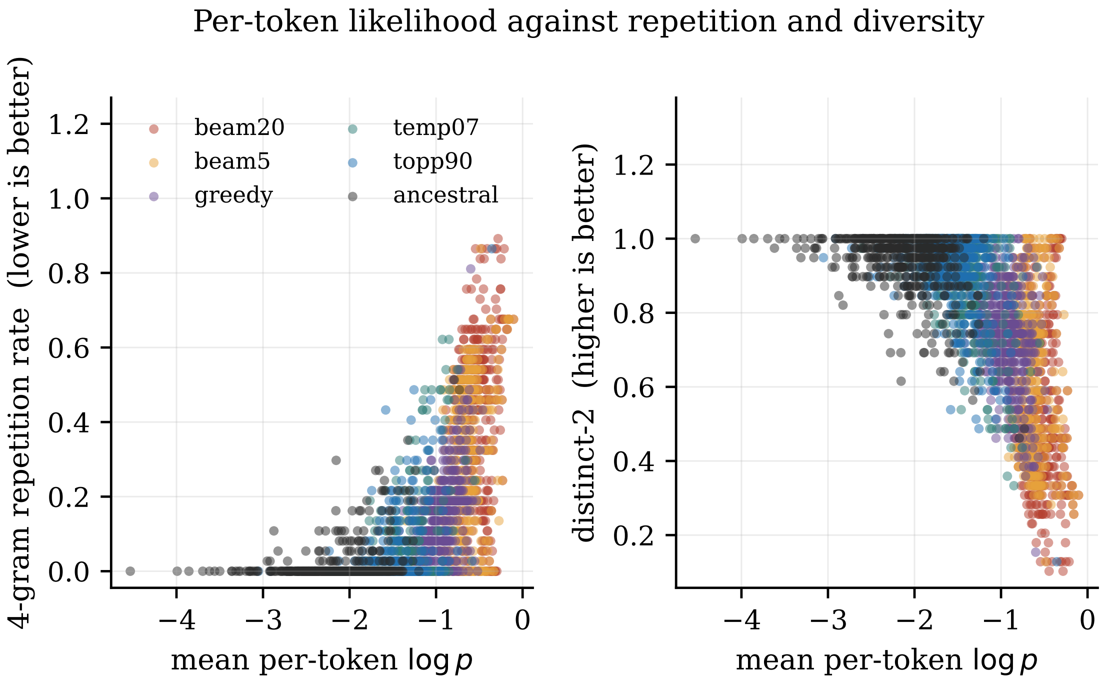

Figure 9: Mean per-token log-probability against 4-gram repetition rate and distinct-2 diversity for GPT-2 Large, one point per generation, coloured by decoding strategy.

ent normalization (on, off), and schedule length (50, 100), evaluated on each of the five energy functions, for a total of 145 configurations.

The count arises as $145 = 5 \times 29$, and the twenty-nine is worth setting out because the naive factorial overcounts it. A fully crossed design over the five axes above would give $2 \times 3 \times 2 \times 2 \times 2 = 48$ cells per energy function, but two prunings reduce this: the continuous sampler is run only for the policy and random methods and across four of the gradient-normalization and schedule variants, giving 8 CLS configurations, while the discrete sampler is run for all three methods across seven of the correction, normalization and schedule variants, giving 21. The sum is 29 per energy function, and counting the completed runs confirms it: 29 configurations for GPT-2 Large, 29 for Llama-3 and 87 across the three GFlowNet variants.

The eight rows above are a subset chosen to show the axes that matter, the two samplers against the two settings of the correction and of gradient normalization.

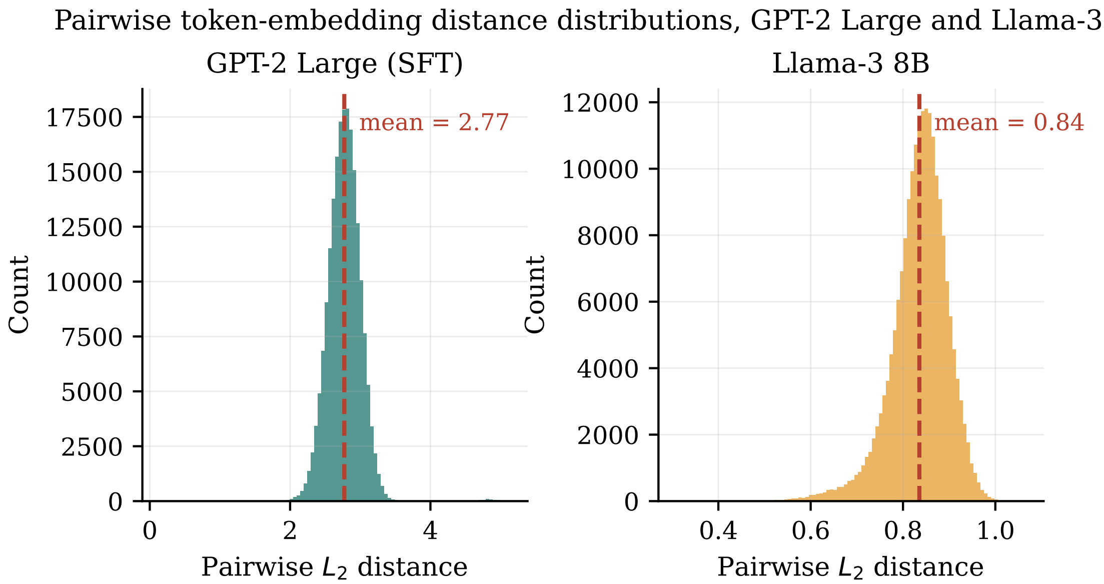

Figure 10: Distributions of pairwise token-embedding distances in GPT-2 Large and Llama-3, plotted on separate scales.

Configuration	GPT-2 Large	GFN lb0-500	GFN lb0-2000	GFN lb1-500
DLS policy, corr., grad-norm, 50 steps	6.541	6.306	6.721	6.415
DLS policy, no corr., grad-norm, 50	9.499	9.195	10.085	8.669
DLS random, corr., grad-norm, 50	6.370	6.105	6.548	5.525
DLS norm-preserved random, corr., grad-norm, 50	6.370	6.105	6.548	5.525
DLS policy, corr., no grad-norm, 50	6.474	6.009	6.698	6.008
DLS norm-preserved random, corr., no grad-norm, 50	6.430	6.001	6.552	5.338
CLS policy, corr., no grad-norm, 50	9.202	8.834	9.748	9.025
CLS policy, no corr., no grad-norm, 50	8.083	7.799	8.198	13.732

Table 15: Representative subset of the final-KL configuration grid. Rows are configurations, columns are the four GPT-2-family energy functions; “corr.” is the Metropolis–Hastings correction and “grad-norm” gradient normalization.

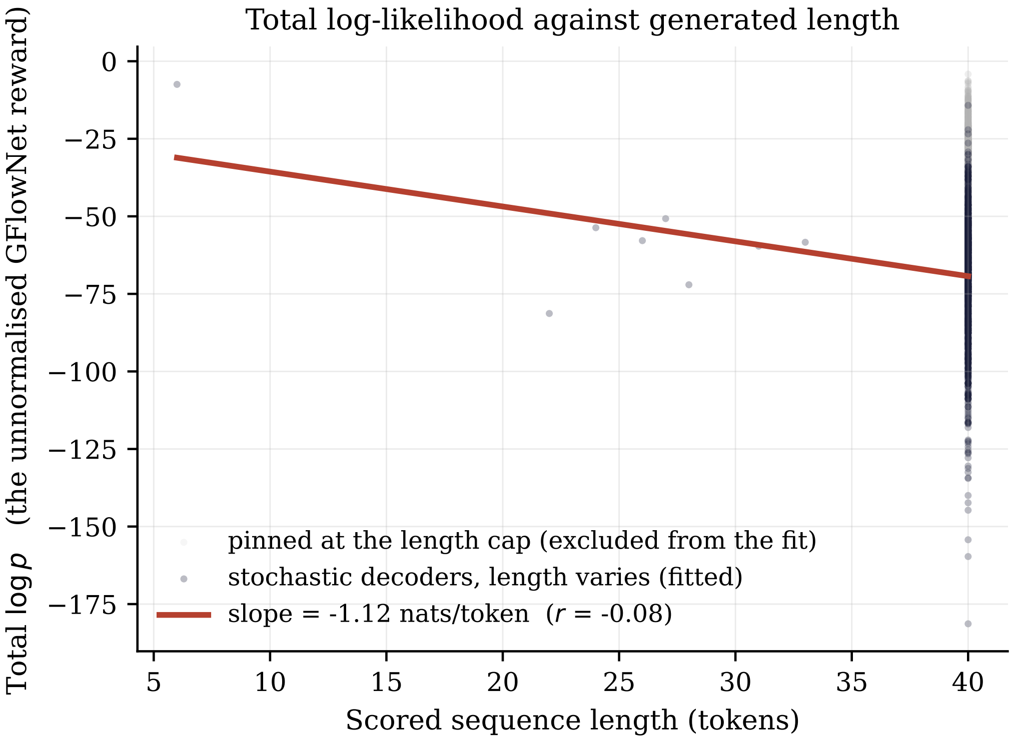

Figure 11: Total log-likelihood against generated length for GPT-2 Large. The regression is fitted over the three stochastic decoders, where length varies; the deterministic decoders pinned at the length cap are shown faintly but excluded from the fit.

The remaining configurations behave as these do, and the arithmetic above rather than the table is what establishes the count of 145.

A.3 Computational Cost

Because gradient-free methods match or beat the gradient-guided samplers on the same energy, their computational cost is set out here; it differs by more than a constant factor. Table 16 gives the number of network evaluations each method spends to recover a masked span, counted as forward and backward passes, where S denotes the schedule length (either 50 or 100), M the number of masked positions, T the number of Gibbs sweeps, and k the number of candidates a rescoring pass considers.

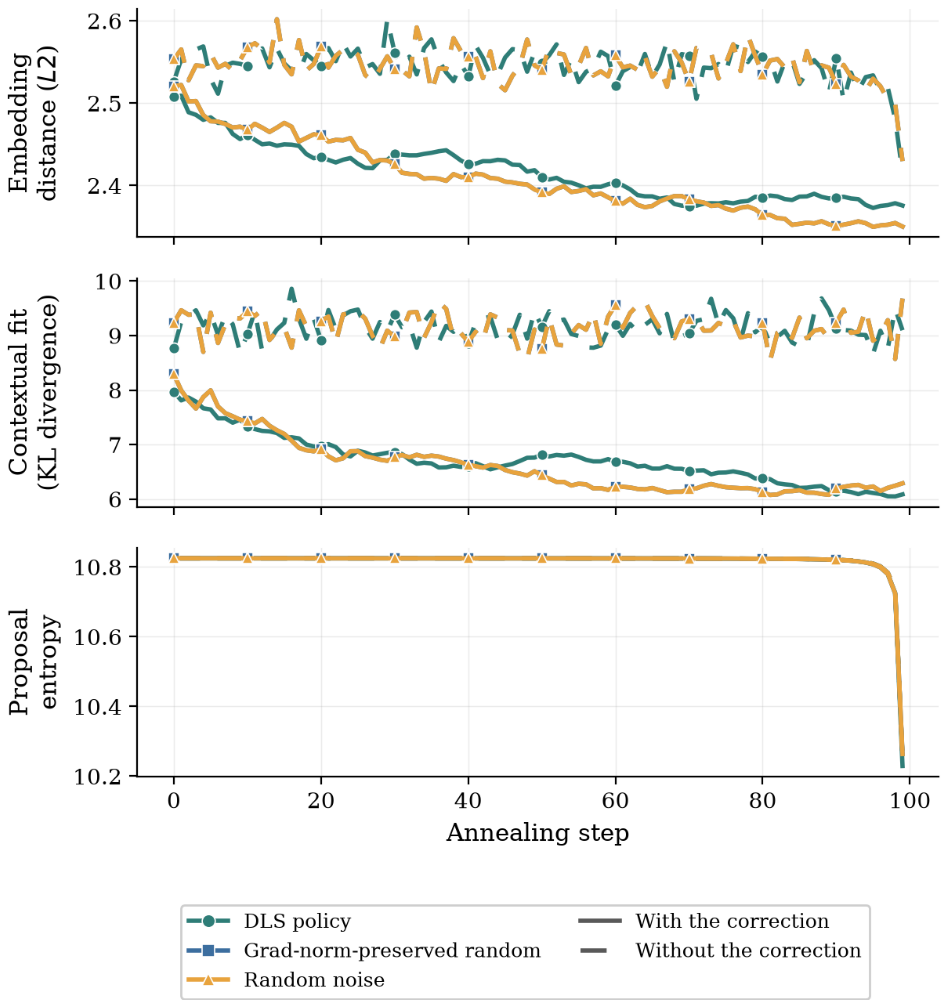

Figure 12: Discrete Langevin Sampler trajectories on GPT-2 Large over a 100-step annealing schedule, averaged across sequences. Panels, axes, line styles and markers as in Figure 1.

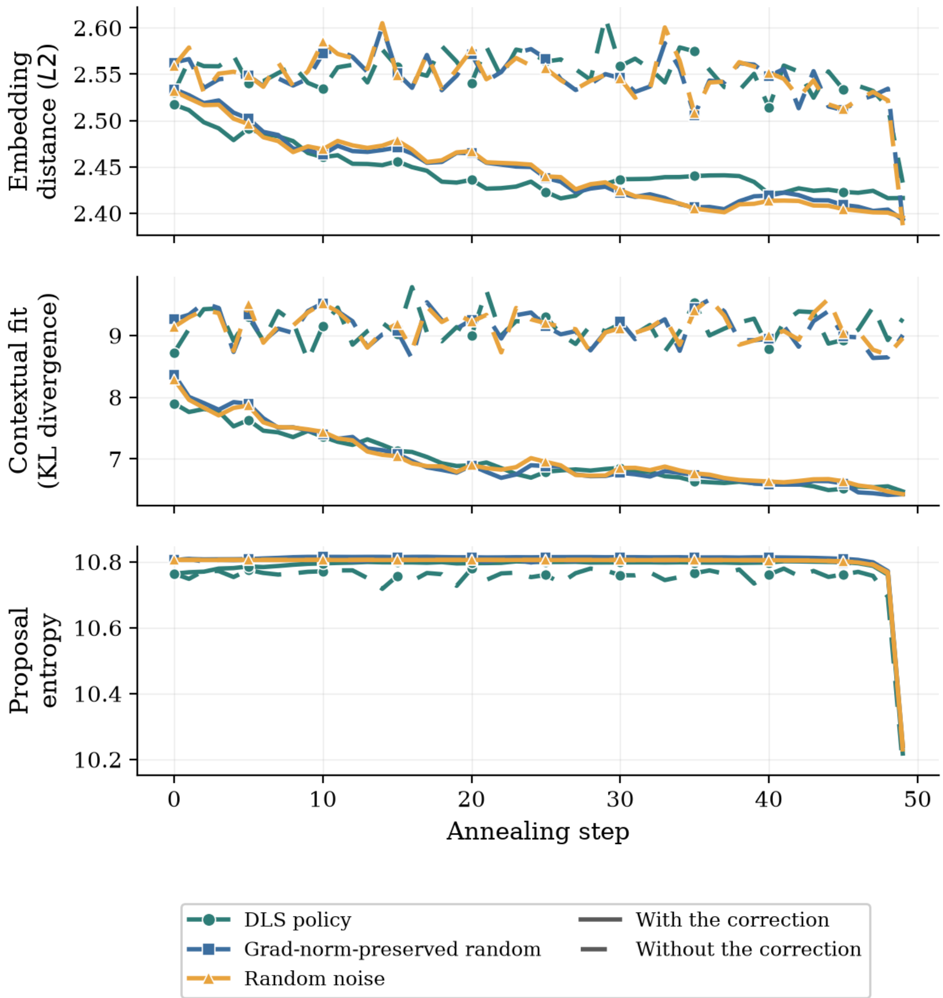

Figure 13: The 50-step schedule with gradient normalization disabled, in which the three proposal arms separate into distinct curves. Panels and axes as in Figure 1.

The measured wall-clock times are consistent with this accounting and give a sense of the absolute scale on the hardware of Section 4.8. The discrete sampler on the free-form continuation task took 2157 seconds for one hundred sequences, about 21.6 seconds per sequence over a fifty-step schedule; the full suite of gradient-free baselines, including the Gibbs sampler, took 215 seconds for two hundred sequences, roughly one second per sequence; and rescoring the six hundred recovered sequences with the external Llama-3 judge, a single forward pass each, took 24.8 seconds in

Method	Forward	Backward	Note
DLS / CLS, no correction	S	S	one gradient per step
DLS / CLS, with correction	$2S$	$2S$	extra gradient at the proposed point
Conditional argmax/sample	1	0	a single forward pass
Top-k rescore	$1 + k$	0	rescore k candidates
Gibbs	MT	0	M positions, T sweeps, no gradient

Table 16: Network evaluations per recovered span, counted as forward and backward network passes.

total. The comparison is stark: the gradient-free baselines reach comparable or better recovery quality at a fraction of the per-sequence cost, because they never pay for the backward pass whose product this thesis finds to be uninformative.

A.4 Geometry of Sampler Trajectories in Embedding Space

This section examines the spatial paths the samplers trace through the embedding space, a concrete counterpart to the acceptance statistics and trajectory summaries of Section 5.2. The primary figure is deliberately *not* a projection, since a two-dimensional projection of a 1280-dimensional space cannot carry the argument; the load-bearing figure reports exact $L2$ distances in the full space, using the full $50{,}257\times 1280$ embedding matrix rather than any subsample, and the projection is demoted to one illustrative panel.

Figure 14 is the primary view. For one trace, index 5, drawn once under a fixed random seed of zero and not reselected, it plots at every step, on a symmetric-log axis so exact zeros remain visible, the $L2$ distance from the sampler's state to the ground-truth token embedding, and for the continuous sampler also the distance to the nearest token of any kind. Because the trace-collection procedure advances the random-number state from one configuration to the next, trace index 5 is the same source sentence in every panel, but its randomly masked position, and hence its ground-truth token, differs by configuration; each panel therefore uses its own exact ground-truth embedding, and the cross-panel claim is about the on-manifold-versus-off-manifold dynamics, which does not require a shared token. The token strip under each axis shows the decoded token at each change, and for the continuous sampler the steps at which the proposal was rejected, inferred exactly from identical consecutive states, are marked as ticks.

The distances, which Section 5.2 summarizes, make the geometry exact. The discrete sampler ends about 2.2 units from the ground truth, a token or so away, and it shows the same dependence on the correction as the continuous one but without the off-manifold cost: with the correction it visits about 5 of 50 cells, and with the correction removed about 49, accepting almost every proposal, yet it never leaves the token manifold, so its wandering is among real tokens rather than through the empty space between them. The continuous state therefore spends its run in essentially empty space where the projected energy is a poor guide, which is why the boundary-crossing correction almost always rejects.

Figure 15 is the secondary, illustrative view: the same trace projected onto the first two principal components of the full vocabulary embedding matrix. It is shown for intuition only, and it cannot be the evidence: the two components capture just 3.3% of the embedding variance, so the projection discards almost all of the geometry the distances above measure exactly. Read for what it can show, it agrees with the distances: the discrete trajectories stay inside the vocabulary cone while the continuous ones escape far outside it.

Read together, the two figures make the central geometric claim visible: the samplers either fail to move in any directed way, the discrete sampler and the free continuous sampler both wandering without bending toward the ground-truth token, or fail to move at all, the corrected continuous sampler frozen far off the manifold, and in neither case does the path converge on the target, which is the spatial signature of a gradient that carries no directional signal.

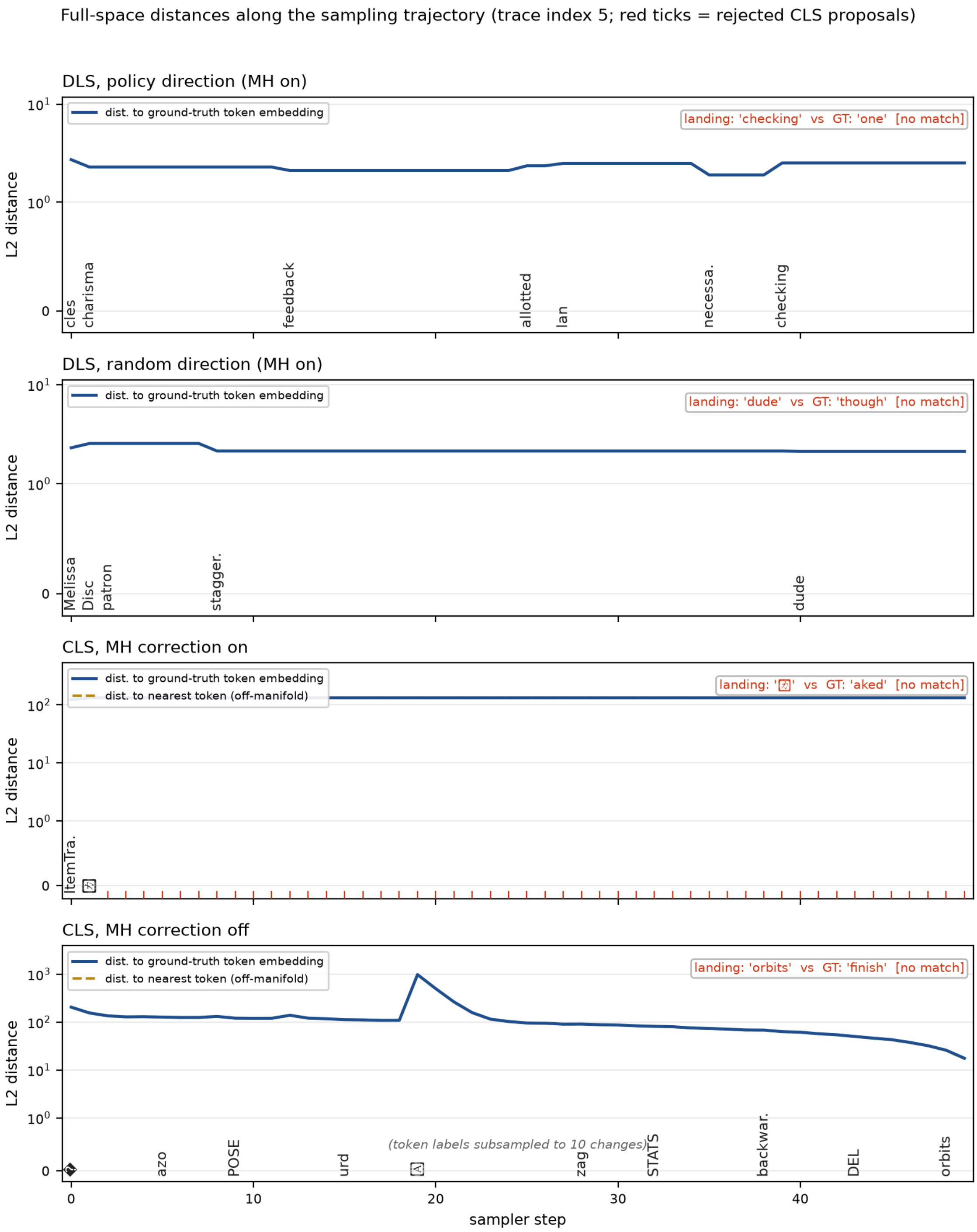

Figure 14: Full-space distances along the sampling trajectory for one trace, index 5, one panel per configuration.

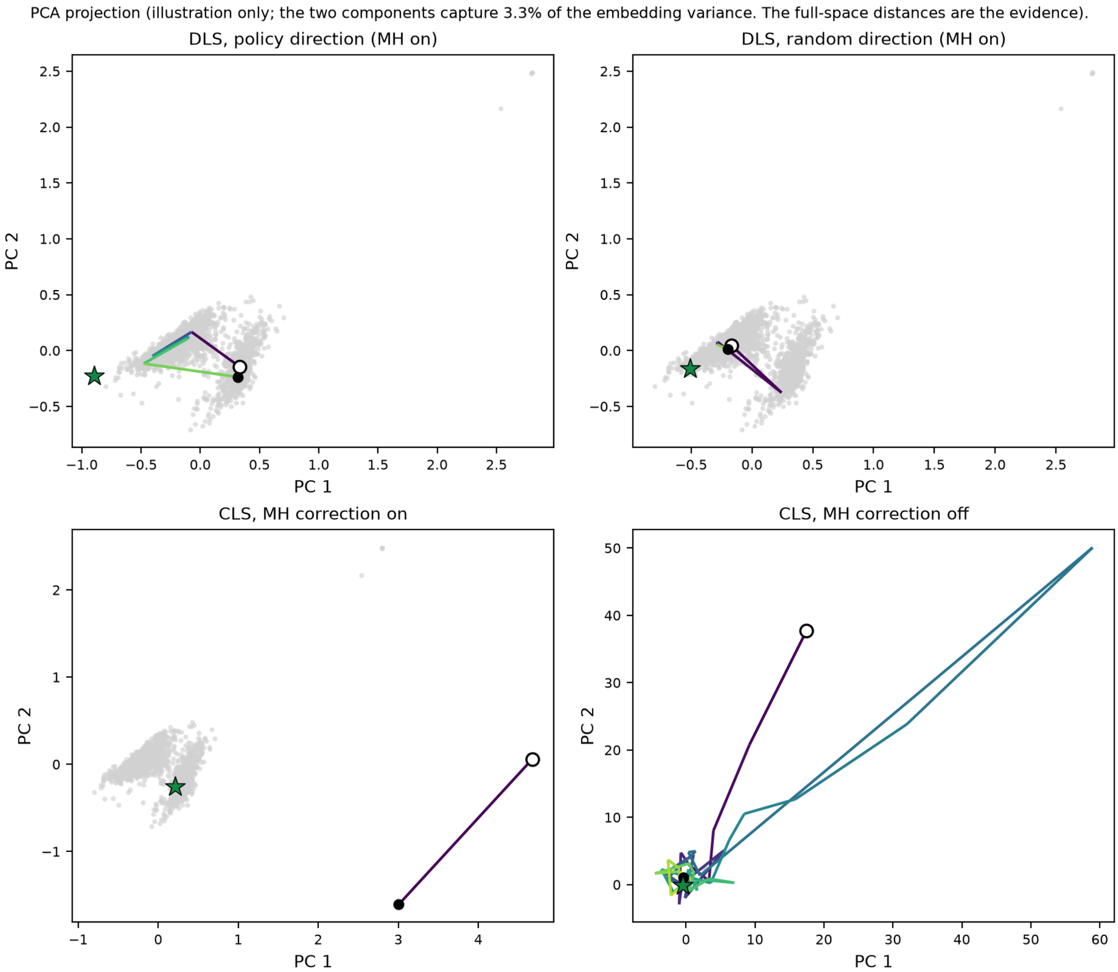

Figure 15: PCA projection of the same trace, one panel per configuration. Grey cloud: the vocabulary embeddings. Open circle: the start state. Filled dot: the end state. Star: the ground-truth token.

A.5 Classifier-Guided Steering: Full Results

This subsection reports in full the exploratory follow-up summarized in Section 5.10. It asks whether the constructive direction of Section 5.6 extends to *control*, which broadens the study from a sampling diagnosis into classifier-guided diffusion and evaluator alignment, and it is reported as exploratory throughout.

A small classifier is trained to read a noisy, partially masked state, the state the diffusion model occupies mid-denoising, by corrupting each labelled example with the model's own forward absorbing process at a noise level drawn uniformly over the schedule. Its held-out accuracy at low noise is 81%. During generation it reweights the diffusion model's top thirty-two candidates at each commitment by the probability of the target label, raised to a guidance strength γ. Steering is scored, exactly as in Section 5.10, by the frozen-model sentiment judge, so the classifier that guides and the judge that grades are strictly separate. Two settings were run: an off-domain continuation setting on prompts neither classifier was calibrated for, and an on-domain setting on held-out SST-2 prompts with a fluency trust region that caps guidance to candidates within $\delta = 5$ nats of the diffusion model's own top choice. Table 17 reports the second, and the sources for the four summary findings are as follows.

γ	target	unguided	guided	judge gain [CI]	self gain	NLL (un→guided)
2	negative	53.0	50.7	-2.3 $[-7.0, 2.3]$	+18.0	7.01 → 6.27
2	positive	47.0	53.7	+6.7 $[1.7, 11.7]$	+9.3	7.01 → 5.30
4	negative	53.0	52.3	-0.7 $[-5.3, 4.0]$	+25.7	7.01 → 7.28
4	positive	47.0	54.3	+7.3 $[2.7, 12.0]$	+17.7	7.01 → 6.01

Table 17: On-domain, trust-region guided diffusion generation on held-out SST-2 prompts. Judge gain is the guided minus unguided target-hit percentage; self gain is the guiding classifier's own verdict.

The claim the data licenses is therefore narrow and setting-contingent: with a single noisy guide at this scale, one-directional transfer to an independent judge is reachable under bounded fluency, while symmetric transfer at both labels is not. The off-domain run, the full agreement ladder, and the per-class confusion analysis follow; Appendix A.7 gives representative guided pairs from both settings.

A.5.1 Off-Domain Guidance Without a Trust Region

The first setting uses prompts from a MuCoLa-style continuation set that neither classifier was trained on, and imposes no fluency constraint. Table 18 reports it. The mechanism works on the instrument it optimizes, moving the guiding classifier's own verdict toward the target at every strength and for both labels by seven to eighteen points. Measured by the independent judge it steers monotonically in the negative direction, from a six-point gain at $\gamma = 1$ to a 16.7-point gain at $\gamma = 4$ whose interval excludes zero, and it does not move the judge in the positive direction. The two agree on only fifty-six to sixty-four percent of these texts, and stronger guidance drives them further off the fluent manifold, the per-token negative log-likelihood rising from 7.1 to 11.0 as γ grows. Guiding hard toward one classifier's notion of a label therefore need not raise the other's, which is the same off-manifold breakdown that Section 5.10 observed for the constraint-only arm.

	Judge target-hit (%)		Judge gain	Self gain	NLL/token
γ, target	unguided	guided	(pts, [CI])	(pts)	(un→guided)
1, negative	49	55	+6.0 $[-1.3, 13.3]$	+9.7	$7.1 \to 8.5$
1, positive	44	39	−5.3 $[-12.3, 1.7]$	+11.0	$7.2 \to 8.5$
2, negative	49	60	+10.7 $[3.3, 18.0]$	+7.7	$7.1 \to 9.7$
2, positive	44	44	+0.0 $[-7.0, 7.0]$	+12.0	$7.2 \to 9.6$
4, negative	49	66	+16.7 $[9.7, 24.0]$	+7.3	$7.1 \to 11.0$
4, positive	44	39	−5.3 $[-13.0, 2.7]$	+18.3	$7.2 \to 11.0$

Table 18: Classifier-guided diffusion generation on the off-domain continuation task. Columns as in Table 17.

A.5.2 The Guide-Judge Agreement Ladder and Per-Class Confusion

The off-domain result leaves two explanations entangled. Guidance might steer only in one fixed direction, or the two independently trained classifiers might disagree on off-domain text they were never calibrated for, with stronger guidance driving the text further off the fluent manifold where disagreement is worst. The continuation run varies domain and fluency together and cannot separate them. The on-domain control holds fluency fixed and moves the instruments on-domain: its prompts are the first three hundred held-out SST-2 validation sentences of at least twelve tokens, on which neither classifier trained, both having seen the split only for an accuracy readout with no parameter update, so it is calibration-clean by construction. The trust region caps guidance to candidates within $\delta = 5$ nats of the diffusion model's top choice at each commitment, so guidance can only reshuffle mass among near-equally-fluent options, and unguided and guided generations share a seed per prompt, so each pair differs only by the intervention.

The agreement between the guiding classifier and the judge climbs a clean ladder as the text they score moves on-domain: from the off-domain band of 56 to 64%, to 71.7% (95% CI $[66.7, 76.7]$) on the unguided on-domain generations, to 79.7% ($[75.0, 84.0]$) on the real held-out sentences themselves, where the judge and the guide reach 88% and 79.7% accuracy against the gold labels. The two classifiers are therefore not miscalibrated in a fixed way; they agree more the closer the scored text sits to the distribution they were trained on, which is what the off-domain diagnosis predicted.

The working direction flips between the two settings: off-domain the judge moved on the negative target and not the positive, on-domain the reverse, with fluency no longer a confound. That flip is what identifies the residual limit. It is not a fluency artifact, because the trust region removed the fluency cost and the asymmetry remained, and it is not a fixed unreachable sentiment, because the reachable direction moved when the setting changed. What remains is an instrument-alignment effect: the two independently trained classifiers still disagree on about 28% of the on-domain generations, and steering toward one classifier's notion of a label registers on the other only where the two align on the generated text.

One explanation of the flip is that the residual disagreement falls asymmetrically by class, so that the guide's positive verdict transfers more reliably than its negative one. Table 19 tests that on the unguided on-domain generations, which are neutral with respect to the intervention, and does not support it. The per-class accuracies rule out a silent label-map flip: the guide is 79.4% accurate on negative and 79.9% on positive held-out sentences, the judge 85.8% and 89.9%, both balanced. On the unguided generations the judge confirms the guide's positive verdict 67.9% of the time and its negative verdict 75.7%, a paired difference of -7.7 points whose 95% confidence interval $[-18.0, +2.6]$ includes zero, so the two conditional agreement rates are statistically indistinguishable. No by-class asymmetry explains the transfer direction, which is therefore reported as observed and setting-contingent, with no class-level mechanism claimed beyond the alignment gap.

		Judge verdict	
		negative	positive
Guide verdict	negative	109	35
	positive	50	106

Table 19: Per-class agreement of the guide and the independent judge on the 300 unguided on-domain generations.

A.6 Raw Steering Gains for the Constrained Extension

Table 20 gives the raw target-hit gains for the constrained-generation extension. They are placed here rather than in Section 5.10 because that section discounts them: every arm, including the pure-fluency and random-direction baselines, flips sign with the target label, which shows that a fixed unconditional sentiment drift dominates the raw gain. The trustworthy statistic is the paired contrast of Table 14, which cancels that shared drift.

A.7 Qualitative Examples

All qualitative material in this thesis is collected here under one selection policy, so that no example shown anywhere was chosen for how well it made a point. Every draw below is taken under a fixed random seed of zero and is unfiltered: dull, crude and failing draws are kept, and the only subsequent pruning is deterministic, a fixed positional subset of the draw, stated at each table. No number in this thesis is taken

Constraint mode	Steering gain	Final KL
Fluency only	0.0	8.558
Constraint only	−8.0	8.308
Combined	−1.0	8.541
Fully random	−7.0	8.512
Randomized constraint	+1.0	8.528

Table 20: Steering effect of the sentiment constraint under the discrete sampler on the infilling task.

from these tables; the statistics are in Table 6, Table 11, Table 17 and Table 19. Garbled bytes that the diffusion model emits are shown as `[?]`.

A.7.1 Classifier-Guided Generations

The pairs below make the guided steering of Section 5.10 visible in text. Table 21 gives the on-domain, trust-region generations, one pair per cell. For each pair g is the guiding classifier's verdict and j the independent judge's, so a pair where they disagree shows the partial instrument alignment directly.

Table 21: On-domain trust-region guided generations, the first pair of each cell.

Cell: $\gamma = 2$, target negative	
Prompt	bleakly funny, its characters all the more
Unguided	bleakly funny, its characters all the more often point to negation of value: will you have a sense of humor? or will you have [g:pos, j:neg; NLL 6.57]
Guided	bleakly funny, its characters all the more so, their than the the- a and b are the moreso. characters than- the the [g:pos, j:pos; NLL 7.92]
Cell: $\gamma = 2$, target positive	
Prompt	you don't have to know about music to
Unguided	you don't have to know about music to be musical– even if you don't know about music Audio 1 11 1 [g:neg, j:pos; NLL 5.36]

Guided	you don't have to know about music to play music you t if you don't listen to music about music music y y y n s y [g:pos, j:neg; NLL 7.10]
Cell: $\gamma = 4$**, target negative**	
Prompt	determined to be fun, and bouncy,
Unguided	determined to be fun, and bouncy, and quirky, and exciting, and colourful, and play-determined to be fun-determined [g:pos, j:pos; NLL 4.57]
Guided	determined to be fun, and bouncy, didn[?] wasd that to to to and havedd andd, wasd of to, [g:pos, j:pos; NLL 10.83]
Cell: $\gamma = 4$**, target positive**	
Prompt	pacino is brilliant as the sleep-deprived
Unguided	pacino is brilliant as the sleep-deprived - is adding already of a sleep-deprived (cowned) bristol is brilliant as is [g:neg, j:pos; NLL 8.40]
Guided	pacino is brilliant as the sleep-deprived-pacino is the sleep-deprivedpacpacpacpacino is brilliant as the - [g:pos, j:neg; NLL 8.92]

A.7.2 Infill Recovery on Seeded Sequences

SELECTION POLICY. The ten sequences were drawn once with a fixed rule and never filtered: ten indices sampled without replacement from the two hundred sequences of the reference set (WikiText-2 validation, one random interior mask, corruption seed zero) under a fixed random seed of zero, then sorted. Four of those ten, the first, fourth, seventh and tenth of the sorted draw, are shown here; the remaining six are not shown. Table 22 gives them. Failures are kept in, and nothing is selected for quality. A check mark means the recovered token equals the ground truth. Numbers cross-reference Sections 5.3.3 and 5.6.

Table 22: Infill recovery on four of the ten seeded sequences, all methods.

Method	Recovered	Hit	Recovered	Hit
Sequence 3, ground truth *fish*, corrupted to *moments. The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star__ and polychaete worms.* **Sequence 34**, ground truth *resided*, corrupted to *By. Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he__ in Meridian prior to 1971.*				
DLS policy (MH, GN)	passages		Client	
DLS random dir	fossils		Client	
CLS flagship (policy, MH, GN, free, s50)	specialization		Although	
Metropolized Gibbs	fish	✓	died	
Conditional argmax	fish	✓	was	
Conditional top-k rescore	fish	✓	served	
Left-conditional (independence MH)	fish	✓	sponsored	
SEDD-small recovery	lings		lived	
SEDD-medium recovery	lings		lived	
Hybrid (SEDD proposal, exact-energy MH)	fish	✓	resided	✓
Sequence 98, ground truth *a*, corrupted to *it. Late in the afternoon, Edson stepped onto__ grenade box and addressed his exhausted troops, saying,* **Sequence 199**, ground truth *1990*, corrupted to *ede. = = = Global security, late @-@__s = = =*				
DLS policy (MH, GN)	1		OR	
DLS random dir	it		o	
CLS flagship (policy, MH, GN, free, s50)	.		[?]	
Metropolized Gibbs	the		;	
Conditional argmax	the		date	
Conditional top-k rescore	the		’	
Left-conditional (independence MH)	the		tech	
SEDD-small recovery	the		@	
SEDD-medium recovery	the		@	

Method	Recovered	Hit	Recovered	Hit
Hybrid (SEDD proposal, exact-energy MH)	the		’	

A.8 Use of AI Tools

The author of this thesis has used AI tools, specifically Google AI Studio (Gemini) and Claude (Anthropic), and documents their use here as required.

They were used in three ways. First, and most substantially, for engineering support: the tools helped build the infrastructure that runs the experiments across several GPUs in parallel, and that generates the manifest files and the queueing mechanism which make the long configuration grids reported here practical to run. Second, for debugging, once the experimental code had been written by the author: the tools were used to check that the implemented samplers agreed with the mathematics they are derived from, and to locate the places where they did not. Third, on the manuscript itself, after the text had been written by the author: the tools were used through an agentic coding interface to edit the LaTeX sources and lay the document out quickly, and in the same pass to paraphrase and copy-edit drafts.

All other parts of the thesis, including the design of the experiments, the synthesis of the background and related work, the analysis and interpretation of the results, and the arguments and conclusions drawn from them, were carried out solely by the author, who has verified all quantitative claims against the underlying experimental output and assumes full responsibility for the entire content of this work.